STATISTICAL MODELLING OF UPPER RESPIRATORY TRACT INFECTIONS

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(*BSc (Hons.), NUS*)

A THESIS SUBMITTED

FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

SAW SWEE HOCK SCHOOL OF PUBLIC HEALTH

NATIONAL UNIVERSITY OF SINGAPORE

2017

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DECLARATION

I hereby declare that this thesis is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any

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01 January 2018

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to my main supervisor, *A/Prof Alex R Cook*, who has been providing constant guidance and unconditional support along my journey to PhD. He has been very responsive and helpful whenever I faced difficult situations in research work. As an inspirational supervisor and mentor, he has been advising on the overall direction of my research, bringing new ideas, helping to solve research problems and celebrating my achievements. Other than research work, he has also provided numerous advices on life and career. Without his guidance and support, I would not become the person I am today.

I would like to express my special thanks to my co-supervisors, *Dr Mark Chen* and *A/Prof Vernon Lee*, who encouraged me pursuing PhD studies, generously shared the research data with me, and have been guiding me throughout my PhD journey. I am exceptionally grateful to *A/Prof Vernon Lee* for providing me an opportunity to have a short-term work with the World Health Organization (WHO) on research agenda for influenza. This experience was very fruitful for me to meet many influenza experts around the world and learn from them.

Thanks to my Thesis Advisory Committee chairman, *Dr Tan Chuen Seng*, for his continuous support and remarkable suggestions. Thanks to my previous workplace, Centre for Infectious Disease Epidemiology and Research (CIDER), and my directors, *A/Prof Hsu Li Yang*, *Prof. Teo Yik Ying* and *Dr Vincent Pang* for supporting and sponsoring me in many great opportunities, such as international conferences and workshops, which helped me grow and enrich my experience.

Thanks to all teachers who have taught me during my PhD studies. Thanks to my ex-colleagues in CIDER and my PhD course mates who shared my happiness and sadness and always made me feel warm when I was working and studying in NUS.

Finally, my deepest gratitude to my family for their understanding and selfless support, especially my husband *Zhang Jinwei*.

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SUMMARY

Influenza infects 5%–30% of the global population every year, causing substantial health and economic burden. It is widely agreed that vaccination is the most effective method for preventing influenza infections and their complications. The formulation of influenza vaccine is revised annually because influenza virus undergoes antigenic drift frequently, emphasizing the need of a simple statistical method for evaluating the performance of vaccine in various settings. Research on the feasibility of using serological data to evaluate vaccine effectiveness is also insufficient. In this thesis, study designs were discussed, and statistical models were developed for evaluating the effectiveness of influenza vaccine using both serological and real-time polymerase chain reaction (PCR) assays in various settings. The test-negative design was found to be the most efficient study design with less bias than other non-interventional studies. A novel Bayesian statistical model was developed based on the serological data to evaluate the performance of seasonal influenza vaccines in the context of interventional studies. The model could adjust for the staggered recruitment of study subjects and the waning effect of vaccine. The results also showed that a titer of 1:40 reduced infection risk by 40%–70% relative to a titer of 1:10, depending on the circulating strain.

The timing of influenza vaccination programme may largely impact its effectiveness. The World Health Organization (WHO) recommends that vaccination programmes should be implemented several weeks ahead of the influenza epidemic season to improve effectiveness and mitigate the impact. However, the lack of understanding of influenza seasonality in the tropics complicates the planning of vaccination programmes. A Bayesian hierarchical model was developed to characterize the dynamics of infection-induced influenza immunity at both individual and population level, and simulations were conducted to examine the association

between herd immunity levels and influenza seasonality in the tropics. The results showed that infection-induced antibody levels peaked 4–7 weeks, and decayed to pre-infection levels approximately in one year, which suggests a need of annual influenza vaccination. The population-level effect of individuals' waxing and waning antibodies were also found to correlate with influenza seasonality in the tropics, providing new insights for better understanding of influenza seasonality.

Estimation of influenza attack rates is crucial to healthcare policy makers to plan for future vaccination programmes and to assess past responses. However, 10%–60% of the influenza infections, based on estimation, are asymptomatic. Serological assessment of influenza infections may provide a more accurate estimation of influenza attack rate because it allows diagnosis of sub-clinical or misdiagnosed influenza cases. A Bayesian bivariate probit model was developed to estimate influenza attack rates using paired sera and to assess the accuracy of current serological methods for determining influenza infection. The results showed that the current serological methods might have substantially underestimated the infection rates. In addition, the estimated infection probability matrix conditioning on antibody levels, derived from the developed model, provides a simple template for more accurate estimation of influenza attack rate if serological data are available.

(484 words)

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LIST OF ABBREVIATIONS

ARI	-	Acute respiratory illness
ARV	-	Attack rate in the vaccinated group
ARU	-	Attack rate in the unvaccinated group
CAR	-	Cumulative attack rate
CI	-	Confidence interval
CrI	-	Credible interval
DALYs	-	Disability-adjusted life-years
FRI	-	Febrile respiratory illness
GMT	-	Geometric mean titers
GP	-	General practitioner
HA	-	Hemagglutinin
HAI	-	Hemagglutination-inhibition
HK	-	Hong Kong
ILI	-	Influenza-like illness
i.i.d.	-	Independent and identically distributed
MCMC	-	Markov chain Monte Carlo
MEC	-	Multi-ethnic Cohort
MLE	-	Maximum likelihood estimation
NA	-	Neuraminidase
NPHL	-	National Public Health laboratory
NPV	-	Negative predictive values
OR	-	Odds ratio
OR_{adj}	-	Adjusted odds ratio
PCR	-	Polymerase chain reaction
pp		Percentage points
PPV	-	Positive predictive values
RCT	-	Randomized controlled trials
RJMCMC	-	Reversible jump MCMC
RT-PCR	-	Reverse transcription polymerase chain reaction
RR	-	Relative risk
SAF	-	Singapore Armed Forces
SCCS	-	Singapore Consortium of Cohort Studies
TIV	-	Trivalent inactivated vaccine
US	-	United States
VE	-	Vaccine effectiveness
WHO	-	World Health Organization

LIST OF PUBLICATIONS

This thesis is based on the following published papers:

1. **Zhao X**, Fang VJ, Ohmit SE, Monto AS, Cook AR, Cowling BJ. Quantifying Protection Against Influenza Virus Infection Measured by Hemagglutination-inhibition Assays in Vaccine Trials: Epidemiology. 2016 Jan;27(1):143–51. [IF: 6.075]
    a. Oral presentation, 1st International Meeting on Respiratory Pathogens, Singapore, 2 – 4 September 2015.

2. **Zhao X**, Ning Y, Chen MI-C, Cook AR. Individual and population trajectories of influenza antibody titers over multiple seasons in tropical Singapore. Am J Epidemiol. 2017 Jun;187(1): 135–143. [IF: 5.595]
    b. Oral presentation, Options IX for The Control of Influenza, Chicago, United States, 24 – 28 August 2016.

3. **Zhao X**, Siegel K, Chen MI-C, Cook AR. Rethinking thresholds for serological evidence of influenza virus infection. Influenza Other Respir Viruses. 2017 May;11(3):202–10. [IF: 2.677]
    c. Poster presentation, Options IX for The Control of Influenza, Chicago, United States, 24 – 28 August 2016.

Relevant work:

1. Ho HP, **Zhao X**, Pang J, Chen MI-C, Lee VJM, Ang LW, et al. Effectiveness of seasonal influenza vaccinations against laboratory-confirmed influenza-associated infections among Singapore military personnel in 2010–2013. Influenza Other Respir Viruses. 2014 Sep 1;8(5):557–66. [IF: 2.677]
    d. Oral presentation, Lunch Symposium at the 8th Singapore Public Health & Occupational Medicine Conference, Singapore, 29 – 30 August 2013.

2. Tan XQ*, **Zhao X***, Lee VJ, Loh JP, Tan BH, Koh WH, et al. Respiratory viral pathogens among Singapore military servicemen 2009 - 2012: epidemiology and clinical characteristics. BMC Infect Dis. 2014 Apr 15;14(1):204. [IF: 2.768]
    e. Poster presentation, Options VIII for The Control of Influenza, Cape Town, South Africa, 5 – 10 September 2013.

*Authors contributed equally to the work

CHAPTER ONE: INTRODUCTION

1.1 History and epidemiology of influenza pandemics

Acute respiratory illness (ARI) is a disease of global interest as it is one of the leading contributor to the loss of disability-adjusted life-years (DALYs) [1]. Influenza is an ARI caused by influenza viruses that are common respiratory pathogens which have circulated amongst humans for thousands of years [2]. Influenza viruses are contagious and can rapidly spread from individual to individual and country to country. Influenza pandemics may happen when a new influenza A virus emerges through antigenic shifts or reassortment of existing strains, and the population has little or no immunity against the novel influenza virus [3,4].

Influenza pandemics occurred several times in the past 100 years resulting in millions of deaths [5]. The first generally agreed and documented influenza pandemic occurred during the summer of 1580 [6]. This pandemic likely originated from Asia, and rapidly spread to Africa, Europe and then North America within six months [6]. All parts of Europe were infected resulting in 8,000 reported deaths in Rome alone and the decimation of local populations in some Spanish cities [7,8].

More influenza data have become available since 1700 due to increased quality and quantity of medical writing. Five major influenza pandemics were recorded in the 18th and 19th centuries [9]. The first influenza pandemic of the 18th century started in April 1729, originated in Russia, and spread to the entire continent of Europe within six months [6]. It then continuously spread to the rest of world over a period of three years infecting people of

all ages. High mortality rates were reported among pregnant women and the elderly [10]. Some 40 years later, the next generally agreed influenza pandemic started in China between 1781 to 1782 with a high attack rate reported [6]. The virus infected mainly young adults, and spread from China westwards to Russia and then to the entire Europe over a period of eight months [6].

In the 19th century, following the influenza pandemics in 1781, another three major influenza pandemics were recorded in periods of 1830–1833, 1889–1890 and 1898–1900. Similar to the influenza pandemics in 1781, the influenza pandemic in 1830–1833 spread rapidly from China to the Philippines, Indonesia, India and Russia, and then reached Europe and North America with a high attack rate and morbidity rate documented [11]. The 1889–1890 influenza pandemic was one of the most widespread influenza pandemics infecting 25% to 50% of the global population, resulting in an estimated deaths of 300,000 worldwide [5]. It was the earliest influenza pandemic with serological data collected, which suggested that influenza A/H2 was the virus responsible. The last influenza pandemic in the 19th century occurred in 1898–1900 due to influenza A/H3 supported by the serological data [5]. The starting place of this pandemic was unknown, but it spread across Europe, Americas and Australia.

Influenza pandemics were documented three times in the 20th century. The pandemic of 1918–1920 during the World War I was the first influenza pandemic recorded in the 20th century, and was also one of the most destructive events in medical history [12]. It was known as the "Spanish flu" because Spain was not involved in the World War I, and this influenza pandemic received a large amount of media attention in Spain, while news of war dominated the media in other European countries [12]. The origin remains unclear; however,

China and North America were considered as the possible places for the origin of the virus [5]. It was known that the pandemic virus was carried by the American military, and spread to France when they reached there in early 1918 [5]. In the next three months, it spread rapidly across Europe and subsequently to Africa, Asia and Oceania in the following six months [5]. The pandemic infected an estimated 500 million people in the world with a high case-fatality rate of > 2.5% leading to a total of 50–100 million deaths [13]. More people died because of pandemic influenza viral infections than those died in the World War I, and almost half of the deaths were among young healthy adults aged between 20 and 40 years old [13]. The 1918 influenza virus was an A(H1N1) strain, and was considered as the genetic ancestor of all three influenza pandemics following the 1918 influenza pandemic because the three influenza pandemics occurred in 1957, 1968 and 2009 were almost all caused by descendants of the 1918 influenza virus which were evolved through virus antigenic drift and/or reassortment [13,14]. Although the Hemagglutinin (HA) and neuraminidase (NA) genes were different between the 1918 and the 1957/1968 influenza viruses, the 1957/1968 influenza viruses have five RNA segments that were preserved from the 1918 influenza virus [14].

The next 1957 influenza pandemic was caused by a reassorted human and avian influenza virus, i.e. A(H2N2), which was named as Asian flu pandemic because it started in Yunnan province, China [12]. It took less than six months for the virus to spread from China to the rest of the world mostly along shipping routes [12]. School-aged children and young adults who were heavily infected showed the highest mortality rate compared to other age groups [15]. Although the case-fatality rate of this pandemic was < 0.5%, which was lower than that of the Spanish flu pandemic in 1918, more than one million people died during the pandemic [16].

Similar to the 1957 Asian flu pandemic, the 1968 influenza pandemic spread from China to the whole world, and was caused by a reassorted human and avian influenza virus, i.e. A(H3N2) [12]. It was known as the "Hong Kong flu" because of its rapid spread in Hong Kong, which infected 500,000 people within two weeks [12]. Vaccine was manufactured two months after the identification of the pandemic virus, but vaccine did not show effective control of the pandemic possibly due to the late deployment of the influenza vaccine [12]. The morbidity and mortality of the Hong Kong flu pandemic was comparable to the 1957 Asian flu pandemic resulting in 0.5 to 1 million deaths [5]. A higher attack rate was observed among children and young adults because the elderly might have previous exposure to the similar influenza virus during the 1898–1900 pandemic supported by the serological data [12].

The most recent influenza pandemic H1N1-2009 in the 21st century began in the state of Veracruz, Mexico in 2009 [17]. It quickly spread to all continents within 9 weeks. Genetic analyses showed that the pandemic A(H1N1) 2009 virus was genetically and antigenically unrelated to the human seasonal A(H1N1) virus that has circulated in humans since 1977, but was genetically related to animal influenza viruses circulating in swine, for example the 1918 influenza virus [18,19]. The attack rate of the pandemic virus ranged from 1% to 33% depending on age group and settings [20]. The greatest attack rate was observed in children and young adults in semi-closed or closed settings, such as boarding schools and military camps [20]. More than 200,000 people died due to the pandemic (50% of the deaths occurred in Southeast Asia and Africa), which was relatively milder in impacts compared with the last influenza pandemic [20]. The new pandemic virus has now replaced the previous A(H1N1)

circulating strain and become one of the seasonal influenza viruses during the influenza epidemic in the Southern hemisphere in the same year [21].

Analyzing past influenza pandemics is important, which may help in better prediction and control of future pandemics. The time interval between pandemics in the 20th and 21st centuries, when there was increased travel and trade within and between countries, suggests a gap of 40–50 years on average, but it can also occur with a small gap, e.g. 10 years [5]. Most of the past influenza pandemics started in China, Russia or other places in Asia where birds, pigs and humans were in close proximity creating more opportunities for virus reassortment of influenza A strains [5]. The reassortment could happen any time of the year even in temperate regions [5]. The transmission of influenza pandemic virus is rapid because the majority of people had little or no pre-existing immunity against the new virus [5]. The attack rate for a pandemic influenza virus could range from 10% to 40% [18,22], and it is more likely to attack children and young adults including healthy ones as the elderly are more likely to have previous exposure to a similar virus in their early years [12].

1.2 Epidemiology and transmission of seasonal influenza

Unlike pandemic influenza that rarely happens, seasonal influenza happens annually with distinct seasonal patterns in temperate regions [23]. It attacks an estimated of 5%–30% of the global population every year [24], and approximately 40%–90% of the infections are symptomatic [25–31]. Typical symptoms of influenza infections are fever, cough, sore throat, running nose and blocked nose [32]. Symptoms usually start to occur in two days after an infection and typically last for 7 to 10 days [33,34]. The severity of seasonal influenza infections varies significantly [35]. For most healthy adults, seasonal influenza infections only cause mild symptoms, and are self-limiting [35]. However, pregnant women, people

with underlying conditions, the very young and very old have a higher risk of severe influenza infections that may lead to death [35].

There are three main types of influenza viruses circulating in humans: influenza A, B and C. Only influenza A and B cause seasonal influenza epidemics annually. Although influenza C also causes influenza symptoms, it is usually very mild and has much less disease burden than influenza A and B have [36]. The seasonal influenza A virus can be further categorized into two subtypes, i.e. A(H1N1)pdm09 and A(H3N2), where A(H1N1)pdm09 and A(H3N2) are the same influenza viruses that caused influenza pandemics in 2009 and in 1968, respectively [37]. The circulating seasonal influenza B virus has two lineages: B/Victoria and B/Yamagata that are named after the places where the two strains were identified [38].

Respiratory droplets and aerosol transmission are suggested to be the main mode of transmission of influenza infections according to previous studies [39–43]. A household modelling study found that approximately half of all transmission events in the household was attributed to aerosol transmission [40]. Droplets and aerosol transmission travel only short distances through air [40]. People who had a close contact with an infected person, usually within 2 meters, are often considered to be at risk of infection [44].

Factors that determine the risk of influenza transmission include susceptibility to infection, pre-existing immunity from vaccination and previous exposure, and personal hygiene [45–47]. Research studies showed that preschool children have a greater risk of influenza infections, probably because of their poorer respiratory hygiene and a lack of pre-existing immunity [48,49]. Moreover, patients with underlying co-morbid conditions and patients who are immunocompromised were found to have a higher frequency of influenza infections and

were usually associated with a higher risk of severe illness because of slower viral clearance [50–52].

Transmission of influenza infections also varies in different settings. The secondary attack rate of household transmission can be up to 38%, and mothers and children are more likely to be infected than fathers [30,53]. In schools, with increased intra- and inter-class activities, a higher attack rate and R_0 than those in community settings were observed in previous studies [54,55], where R_0 represents the basic reproduction rate that measures the number of secondary infections produced by a typical case. Semi-closed and closed environments, such as military camps, transport facilities and conferences, also have high attack rates of influenza infections, and could have 40%–70% of people infected in an outbreak [56–59]. Influenza transmission was thought to be efficient in hospitals because hospitalized patients with influenza have been observed to be more infectious than those in community cases [60], and influenza virus shedding correlates with influenza severity and duration [26,27,30,61,62], which may increase transmission of influenza in hospitals. However, a recent study in Australia has found that hospital-acquired influenza infections are uncommon but are usually associated with severe complications due to patients' pre-existing conditions [63]. Similar results were found in a UK study, but the results may only apply to developed countries [63].

1.3 Seasonality of seasonal influenza infections in different regions

Seasonal patterns of seasonal influenza epidemics vary in different regions, and the mechanism of the patterns is not well understood. Seasonal influenza epidemics occur during colder winter months with a single peak in temperate countries [64]. However, in the tropics, which lack defined seasons, influenza can have a high baseline level of activity with multiple

peaks [65–69]. New studies have found that semi-annual influenza activities with peaks occurring in winter and summer have been observed in several temperate locations and many tropical locations in Southeast Asia, with year-to-year variation in intensity [70–72].

In previous studies, different seasonality patterns were hypothesized to relate to the timing of the global circulation of influenza viruses [73]. Influenza viruses continuously circulate in the tropics, especially in South-East Asia, resulting in overlapping epidemics and seeding of epidemics in temperate regions in both northern and southern hemispheres [73]. More recent studies found that the seasonality of influenza activities may also change with latitude and climate [67–69,74]. In Asia, countries located above 30°N have influenza A epidemics during winter seasons; those below 30°N and above 10°N have seasonality during monsoon seasons; while those near the equator have year round circulation [67–69,74]. The seasonality of influenza B epidemics coincides with the seasonality of influenza A in areas above 30°N, but there is a year round circulation of influenza B activities in areas below 30°N [69]. Reasons for the difference in the seasonality of influenza A and B are unclear. In Africa, where seasonal influenza characterization is limited, the main season of influenza activities occurs between October and January in northern Africa, between April and August in Southern Africa, between July and October in East Africa, and throughout the year in tropical Africa [75,76]. Environmental factors have also been associated with the seasonality of influenza [67]. Cold temperature, low specific humidity and minimal solar radiation facilitate influenza transmission in temperate regions, while high humidity in the rainy season is associated with influenza activities in some tropical or subtropical regions [77,78]. Studies have shown that the viability of the influenza A virus is the highest if the humidity is close to 100% or, conversely, below 50% [79]. Among these factors, the annual variation in climatic factors was found to be the most important in influencing the seasonality of influenza [80].

There are still knowledge gaps in the seasonality of influenza transmissions, particularly, in the tropics because of year-round influenza activities and a lack of climate patterns in these regions [81]. Understanding the pattern and mechanisms of the spread of influenza epidemics may help to plan and improve the effectiveness of vaccination campaigns.

1.4 Public health and pharmaceutical interventions to limit influenza transmission

1.4.1 Public health interventions

Public health interventions including both individual-level and community-level interventions have been implemented widely in limiting influenza transmissions [82]. They have been shown to be useful, particularly, for slowing the spread of pandemic influenza viruses and controlling influenza pandemics in which influenza vaccines are not sufficient in quantities within short period of time [82].

Practicing good personal hygiene, such as washing hands frequently and avoiding touching eyes, nose and mouth with unwashed hands, (has long been recommended) as an individual-level intervention to prevent the droplet-borne transmission of influenza infections. Research evidence has shown that hand hygiene interventions could decrease respiratory tract infections by 6%–44% in various settings [83]. An internet-delivered handwashing intervention study also reported significant decrease in the transmission of respiratory tract infections [84]. Mask use is also recommended, and is best to be undertaken as part of a package or ‘bundle’ of personal protection including hand hygiene [85]. A significant decrease in the transmission of influenza infections was observed in a Hong Kong household

study if surgical masks and hand hygiene were implemented within 36 hours of the disease onset [86]. However, the effectiveness of masks/respirators is likely linked to consistent, correct usage and compliance, which remain a major challenge [86]. Ongoing training and education on the importance of personal hygiene and the correct use of mask for the public and healthcare workers to prevent influenza infections are warranted [87].

Community-level measures that include contact tracing and quarantine of close contacts, school and workplace closures, reduction of mass gatherings, border temperature screening, travel restrictions and surveillance are often considered and used by policy makers in mitigating influenza transmissions, particularly, during influenza pandemics when a large number of people are infected and significant health and economic burdens are imposed [88]. Since the influenza pandemic in 2009, there has been an increasing number of publications documenting the type of public health interventions used in the pandemic and the effectiveness of these interventions. Contact tracing and quarantine of contactees have been used to contain influenza transmission, however, their effectiveness remain unclear. The closure of all schools and workplaces is another approach that would reduce the overall attack rate by 95%, however, this measure had significant economic impact that could exceed the impact of an influenza pandemic [89,90]. Moreover, research studies have shown that school closures will not be effective if it is implemented after ILI infections are widespread [91–94]. Currently there is still insufficient evidence to recommend a particular school closure policy (e.g. proactive or reactive) over another. The timing and duration of closure required to produce an effect in delaying the spread and overall impact of influenza are also unclear. Reducing mass gatherings have also been recommended to prevent and contain influenza transmissions during influenza epidemics and pandemics [95]. The type, size, duration and setting (i.e. indoor/outdoor) of such events may play an important role in the

risk of influenza transmission [95]. Mass gathering facilitates the transmission of influenza if there are more than 5 people per square meter and if participants stay close together for prolonged periods such as during the hajj, musical festivals and large international conferences [95]. However, major sporting events such as the World Cup and Olympics may not have a substantial effect on the spread of influenza infections because of the open-air setting, good between-seat spacing and prompt management of influenza transmissions, e.g.. use of oseltamivir and active surveillance of influenza infections [95–98]. Additionally, establishing surveillance systems for tracking influenza activities has been shown to be important to understand the epidemiology and key features of circulating influenza viruses and to inform optimal interventions to assist epidemic/pandemic preparedness planning [99–101]. Many countries have established surveillance networks for detecting influenza/ILI infections, and there was a large increase in the number of publications on influenza and ILI surveillance since the 2009 A(H1N1) pandemic [99–101]. These published studies have found that the ILI surveillance using electronic health records shows high sensitivity and specificity, and exhibits similar quality to manual sentinel surveillance and laboratory-confirmed virologic data [99–101]. The development of biologically complex models that track the evolution of influenza viruses in animals can also inform optimal interventions to assist pandemic preparedness planning [102].

1.4.2 Pharmaceutical interventions

Pharmaceutical interventions to contain influenza epidemics and pandemics include anti-viral drugs and influenza vaccination [103]. Oseltamivir is an example of anti-viral drugs that is effective in releasing symptoms of influenza A and B infections, which can also prevent influenza infections after exposure [104]. A Singapore study has shown that the use of

oseltamivir among those who have been exposed to influenza infections was effective in reducing influenza activities, together with active surveillance and isolation of infected cases in semi-closed settings during the influenza A(H1N1) pandemic in 2009 [105]. However, oseltamivir might cause nausea and vomiting headache and psychiatric syndromes [106]. Influenza vaccination is another type of pharmaceutical intervention, and is widely agreed to be the best way to prevent influenza infections and annual seasonal influenza epidemics [35]. It works by stimulating antibodies in the human body to protect against the influenza viruses that are in the vaccine [107]. The vaccine usually takes effect about two weeks after injection, and its effectiveness gradually decay to approximately half of the peak level in six months [108]. By one year, the vaccine-induced antibodies will be back to pre-vaccination levels [109] which may suggest an annual vaccination for optimal protection. In addition, influenza viruses often undergo antigenic drift frequently [110]. The existing repertoire of antibodies against the previous circulating strain may not be specific enough to sufficiently neutralize the current strain. Thus, the formulation of influenza vaccine is reviewed and updated every year to keep up with changing influenza viruses [110]. However, this process is imperfect because influenza experts need to pick the viruses to be included in the vaccine many months in advance for on-time production and delivery of influenza vaccine [111]. This leads to a year to year variability in the efficacy and effectiveness of the vaccine formulation. Annual influenza vaccination is recommended to people who have higher risk of influenza infections and those who are more likely to develop severe complications after an influenza infection, such as children aged 6 months–4 years old, people aged 50 years and older, people who are immunosuppressed, people with underlying chronic conditions, and pregnant women [112].

The traditional seasonal influenza vaccines, i.e. trivalent influenza vaccine, are made to protect against three seasonal influenza viruses including influenza A(H1N1)pdm09,

A(H3N2) and B. The formulation of the seasonal influenza vaccines is reviewed and updated every year to better match the circulating influenza viruses because influenza viruses are constantly changing from year to year ("antigenic drift") [113]. Antigenic drift occurs when there are some minor changes in the protein structure of a seasonal influenza virus due to genetic mutation [114]. Through this process, a new seasonal influenza strain will be created that is often associated with seasonal influenza epidemics because the small change may be just enough for human antibodies, which are generated against the older viruses, to fail in recognizing and responding to the newer virus [114]. Recent studies have shown that influenza vaccine effectiveness was about 40% to 60% during influenza epidemic seasons when the vaccine strains are well-matched with the circulating influenza strains [113]. However, studies that evaluate influenza vaccine effectiveness are required to be conducted each year to assess the performance of the vaccine and to evaluate the benefits of influenza vaccination programs because the influenza vaccine is adjusted annually.

1.4.3 Planning and selection of public health and pharmaceutical interventions

The selection of interventions used for an influenza epidemic or pandemic is challenging for policy makers and public health professionals because the selected interventions should be effective, responsive to the setting, i.e. seasonal epidemic vs pandemic, and features of the influenza viruses, and should not lead to greater impact in terms of social and economic burden compared to the epidemic/pandemic itself. For example, if transmission occurs mainly in adults during an influenza pandemic, school closure will have much less effect than in the situation where majority of transmission occurs in young children [115], and it may also create significant societal and economic burdens.

The transmission dynamics, the severity of the influenza infection and the extent of spread are important factors that need to be considered to determine the urgency of response and appropriate intervention strategy. For example, more extreme and comprehensive interventions should be warranted for an epidemic caused by an influenza virus with higher morbidity/mortality and attack rate than those for a relatively milder pandemic. These factors, however, are largely unknown at the early stage of an influenza pandemic or an epidemic caused by a new seasonal influenza strain. A good and effective surveillance system coupled with appropriate statistical models may allow for quick assessing and understanding of key features for consideration in selecting interventions to be recommended.

The timing of interventions has impact on the effectiveness [116]. Previous studies showed that anti-viral pre-exposure prophylaxis on healthcare workers was only effective if the deployment of prophylaxis was timed against the peak of the pandemic [116]. Recent studies also found that school closures may only have significant impact if these interventions are implemented before widespread infections [91,93]. Moreover, the timing of vaccination is important to maximize its effectiveness, and studies have shown that optimal vaccination allocation should be before the onset of epidemic or pandemic [117]. However, the lack of understanding of the seasonality of influenza epidemic, particularly, in the tropical regions, unavailability of sufficient vaccine and antivirals, and poor public health and clinical infrastructures for rapid deployment of vaccination always limit the effectiveness of vaccination in alleviating the burden of an influenza epidemic/pandemic [118].

Furthermore, social demographics and social-cultural characteristics are also important factors that should be studied in the local context because public perceptions on interventions

vary from country to country [119,120]. Recent studies found that hand and respiratory hygiene are well-accepted concepts, while personal distancing, wearing of masks and vaccination in some contexts may cause concerns [120,121]. Personal and cultural beliefs, misconceptions in terms of transmission and preventive measures are the key barriers in adopting non-pharmaceutical interventions, while the concern on the safety of vaccine and antivirals, and religious incompatibility are associated with negative sentiment towards pharmaceutical interventions [119–121]. Knowing the social demographics and social-cultural characteristics in the local context, therefore, will guide appropriate interventions and increase the compliance with selected interventions.

1.5 Overview of Thesis

1.5.1 Knowledge gaps and motivation

As discussed in section 1.4, many preventive measures are available today, which can be used to limit influenza transmission during an influenza epidemic/pandemic. Of these, vaccination has been demonstrated to be both medically and economically compelling for all age groups in mitigating the overall impact of influenza epidemics [122–127]. Annual vaccination is, therefore, strongly recommended to protect general populations from influenza infections especially for people at higher risk of complications of influenza [128]. A simple and fast approach for assessing influenza vaccine effectiveness in various settings, therefore, is necessary to evaluate the performance of vaccine and to demonstrate the importance of influenza vaccination.

In situations when only serological data are available to evaluate influenza vaccine effectiveness, hemagglutination inhibition (HAI) titers are often used to study the vaccine-

induced protection [129]. The dynamism of HAI titers against the circulating strain, however, may complicate the understanding of their protective effect, for they may change substantially over the course of a single study [130]. Due to limited resources, only 3–4 blood samples, e.g. pre-vaccination, post-vaccination and end-season blood samples, are collected in most studies [129,131–133]. This set of samples may not adequately characterize the antibody levels at the time of exposure to infection to quantify the protection conferred by antibodies. Future research is needed to understand and to model the evolution of HAI titers post-infection/vaccination.

Moreover, evaluating the performance of an influenza vaccine does not only depend on the vaccine coverage (the proportion of vaccinees in a population) and vaccine formulation. Selecting the best timing of vaccination campaigns, which should be several weeks before the onset of influenza outbreaks, is important to improve vaccine effectiveness [134]. Timing of vaccination programmes is relatively straightforward in temperate regions where influenza epidemics routinely occur in the winter [78]. In contrast to temperate regions, tropical regions have no winter season, and influenza epidemics occur year-round in the tropics, complicating the planning of vaccination programmes. The seasonal mechanisms on the equator remain unclear. A possible explanation of the lack of seasonality could be that the timing of influenza epidemics may be driven more by changes in herd immunity rather than by the synchronization with climatic patterns. Therefore, knowing the protection conferred by different antibody levels as measured by hemagglutination-inhibition (HAI) titer or a similar assay, may shed light on the seasonality of influenza in tropical/sub-tropical regions [78,135].

Finally, in the event of an influenza pandemic or a seasonal influenza epidemic due to a new influenza strain, the world may require timely production and deployment of influenza

vaccine in sufficient quantities to control rapid influenza transmission. The total number of doses of influenza vaccine that needs to be distributed mainly depends on the anticipated overall attack rate of influenza infections in a population. However, statistical models that allow quick estimation of influenza attack rate during an epidemic/pandemic using epidemiological data are limited. The development of fast and efficient statistical methods is important to advise the most effective allocation of influenza vaccines.

1.5.2 Aim and objectives

In this PhD thesis, the aim is to provide a simple approach to estimate the effectiveness of the seasonal influenza vaccine using ILI surveillance data in an observational study and to develop statistical models to estimate vaccine effectiveness using serological data in randomized controlled trials (RCTs). Moreover, this PhD thesis also develops novel statistical models to study the correlation of the dynamics of herd immunity and the seasonality of influenza epidemics, and to estimate influenza attack rate to guide future planning of prevention programme in the tropics.

This proposed thesis fills current knowledge gaps and assists in evaluating and optimizing influenza vaccination to guide effective vaccination policy.

The specific aims of the following four chapters are:

1. Chapter 2: To discuss various study designs in estimating influenza vaccine effectiveness, and to demonstrate methods of estimating influenza vaccine effectiveness in a test-negative design using statistical software. The example used in this chapter has been

published in *Influenza and Other Respiratory Viruses*, Year 2014, Volume 8, Issue 5, page 557–66.

2. Chapter 3: To quantify protection against influenza virus infection adjusting for waning in immunity, and to evaluate the performance seasonal influenza vaccine using serological data in RCTs. This chapter has been published in *Epidemiology*, Year 2016, Volume 27, Issue 1, Page 143–51.

3. Chapter 4: To model the individual and population trajectories of influenza antibody titers over multiple seasons in Singapore, and to develop a simulation model to understand the seasonality of influenza epidemics in tropical countries. This chapter has been published online in *American Journal of Epidemiology*, Year 2016, Volume 187, Issue 1, Page 135–143.

4. Chapter 5: To develop a bivariate probit model for quantifying influenza infection rates using paired sera, and to evaluate the accuracy of thresholds currently used for influenza epidemics. This chapter has been published in *Influenza and Other Respiratory Viruses*, Year 2017, Volume 11, Issue 3, Page 202–210.

The last chapter of this PhD thesis, i.e. Chapter 6, summarized the major findings of each project, and discussed the limitations and contributions to the current literature.

1.5.3 Statistical analysis

Frequentist approach

In Chapter 2, all statistical analyses were conducted using a frequentist approach. The association between the exposure, i.e. vaccination, and the outcome, i.e. an influenza infection, was assessed using multivariable logistic regressions adjusting for potential confounding factors. In frequentist approach, regression coefficients were considered as unknowns but fixed parameters with uncertainty intervals, i.e. 95% confidence interval (CI), and were calculated using the maximum likelihood estimation (MLE) method. The MLE is a method of finding parameter values that maximize the likelihood of getting the observed empirical data ($\boldsymbol{D}$) given the parameters ($\boldsymbol{\theta}$), $P(\boldsymbol{D}|\boldsymbol{\theta})$ [136].

Bayesian approach

Parameters in a model that describes a process are often unknown and can be estimated given the observed data. Many people found that frequentist inferences about an unknown parameter, such as confidence interval and p-value, are difficult to explain. In reality it would be more reasonable and natural to estimate the values of unknown parameters by maximizing the probability of getting the unknown parameters given observed data, i.e. $P(\theta|D)$. Bayesian approach, which is a state-of-the-art method, adopts the same concept and is interested in estimating $P(\theta|D)$. Despite Bayesian approach has a more natural interpretation, it was less popular than the frequentist approach due to the lack of computing power in the past, as well as the influence by R. A. Fisher and the success publication of his book *Statistical Methods of Research Workers* in 1925 [137]. Bayesian now becomes increasingly popular because computers are much faster than before, and it excels at synthesizing information from

multiple sources and estimating a large number of unknown parameters for a population-based model when algorithms are often not available.

In Bayesian, the term $P(\theta|D)$ is the posterior distribution (or "posterior) for θ that can be derived by applying the Bayes theorem over the range of θ using the formula $P(\theta|D) = \frac{P(\theta)L(D|\theta)}{P(D)} = \frac{P(\theta)L(D|\theta)}{\int_a^b P(\varphi)L(D|\varphi)d\varphi}$, where $P(\theta)$ is the prior distribution (or "prior") for θ, $L(D|\theta)$ is the likelihood function, and the unknown parameter θ is considered as a random variable that follows a probability distribution ranging from a to b [138]. The prior represents the information about the unknown parameter θ that is independent from the observed data. The most common choice of a prior distribution for a given unknown parameter is a uniform distribution over the support region of the unknown parameter. By specifying a prior distribution for the unknown parameter, it has the advantage over the frequentist approach in preventing extreme events. For example, if some values fail to occur in the observed data, which is very likely when the sample size is too small, the estimation of maximum likelihood will be biased, for instance, the separation issue in logistic regression [139]. In contrast, Bayesian approach takes values from available sources as a prior and then updates prior beliefs in the evidence of new data. Moreover, Bayesian inference is flexible to integrate various types of information, and it provides a template for complex models, i.e. setting up a prior and a likelihood function to maximize posterior. All statistical models in Chapter 3, 4 and 5 were, therefore, developed through a Bayesian approach.

In Bayesian, the challenge is to compute the integral $\int_a^b P(\theta)L(D|\theta)d\theta$ that is for searching the parameter space to evaluate the posterior distribution. Several methods are available and widely used including Monte Carlo sampling, importance sampling and Monte Carlo Markov Chain (MCMC).

Monte Carlo Sampling

Monte Carlo sampling is good for simple problems when a conjugate prior is available for the likelihood function [140]. A conjugate prior is defined as a prior distribution that results in a same form of distribution for the posterior [141]. For example, the conjugate prior for a Bernoulli/Binomial likelihood function is Beta distribution, and the conjugate prior for a normal distribution with known variance is normal distribution itself [142]. A sample of the unknown parameters can be then easily generated from the posterior distribution. The expected value of the unknown parameter equals to the posterior mean, and the equal tail 95% uncertainty interval is the 2.5^{th} and 97.5^{th} percentile of the posterior distribution.

Importance Sampling

For most of the real-world problems, however, it is difficult to work out the form of the posterior distribution, or the posterior distribution does not belong to any parametric families so that it is impossible to generate samples directly from the posterior distribution. Importance sampling is a method that evaluates the posterior distribution by simulating from a different distribution (the proposal distribution) that ideally has a similar shape to the posterior distribution but has thicker tails and re-weighting the samples using the ratio of posterior density $f(x)$ to the proposal density $g(x)$, which is called "importance weights", to adjust for under- or over-sampling. This gives a Monte Carlo estimator: $\int_a^b f(x)dx = \int_a^b \frac{f(x)}{g(x)} g(x)dx = E_{g(X)}\left[\frac{f(X)}{g(X)}\right] = \frac{1}{m}\sum_{i=1}^{m} \frac{f(X_i)}{g(X_i)}$, where $X_i \sim g(x)$ and m is the total number of independent draws. A good importance sampling function should be: (1) more than 0 whenever the posterior density does not equal to 0; (2) close to the posterior distribution with

a thicker tail; (3) easy to simulate from the proposal distribution; and (4) easy to compute the proposal density for any realized values of x [143].

MCMC

In importance sampling, random draws are simulated independently from a "wrong" proposal distribution and then the integral is re-weighed using importance weights. This process may induce inefficiency, and does not work well with high dimensions. MCMC is an extension of importance sampling that is widely used, particularly, when independent and identically distributed (i.i.d.) simulation is costly, and the dimension of unknown parameters is large [144]. It draws a sequence of simulations by setting up a Markov chain with a stationary distribution. A Markov chain is a process in which the transition probability to the next state θ_{t+1} given previous states $\theta_1, \theta_2, \dots, \theta_t$ only depends on the current state θ_t. A Markov chain has a stationary distribution if and only if the marginal distribution of θ_t does not depend on the value of t, i.e. irreducible, aperiodic and ergodic [138]. The Metropolis-Hastings algorithm is the most commonly used MCMC routine, which constructs a Markov chain mechanism with a specified stationary distribution $\pi(x)$, where $\pi(x) = \frac{p(x)}{K}$ and K is a normalizing constant. It starts with an initial value, and the Markov chain will eventually converge to the target posterior distribution after enough burn-in iterations.

The Metropolis-Hastings algorithm is outlined as follows.

Step 1: Generate an initial value θ_0 from the target posterior density such that $p(\theta_0|D) > 0$, and then calculate the log- posterior $\log p(\theta_0|D)$.

Step 2: Set $i = 0$.

Step 3: Generate a candidate point θ^* from a proposal distribution $q(\theta^*|\theta_i)$, and then calculate $\log p(\theta^*|D)$.

Step 4: Calculating the acceptance probability in the log scale $\log \alpha = \log \pi(\theta^*) + \log q(\theta_i|\theta^*) - (\log \pi(\theta_i) + \log q(\theta^*|\theta_i)) = \log p(\theta^*|D) - \log p(\theta_i|D) + \log q(\theta_i|\theta^*) - \log q(\theta^*|\theta_i)$. The most common choice for the proposal distribution is $\theta^* \sim N(\theta_i, \sigma_\theta)$, i.e. a normal distribution that is symmetric around the current value θ_i. The advantage of selecting a symmetric distribution is that the term $\log q(\theta_i|\theta^*)$ equals to the term $\log q(\theta^*|\theta_i)$ resulting $\log \alpha = \log p(\theta^*|D) - \log p(\theta_i|D)$.

Step 5: Generate a random variable $u \sim U(0, 1)$. Accept θ^* and set $\theta_{i+1} = \theta^*$ if $\log u \leq \log \alpha$; otherwise reject θ^* and set $\theta_{i+1} = \theta_i$.

Step 6: Set $i = i + 1$ and iterate Steps 3–6 until the desired sample size is achieved.

To assess the convergence of the simulated estimates, trace plots and the Geweke diagnostic are often used [145].

The Gibbs sampler is a special case of Metropolis-Hastings algorithm. It can be more efficient than the Metropolis-Hastings algorithm because all proposed samples are accepted, while the Metropolis-Hastings algorithm corrects the "wrong" proposal distribution by accepting or rejecting the simulated samples. To use the Gibbs sampler, it requires that there is a mathematical expression for the conditional distribution of each unknown parameter given all

other variables in the joint distribution and it is possible to draw random samples from this conditional distribution. However, in many scenarios, it may be challenging to obtain a closed-form expression for all the conditional distributions.

The Gibbs sampler is outlined as follows.

Step 1: Generate an initial state $\boldsymbol{\theta}^0$.

Step 2: Set $t = 0$.

Step 3: For each dimension, $i = 1, 2, \dots n$, generate a random sample θ_i from the conditional distribution $p(\theta_i|\theta_1, \theta_2, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_n)$.

Step 4: Set $t = t + 1$ and iterate Steps 3–4 until a desired sample size is achieved.

Reversible jump MCMC (RJMCMC)

RJMCMC is an extension of MCMC that allows changes in the dimension of the parameter vector when simulating the posterior distribution [146]. It also uses the Metropolis-Hastings algorithm to estimate the target posterior density. In Chapter 4, RJMCMC was adopted to compute the posterior density accounting for the change in the parameter dimension due to the transitions from an influenza infection state to a non-infection state and vice versa. In RJMCMC, the Markov chain is constructed by a diffeomorphic (invertible and differentiable) function h such that $(\boldsymbol{\theta}', \boldsymbol{u}') = h(\boldsymbol{\theta}, \boldsymbol{u})$, where $\boldsymbol{\theta}$ is the current state, $\boldsymbol{u}'$ are multi-dimensional random variables generated from a known distribution g'. The reverse move from $\boldsymbol{\theta}'$ to $\boldsymbol{\theta}$,

therefore, uses the inverse function h' of h with the acceptance probability $\alpha = \min(\frac{p(\boldsymbol{\theta}'|D)g'(\boldsymbol{u}')}{p(\boldsymbol{\theta}|D)g(\boldsymbol{u})}|\frac{\partial(\boldsymbol{\theta}',\boldsymbol{u}')}{\partial(\boldsymbol{\theta},\boldsymbol{u})}|, 1)$, where $|\frac{\partial(\boldsymbol{\theta}',\boldsymbol{u}')}{\partial(\boldsymbol{\theta},\boldsymbol{u})}|$ is the Jacobian of the transition from $(\boldsymbol{\theta},\boldsymbol{u})$ to $(\boldsymbol{\theta}',\boldsymbol{u}')$. It then follows the Metropolis-Hastings algorithm—either reject or accept the proposed parameter vector using the acceptance probability and repeat these steps for many iterations until the desired sample size is reached.

Geweke's convergence diagnostic

Using an MCMC approach for parameters estimation is important to estimate the convergence of the posterior distribution. Geweke's diagnostics is a convergence diagnostic test for posterior distributions proposed by Geweke in 1992 [147]. It tests the equality of the means of the first 10% and last 50% of the posterior distribution. If the mean of the first 10% equals to the last 50% of the posterior distribution, we could conclude that the chain has converged, and the two samples are from the same distribution. The Geweke's diagnostic uses spectral densities to estimate the two sample variances to account for the correlation between the first and last samples, and the test statistic is a standard Z-score. The formula is as follows:

$$\mathrm{Z_n} = \frac{\mu_a - \mu_b}{\sqrt{\frac{1}{n}s_a + \frac{1}{m}s_b}} \rightarrow N(0,1) \quad \text{if } n, m \rightarrow \infty$$

where μ_a and μ_b are samples means, and n and m are number of chains for the first 10% and last 50% of the posterior distribution, respectively.

CHAPTER TWO: ESTIMATION OF VACCINE EFFECTIVENESS IN A TEST POSITIVE TEST NEGATIVE DESIGN

2.1 Abstract

Vaccination is one of the most effective control measures to prevent influenza outbreaks. Annual update of influenza vaccine is necessary because of frequent antigenic drift in seasonal influenza strains. A simple biostatistical method that allows timely estimation of vaccine effectiveness (VE) is needed for annual assessment of vaccination policies. In this chapter, we will show how test-negative designs can minimize bias and provide fast estimation of influenza VE compared with other observational studies, and we will also explain how to conduct multivariable logistic regression analysis to estimate influenza VE in test-negative designs using two widely used statistical softwares: R (version 3.3.0) and STATA (release 14).

2.2 Introduction

Influenza A(H1N1)pdm09, A(H3N2), and B are currently the main influenza subtypes circulating in humans and create significant disease and economic burdens worldwide [148]. Although seasonal influenza infections are usually self-limiting and rarely fatal, they are the most frequent reason for medical attention and school/work absence [149]. Vaccination has been proven to be one of the most effective measures to prevent influenza infections and to reduce the spread of the disease [150]. However, annual reviews and updates of the seasonal influenza vaccine are necessary because influenza viruses are frequently subject to antigenic drift [151]. As a result, a year-to-year variation in vaccination effectiveness (VE) is to be expected, i.e. the estimated VE for a particular influenza season might not accurately presage the VE for the next season. A simple and fast biostatistical method that allows the estimation of VE is therefore crucial for the annual assessment of current vaccination policies, which in turn is important as it helps to determine the benefits of the vaccine, promote immunization, and sustain the uptake and stimulate the development of improved vaccines.

Vaccine efficacy and effectiveness are two distinct concepts that can easily be confused [152,153]. Vaccine efficacy is a measure of the effect of a vaccine under certain optimal conditions, such as double-blinded, randomized, placebo-controlled trials [154,155]. It usually has to be assessed before the vaccine is approved and licensed for routine use. It is designed to maximize internal validity, representing the "best case scenarios" of the vaccine's protection against infections, but this comes at the expense of generalizability. On the other hand, vaccine effectiveness (VE) is an estimate of vaccine impact in "real world" settings, e.g. primary-care settings, evaluating how the vaccination will reduce the disease in a population that does not have any randomization to account for other influences [152]. It is therefore possible to obtain a poor VE even though a high vaccine efficacy has been achieved in a prior randomized placebo-controlled vaccine trial. Observational studies including cohort studies and case-control studies have been used to estimate VE, usually with adjustment for confounding factors [156,157]. None of the observational studies is perfect, however. Bias is an important issue, and one common to all observational studies, that should be considered and minimized. The test-negative design, which is a special form of the traditional case-control study, has been recognized as the standard study design for timely estimation of VE because it is less susceptible to bias and may provide better and more accurate estimates of VE in both inpatient and outpatient settings [158–163].

In this chapter, the various study designs that have been used for estimating influenza VE will first be discussed. We will then summarize and illustrate the statistical methods that are commonly used to estimate vaccine efficacy in double-blinded, randomized, placebo-controlled trials as well as VE in a test-negative design, where the outcome is defined using laboratory-confirmed infections, e.g. reverse transcription polymerase reaction (RT-PCR).

The purpose of this chapter is to provide a good understanding of the statistical methods and to demonstrate how statistical software can be used to estimate vaccine effectiveness in observations studies, with concrete examples.

2.3 Study designs

Double-blinded, randomized, placebo-controlled trial

Double-blinded, randomized, placebo-controlled trials are considered the "gold standard" for evaluating vaccine efficacy, and are usually conducted for pre-licensure evaluation [164]. With this design, study subjects are allocated to one of the intervention groups (vaccine or placebo group) by randomization. The benefit of randomization is that it eliminates selection bias and confounding, and maximizes internal validity. Influenza is usually determined and diagnosed by RT-PCR because of its high sensitivity [165]. Vaccine efficacy is then measured by comparing the risks of influenza for the vaccine and placebo groups under the assumption that there is an equal amount of exposure to infection in both intervention groups. The very first mathematical formula for deriving vaccine efficacy was proposed by Greenwood and Tule in 1915, more than 100 years ago [166]. The commonly used formula for vaccine efficacy today is simple; it can be expressed as (1-RR) × 100%, where RR is relative risk, namely the ratio of the influenza attack rate in the vaccinated group (ARV) to the influenza attack rate in the unvaccinated group (ARU). The ARV and ARU are the probabilities of being a case in the vaccinated and unvaccinated group, respectively. To obtain the 95% confidence interval (CI) for RR, the common practice is to take the natural logarithm of RR because it computes the uncertainty interval on the natural logarithm scale,

assuming a normal distribution. The exponential function could be used to find the uncertainty interval on the original scale [167].

Observational studies

There are many observational study designs for estimating VE, including cohort studies and case-control studies [152,168]. The preferred study design may depend on context and study objectives.

Cohort study

Cohort studies are useful for assessing multiple outcomes for a single exposure, and for estimating vaccine effectiveness in institutional settings, e.g. schools and childcare centers, where a high attack rate is expected, with good vaccination records available [169]. The study population will first be divided into two groups depending on vaccination status (vaccinated vs. unvaccinated). Similar to randomized controlled trials, VE is calculated as the ratio of the incidence rate of influenza infections in the vaccinated group to the incidence rate in the unvaccinated group, i.e. $VE = 100\% \times \left(1 - \frac{I_v}{I_u}\right)$, where I_v and I_u are the incidence rates in the vaccinated and unvaccinated groups, respectively. Since the incidence rate can be easily measured in cohort studies, estimation of the burden of influenza in the source population will become possible. However, cohort studies can be expensive and time-consuming. Specifically, a large sample size is required for rare diseases, and cohort studies may take a long follow-up period to generate useful results. Cohort studies are also prone to poor representativeness of the study population, bias in the ascertainment of vaccination status and influenza infection outcomes, high rates of miss follow-ups, and changes in vaccination status during the follow-up period [168,169]. Moreover, a Kenya healthcare-

workers cohort study of VE against influenza A(H1N1)pdm09 infections found that, even with accurate ascertainment of exposure and outcomes, cohort studies were also prone to healthcare-seeking bias between vaccinated and unvaccinated groups [170].

Case-control study

Another commonly used observational study design for estimating influenza VE is the case-control study, based on sentinel general practitioner (GP) networks. The main advantages of case-control studies over cohort studies are (1) they are inexpensive, (2) a smaller sample size is required for rare outcomes, and (3) the study duration is shorter. Case-control studies can be further categorized depending on how controls are selected. Generally, cases are defined as patients who report influenza-like illness (ILI) symptoms in GP clinics with laboratory-confirmed influenza infections. However, there are different ways to select controls, and the selection of controls is the greatest challenge for case-control studies. The ideal controls should be representative of the source population that generates the cases, and the selection of controls should also be independent of their influenza-vaccination status.

One possible approach is to select controls randomly from the general community. This may, however, lead to underestimation of VE, because previous research has found that the vaccine coverage in the general community is expected to be lower than the vaccine coverage among GP patients [171].

The traditional case-control study is another type of case-control study for which controls are selected from GP patients who have no ILI symptoms. This study design has the advantage of minimizing the difference in vaccine coverage between the cases and controls, but this may

limit the validity of the VE estimate because it will be challenging to ensure the susceptibility of controls, particularly when there is a high ILI attack rate in the community [172].

The test-negative design, which was originally developed by Canadian researchers for evaluating the VE of the trivalent influenza vaccine in the 2004–05 season [173], is an efficient form of traditional case-control study for estimating influenza VE. The two study designs differ with regard to the point in time when cases and controls are identified. In traditional case-control studies, cases and controls are identified at the time of enrollment. In test-negative design studies, however, consenting patients who seek medical consultations for ILIs are recruited, and cases and controls are identified using laboratory tests such as RT-PCR [154,158,161,168,173]. Subjects whose biological samples test positive against influenza infections are defined as cases, and those whose samples test negative against influenza infections are defined as controls. The benefit of selecting controls from ILI patients is that the controls are generated from the same communities as the cases, thereby overcoming the risk of health user bias—people who have been vaccinated are more likely to seek care if they contract an illness [158,168,173]. Moreover, using laboratory methods, such as RT-PCR, to determine influenza infections in the test-negative design reduces disease misclassification [174], and studies can be easily implemented as long as vaccination records and laboratory confirmation data are available for ILI patients [162]. The test-negative design has been conducted frequently to routinely monitor influenza VE using existing surveillance systems, because it estimates VE in a timely manner, requires less work in recruiting controls, minimizes selection bias, and does not require substantial additional resources.

However, although the test-negative design has notable advantages over other types of observational studies, there are a few limitations to be considered when using this design to

estimate influenza VE. For instance, potential bias due to virus interference may be an important issue for test-negative designs. Preventing influenza infections through vaccination has been shown to increase the risk of non-influenza respiratory virus infections [175]. This may result in a higher vaccination rate among controls than cases, which will result in overestimation of influenza VE. However, this is not always true: a meta-analysis published recently compared VE estimates from three types of control groups: (1) patients testing negative for influenza, (2) patients testing negative for influenza but positive for other respiratory viruses, and (3) patients testing negative for all respiratory viruses [176]. The meta-analysis showed that VE estimates were very similar in the three groups, suggesting that test-negative designs are not biased by virus interference. Nevertheless, previous research has also found that VE estimates might be biased with test-negative designs in inpatient settings if chronic respiratory conditions are not adjusted for in the analyses [163]. Patients with chronic respiratory conditions may have increased risk of hospitalization for respiratory symptoms, and may be more likely to receive influenza vaccinations. This will result in overestimation of influenza VE if chronic respiratory conditions are not controlled for in the analyses. The advantages and disadvantages of each study design for estimating influenza VE are summarized in **Table 1**.

2.4 Model specifications for Test-negative designs

Data required

A good URTI surveillance network, including laboratory-confirmed influenza infections as the outcome, is needed for measuring VE. Swab samples for the confirmation of influenza infections should be collected as soon as possible after the onset of disease, because studies have shown that seasonal influenza viruses may stop shedding after seven days [177]. It is

recommended that the dates of disease onset and swabs be collected to take into account any delays in swab collection.

Table 1. A summary of the advantages and disadvantages of various observational study designs for estimating influenza vaccine effectiveness.

Study design		Pros	Cons
Cohort study		Possibility of assessing multiple outcomes; Minimized selection bias; More information on confounding	Expensive; Large sample required for rare diseases; Time-consuming; Possibility of high rate of loss to follow-ups and continuous change of vaccination status;
Case-control study	Community controls	Easy to recruit cases and controls; Timeliness	Additional work on recruiting controls; Poor representativeness of the controls (vaccine coverage rate may be lower in the community); Misclassification of influenza infection; Misclassification of vaccine history
	Traditional case-control study	Easy to recruit cases and controls; Timeliness; Minimized selection bias;	Additional work on recruiting controls; Poor representativeness of the controls; Misclassification of influenza infection; Misclassification of vaccine history
	Test-negative design	Easy to recruit cases and controls; Timeliness; Minimized selection bias; Simplicity; Reduced misclassification of influenza infection	Misclassification of vaccine history; Chronic respiratory conditions need to be adjusted for in the analysis in inpatient settings

The type and date of influenza vaccination should be recorded because VE could also be affected by how accurately the influenza vaccination status is determined. VE might be underestimated if subjects who had disease onset within 14 days after vaccination were

placed in the vaccinated group, because a vaccine may take 14 days on average to take effect [148]. Studies have shown that vaccine-induced immunity may drop substantially after six months of administration [178]. Thus, it is recommended that individuals who have had influenza vaccination during the current season (less than six months after the disease onset and more than 14 days before the disease onset) be placed in the group to be vaccinated.

Observational studies, including test-negative designs, are prone to confounding factors because they do not involve any randomization. Factors that may relate either to being vaccinated or to the likelihood of the occurrence of influenza infections should be adjusted for when calculating VE. The common potential confounding factors, such as age, gender, comorbidity, including pregnancy, the interval between disease onset and swab, healthcare settings, and prior exposure to influenza infections, should be collected through questionnaires or interviews if this information is not available in patients' medical records [162].

Statistical model

Influenza VE can be estimated with the use of the adjusted odds ratio (OR_{adj}) for developing influenza infections compared between intervention groups; the odds ratio is the ratio of the odds of being infected in the vaccinated group to the odds of being infected in the unvaccinated group, and the odds is the probability of an event (being infected, in this case) occurring relative to the probability of an event not occurring [179]. The formula for calculating VE is (1-OR_{adj}) × 100% [180], and the OR_{adj} obtained from test-negative designs is a valid estimate of RR in RCTs. As mentioned earlier, RR can be calculated as $\frac{\mathrm{ARV}}{ARU} = \frac{\mathrm{A}}{A+C} / \frac{B}{B+D}$ in RCTs, where A and B are the number of cases who have been vaccinated and

unvaccinated, respectively, and C and D are the number of all other non-cases in the study population who have been vaccinated and unvaccinated, respectively. Assuming an equal chance of having non-influenza illness (θ) in both intervention groups and that the likelihood of an ILI patient seeking care from local GPs is p_1 and p_2 in the vaccinated and unvaccinated groups, respectively, in test-negative designs the odds ratio is, therefore, $\mathrm{OR} = \frac{p_1 A}{p_2 B} / \frac{p_1 \theta C}{p_2 \theta D} = \frac{\mathrm{A}}{C} / \frac{B}{D}$, which approximates RR obtained from RCTs. The crude OR may not indicate the causal effect between vaccination and the risk of developing infection because the effect could be confounded by various confounding variables, such as age and gender [179]. To overcome the confounding effect, statistical adjustment for confounders using either stratification or multivariable logistic models (conditional logistic models should be used if there has been any matching) is required. In regression models, the outcome variable should be a dichotomous variable indicating infection status in the form of 0 or 1, where 1 indicates the presence of infections and 0 indicates no infections, and vaccination status and the relevant confounding variables should be included as the explanatory variables. The $\mathrm{OR}_{\mathrm{adj}}$ is then calculated by taking the exponential function of the estimated regression coefficients for vaccination status.

2.5 Illustrative examples

In this section, a real-application example of vaccine-effectiveness estimation using a test-negative design will be presented.

For this example, we will use the febrile respiratory illness (FRI) surveillance data collected from the Singapore Armed Forces (SAF) from 2009 to 2012 to illustrate the method [32,148,181]. This study has been published as the work of Ho HP, Zhao X, Pang J, Chen

MI-C, Lee VJM, Ang LW et al. ('Effectiveness of seasonal influenza vaccinations against laboratory-confirmed influenza-associated infections among Singapore military personnel in 2010–2013.' *Influenza Other Respir Viruses*. 2014 Sep 1;8(5):557–66.). An FRI surveillance programme was established in four SAF camps, including a new recruit camp and three other camps, just before the H1N1 pandemic in 2009. Patients who visited medical centers in any of the four camps and reported FRI infections were asked to participate in the study. Demographic information, such as age and gender, and influenza vaccination history were collected from all consenting FRI cases. Nasal swab samples were also collected and were sent to a laboratory to be tested against various upper-respiratory pathogens using RT-PCR within 24 hours.

In this example, we will only use the data on influenza A(H1N1)pdm09 infection during the period when the trivalent influenza vaccine was introduced to the new recruit camp, which was from 1 November 2010 to 30 September 2011, for illustration [148]. Cases were defined as FRI patients with laboratory-confirmed A(H1N1)pdm09 infections, and controls were defined as subjects without A(H1N1)pdm09 infections identified using RT-PCR. The vaccination status for each subject was coded as 1 if he/she had received the trivalent influenza vaccine, and 0 if not. It was found that only the camp group (recruit camp vs. other camps) was an effect modifier that modified the association between vaccination and the risk of influenza infection. Although age and gender are confounders that are usually adjusted for in observational studies, we did not find significant effects related to these two factors because the majority of the study subjects were male (>95%) and aged between 20 and 24 years (>90%). Thus, the multivariable logistic model would be $logit(\text{infection}_\text{i}) = \beta_0 + \beta_1 \text{vaccination}_\text{i} + \beta_2 \text{vaccination}_\text{i} \times \text{camp group}_\text{i} + \beta_3 camp\ group_i$, where $\beta_i, i = 1,2,3$, is the regression coefficient. The VE for the trivalent influenza vaccine in the recruit camp would be $(1 - e^{\beta_1+\beta_2}) \times 100\%$ and $(1 - e^{\beta_1}) \times 100\%$ in other camps.

The calculation of VE has now become increasingly easy with the use of statistical software, with R, STATA, and SPSS being the most widely used. Regardless of which statistical software is used, the very first step is to import data before conducting any statistical analysis. The dataset used for this analysis is saved in CSV format, and **Table 2** below shows an example of the first five rows in the dataset. **Figure 1** and **Figure** *2* below present the commands used in R (version 3.3.0) [182] and STATA (release 14) [183] for calculating influenza VE, respectively.

Table 2. The first five observations in the dataset used.

Serial No.	**Infection**	**Vaccination**	**Camp group**	**Age**	**Gender**	**…**
1	0	0	Recruit	23	Male	…
2	1	0	Recruit	20	Male	…
3	0	0	Recruit	22	Male	…
4	0	0	Recruit	22	Male	…
5	1	0	Recruit	22	Male	…
…	…	…	…	…	…	…

The adjusted odds ratio comparing the risks of influenza infection for the vaccinated and unvaccinated groups was 0.19 (95% CI: 0.10 to 0.36) for the recruit camp and 0.95 (95% CI: 0.28 to 3.22) for other camps. Therefore, the influenza VE was 81% (95% CI: 64% to 90%) and 5% (95% CI: -222% to 82%) for the recruit camp and other camps, respectively. A

strong influenza VE was found in the recruit camp only during the period of 1 November 2010 to 30 September 2011.

```
fluData=read.csv('data.csv',as.is = T)
model=glm(Infection~Vaccination+Vaccination*Camp.group,data=fluData,family="binomial")
###Recruit camp
### Adjusted log(OR)
logOR.mixrural <- c(0,1,0,1)%*%coef(model)
### Standard error for the adjusted log(OR)
se.OR <- sqrt(c(0,1,0,1)%*%vcov(model)%*%c(0,1,0,1))
###VE for the recruit camp
1-exp(logOR.mixrural)


##           [,1]
## [1,] 0.8117224

###95% confidence interval (CI) for VE for the recruit camp
1-exp(logOR.mixrural + c(qnorm(.975), qnorm(.025))*se.OR)


## [1] 0.6409149 0.9012812

####othere camps
###VE for other camps
1-exp(coef(model)[2])


## Vaccination
##  0.04761905

###95% CI for VE for other camps
1-exp(confint.default(model)[2,])


##      2.5 %     97.5 %
##  0.7179037 -2.2153181
```

Figure 1. An illustration of the use of R commands to estimate vaccine effectiveness in a test-negative design using the SAF data.

```
. import delimited C:\data.csv
(6 vars, 1784 obs)

. label define infection 0 "Negative" 1 "Positive"

. label define vaccination 0 "No" 1 "Yes"

. label values infection infection

. label values vaccination vaccination

. encode campgroup, generate(campgroup2)

. logit infection vaccination##campgroup2

Iteration 0:   log likelihood =   -248.94238
Iteration 1:   log likelihood =   -239.12624
Iteration 2:   log likelihood =   -234.71537
Iteration 3:   log likelihood =   -234.69625
Iteration 4:   log likelihood =   -234.69624

Logistic regression                             Number of obs     =          1,784
                                                LR chi2(3)        =          28.49
                                                Prob > chi2       =         0.0000
Log likelihood =  -234.69624                    Pseudo R2         =         0.0572

------------------------------------------------------------------------------
   infection |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
 vaccination |
         Yes |  -.0487902   .6207851    -0.08   0.937    -1.265507    1.167926
             |
  campgroup2 |
     Recruit |    1.03229   .4056828     2.54   0.011     .2371661    1.827414
             |
 vaccination#|
  campgroup2 |
Yes#Recruit  |  -1.621048   .7027721    -2.31   0.021    -2.998456   -.2436398
             |
       _cons |  -3.555348   .3585686    -9.92   0.000     -4.25813   -2.852567
------------------------------------------------------------------------------

. lincom 1.vaccination + 1.vaccination#2.campgroup2,or

 ( 1)  [infection]1.vaccination + [infection]1.vaccination#2.campgroup2 = 0

------------------------------------------------------------------------------
   infection | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         (1) |  .1882776   .0620215    -5.07   0.000     .0987188    .3590851
------------------------------------------------------------------------------

. display 1-.1882776
.8117224

. display 1-.3590851
.6409149

. display 1-.0987188
.9012812

. lincom 1.vaccination,or

 ( 1)  [infection]1.vaccination = 0

------------------------------------------------------------------------------
   infection | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         (1) |   .952381   .5912239    -0.08   0.937     .2820963    3.215318
------------------------------------------------------------------------------

. display 1-.952381
.047619

. display 1-3.215318
-2.215318

. display 1-.2820963
.7179037
```

Figure 2. An illustration of the use of STATA commands to estimate vaccine effectiveness in a test-negative design using the SAF data.

2.6 Conclusion

In this chapter, various observational studies were compared and discussed, showing that the test-negative design provides a faster and more accurate estimation of influenza VE with less bias. The minimum data fields required and the standard statistical model for estimating influenza VE were also described. Influenza VE can be estimated using the formula $\left(1 - \mathrm{OR}_{\mathrm{adj}}\right) \times 100\%$, where $\mathrm{OR}_{\mathrm{adj}}$ is calculated from multivariable logistic regression models. The estimation of influenza VE in test-negative designs using R (version 3.3.0) and STATA (release 14) was also presented. It is hoped that this chapter establishes a standard procedure, such as study design, data collection, and statistical analysis, for the timely evaluation of influenza VE.

2.7 Motivations for next chapter

Research has shown that vaccine-induced protection against influenza wanes over time [184]. The risk of influenza virus infection, therefore, may increase over time after vaccination. The approach and methodology proposed for estimating influenza VE in this chapter, however, cannot account for this waning effect, because the analysis is purely based on RT-PCR data. The next chapter, i.e. Chapter 3, showed how to synthesize both RT-PCR and serological data to estimate influenza VE using a Bayesian hierarchical model, which adjusts for waning in vaccine-induced protection.

CHAPTER THREE: QUANTIFYING PROTECTION AGAINST INFLUENZA VIRUS INFECTION MEASURED BY HEMAGGLUTINATION-INHIBITION ASSAYS IN VACCINE TRIALS

This work has been published as: Zhao X, Fang VJ, Ohmit SE, Monto AS, Cook AR, Cowling BJ. Quantifying Protection Against Influenza Virus Infection Measured by Hemagglutination-inhibition Assays in Vaccine Trials: Epidemiology. 2016 Jan;27(1):143–51.

3.1 Abstract

Background

Correlations between hemagglutination-inhibition titers (hereafter 'titers') and protection against infection have been identified in historical studies. However, limited information is available about the dynamics of how titer influences protection.

Methods

Titers were measured in randomized, placebo-controlled vaccine trials in Hong Kong (HK) among pediatrics during September 2009 – December 2010 and the United States (US) among adults during Oct 2007 – April 2008. Intermediate unobserved titers were imputed using three interpolation methods. As participants were recruited at different times leading to varying exposure to infection relative to entry, a modified proportional hazards model was developed to account for staggered entry into the studies and to quantify the correlation of titers with protection against influenza infections, adjusting for waning in titers. The model was fitted using Markov chain Monte Carlo and importance sampling.

Results

A titer of 1:40 reduced infection risk by 40%–70% relative to a titer of 1:10, depending on the circulating strain; the corresponding protection by a titer of 1:80 was 54%–84%. Results were robust across interpolation methods. The trivalent inactivated vaccine reduced cumulative incidence of influenza B and influenza A(H3N2) infections by 6 percentage points (pp) (95% credible interval (CrI): 2pp, 10pp) and 1pp (95% CrI: 0.3pp, 2pp) respectively, but not for influenza A(H1N1)pdm09. The live attenuated vaccine showed little efficacy against influenza A(H3N2) infections.

Conclusions

Titers are correlated with protection against influenza infections. The trivalent inactivated vaccine can reduce the risk of influenza A(H3N2) and influenza B infections in the community.

Word count = 245

3.2 Introduction

Long-term immunity against influenza virus infection is thought to be primarily conferred by the adaptive immune system, and specifically antibody-mediated humoral immunity [185]. Important historical studies identified a correlation between the antibody titer measured by the hemagglutination-inhibition assay with protection against influenza virus infection [185], suggesting that an hemagglutination-inhibition titer (hereafter referred to as 'titer') of 40 or higher could provide ~50% protection against infection [186]. Subsequent studies support this finding [187]. Titers are rapidly boosted by infection or vaccination peaking within 2–4

weeks of either event [188], and then wane over time [189]. This leads to fluctuations in the level of herd immunity that may drive the timing and magnitude of epidemics [190].

Knowing the protection conferred by antibodies corresponding to a given titer may shed light on the seasonality of influenza in tropical/sub-tropical regions, where outbreaks may occur year round or several times a year, in contrast to temperate regions [78,135]. Identifying the degree of correlation of titers with protection against infection is also important for vaccine development, because annual reformulations of inactivated influenza vaccines can now be approved based entirely on immunogenicity data [191]. The dynamism of titers complicates understanding of their protective effect, for they may change substantially over the course of a single study, in which case the titer pre-season, or post-vaccination, may not adequately characterize the antibody levels at the time of exposure to infection. Furthermore, if entry to the study is staggered, individuals will experience neither the same total quantum nor the same timeline of exposures. Such heterogeneities in exposure add complexity to statistical analysis, for instance, by violating the assumption of proportional hazards in Cox models.

Here, we developed a statistical methodology allowing staggered entry and used this to investigate how fast antibody titers waned over time, and how adjustment for antibody waning affected the correlation between titers and protection against confirmed influenza virus infection, in vaccine trials in Hong Kong and the United States.

3.3 Methods

Sources of data

Placebo-controlled trials of influenza vaccine (with compositions tabulated in **Table 3**) [131,192] were conducted in Hong Kong during September 2009 – December 2010 and in

the United States during Oct 2007 – April 2008 [129,131,192].

Table 3. Description of vaccines against each subtype as well as circulating strains in both trials.

HK trial			
Trivalent inactive influenza vaccine 08/09[1]	A/Brisbane/59/2007(H1N1)-like virus	A/Brisbane/10/2007(H3N2)-like virus	B/Brisbane/60/2008-like virus
Circulating Strain	A/California/7/2009(H1N1)-like virus	-	B/Brisbane/60/2008-like virus
US trial			
Trivalent inactive influenza vaccine 07/08[2]	A/Solomon Islands/3/2006 (H1N1)-like virus	A/Wisconsin/67/2005(H3N2) -like virus	B/Malaysia/2506/2004-like virus
Live attenuated influenza vaccine 07/08[3]	A/Solomon Islands/3/2006 (H1N1) -like virus	A/Wisconsin/67/2005(H3N2) -like virus	B/Malaysia/2506/2004-like virus
Circulating Strain	-	A/Uruguay/716/2007(H3N2)-like virus	-

[1]Each 0.5ml dose of the TIV contained 15μg of hemagglutinin from each of the following strains: A/Brisbane/59/2007(H1N1)-like, A/Brisbane/10/2007(H3N2)-like, and B/Brisbane/60/2008-like (Victoria lineage) strains

[2]Each 0.5ml dose of the TIV contained 15μg of hemagglutinin for each of A/Solomon Islands/3/2006 (H1N1), A/Wisconsin/67/2005(H3N2), and B/Malaysia/2506/2004 (B/Victoria lineage)

[3]Each 0.2ml dose of the live attenuated vaccine contained $10^{6.5–7.5}$ fluorescent focus units of live attenuated influenza virus reassortants of the same strains

In Hong Kong, 773 children aged 6–17 years old were randomly assigned in a 3:2 ratio to receive either 2008–09 seasonal trivalent inactive influenza vaccine (Vaxigrip, Sanofi Pasteur) or saline placebo from late August 2009 to January 2010, coinciding with the epidemics of influenza A(H1N1)pdm09; they were followed up throughout the subsequent influenza B(Victoria lineage) epidemic until the end of the study in August – December 2010 [131]. Influenza A(H3N2) was excluded due to the small sample size (n = 8). Participants provided a series of serum samples to determine titers against A(H1N1)pdm09

(A/California/7/2009(H1N1))—which was not included in the trivalent influenza vaccine—and B (B/Brisbane/60/2008-like)—which was [193]. The circulating and vaccine influenza B strains had a close antigenic match according to phylogenetic analysis of the HA gene [193]. Sera were collected before vaccination, one month after vaccination, and at the end of the study 9–12 months later. ~36% participants ($n = 281$) were randomly selected to provide an extra blood sample from mid-April to mid-May after the winter influenza season. All participants were regularly contacted and requested to report if they presented acute respiratory illnesses, and home visits were arranged immediately to collect nasal and throat swab samples. The samples were then tested using reverse transcription polymerase chain reaction (RT-PCR). Written consent was collected from parents or legal guardians of the participants; additional written consent was obtained from participants aged 8-17y. The study was approved by the Institutional Review Board of the University of Hong Kong.
A network of sentinel general practitioners reported weekly influenza-like illness rates, and the Public Health Laboratory Service reported weekly rates of RT-PCR confirmed influenza infections among specimens submitted from the sentinels and local hospitals. The product of influenza-like illness rates and the proportion RT-PCR positive samples was used as the proxy measure of incidence [194].

The United States trial was carried out during the 2007–2008 influenza season and involved 1952 adults aged 18–49 years old. From October to early November 2007 (the beginning of the influenza season), participants were randomized to receive either trivalent inactivated vaccine (Fluzone, Sanofi Pasteur), live attenuated vaccine (FluMist, MedImmune Inc.) or saline placebo, in a 2:2:1 ratio [129,192]. This analysis is limited to A(H3N2) infection, the dominant influenza strain detected among study participants during the study period. Serum samples were taken from all participants, of which a subset (n = 728, 37.3%) were assayed

for titers against the circulating A(H3N2) strain (A/Uruguay/716/07) before vaccination, 30 days post vaccination, and after the 2007–08 United States winter season. The circulating A(H3N2) strain was not included in both vaccines, however, it was antigenically similar to the A(H3N2) vaccine strain (**Table 3**) [129]. All participants were regularly followed up to identify acute respiratory illness. Throat swabs were collected as soon as possible to identify influenza types/subtypes by RT-PCR. As in Ref. [192], we included titers from 658 of the 728 assayed participants: those with RT-PCR confirmed A(H3N2) (n = 105) and those without any RT-PCR confirmed infection were included (n = 553), while those with a laboratory confirmed A(H1N1)pdm09 or B infection or without postseason sera were excluded. Also excluded were those with 4-fold rises in titers between the post-vaccination and post-season samples who did not also have infection confirmed by virus isolation or RT-PCR [129]. Written consent was obtained from all participants. The study was approved by the institutional review board at the University of Michigan Medical School. The daily percentage of visits for influenza-like illness reported by Michigan Sentinel Providers was used as a proxy for influenza infections in the community [195].

Statistical analysis

The two primary endpoints in the analysis are (i) RT-PCR confirmed infection and, for those in the placebo arms, (ii) a four-fold or greater rise in antibody titers measured in HAI assays. The protection against infection associated with higher antibody titers, as measured by the HAI assay, was analysed separately by trial and influenza subtype. HAI titers were measured at 3–4 time points, and HAI titers throughout the follow-up period were imputed using three alternative methods (**Figure 3**). The interpolated titer $\hat{y}(t)$ at time t was taken to be one from the following:

- Method 1: carrying forward the post-vaccination titer. $\hat{y}(t) = y_{t_{1i}} I(t \leq t_i^v + 14) + y_{t_{2i}} I(t > t_i^v + 14)$ where t_i^v is the day that subject i was vaccinated, t_{1i} is the time s/he had a first serum sample (prior to vaccination), t_{2i} the time of 2nd serum sample (i.e. post-vaccination), and $I(A)$ the indicator function equal to 1 if A is true and 0 otherwise. A cut-off of 14 days was used to adjust for the time before vaccine protection [196].

- Method 2: carrying forward the most recently measured titer. $\hat{y}(t) = y_{T_i(t)}$ where $T_i(t)$ is the most recent time individual i was sampled prior to time t.

- Method 3: waning antibody titers. $\hat{y}(t) = y_{T_i(t)} - b[t - T_i(t)]$ where b characterises the speed at which titers wane, which was estimated by fitting a linear regression model on the pooled changes in observed titers versus time on a log-scale. The cases of a 4-fold rise of any two consecutive observed titers were excluded when the overall average decline rate was calculated. The crude average slopes against A(H1N1)pdm09, B and A(H3N2) were −0.003, −0.005 and −0.005 respectively corresponding to a 20%, 35% and 34% decrease over three months respectively.

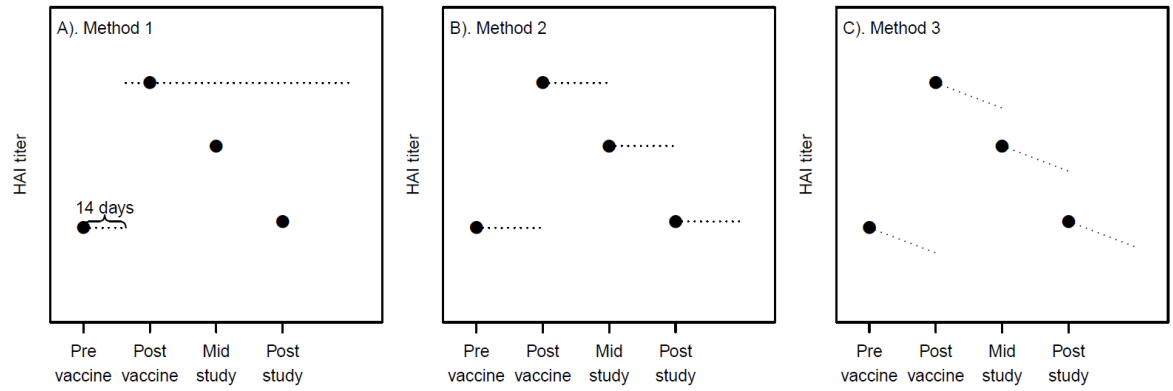


Figure 3. Summary of interpolation methods for imputing titer scores at unobserved time points.
Points represent the observed titer scores, while dotted lines represent the imputed titer scores. Method 1 (A) carries forward the postvaccination titer for the duration of the study, method 2 (B) carries forward the most recent titer, while method 3 (C) applies a gradient representing the average decline in titers across all participants that is estimated using linear regression, as described in the text.

We developed a parametric proportional hazards model to quantify the association between hemagglutination inhibition titers and protection against A(H1N1)pdm09 and B using the Hong Kong data and against A(H3N2) using the United States data. This model was motivated by the need to account for the staggered entry of participants into the trials, which led to exposure to infection at different times relative to entry, which would otherwise violate the assumption of proportional hazards. Rather than being unspecified, as in basic Cox proportional hazards models as detailed elsewhere [197], the baseline hazard was set to be proportional to a proxy of influenza activity based on community surveillance data, with the constant of proportionality estimated as a parameter in the model.

The instantaneous hazard for RT-PCR confirmed infection is therefore $\alpha p_i(t)\exp[\beta\hat{y}_i(t) + \gamma v_i\hat{y}_i(t)]$, where $p_i(t)$ is the proxy for community influenza activity t days after individual i's entry to the study, $\hat{y}_i(t)$ is the imputed log titer for individual i on day t, v_i is a dummy variable indicating vaccination status (an additional dummy variable was introduced for the

three arm United States trial) and (α, β, γ) are parameters governing the baseline risk of RT-PCR confirmed infection, the protection conferred by titers, and the interaction between titers and vaccination, respectively. The coefficient of vaccination alone was excluded as the model would be over-parameterized. Parameters were estimated using a Markov chain Monte Carlo routine [138], with improper flat priors and the likelihood following from the definition of the hazard above derived using standard properties of survival analysis [198]. The likelihood function L was, therefore, written as $L = \prod_{i=1}^{n} L_i = \prod_i \lambda(t_i)^{d_i} S(t_i)$, where d_i is a death indicator at time t_i for individual i taking the value of 1 if individual i died and 0 otherwise, $\lambda(t_i)$ is instantaneous hazard function described above, and $S(t_i)$ is the survivor function if the individual i lived up to time t_i. 10 000 draws from the posterior were obtained for each arm/virus combination from six independent chains, and convergence was assessed using the Geweke diagnostic [145]. As the hemagglutination inhibition assay returns an interval censored titer (e.g. positive at a 1:40 titration, negative at a 1:80 titration, or $y \in (40,80)$) and as not all of the participants in the trials had titers determined, either due to the study design or dropout, the exact uncensored titer at each observation for each individual was also included as an estimand in the Markov chain Monte Carlo routine. This permitted comparison across the studies, which differed in the dilution assayed.

The use of RT-PCR confirmed infection as the endpoint means the potential for under-ascertainment, if some infections were not virologically confirmed. The hazard above (i.e. the risk of RT-PCR confirmed infection) was scaled to obtain the risk of infection, $\theta p_i(t)\exp[\beta y_i(t) + \gamma v_i y_i(t)]$. The parameter θ was estimated using importance sampling [199], by sampling θ from a uniform distribution over a range spanning the values with high posterior support, while the parameters (α, β, γ) were sampled from their posterior distribution obtained from Markov chain Monte Carlo. For each draw, the survival function

was calculated to obtain a likelihood contribution for individuals receiving the placebo based on serologic evidence of infection, which provided the importance weights accordingly. To estimate the proportion of infections that were RT-PCR confirmed, we used serologic data only from the placebo group in each study, because of potential antibody ceiling effects among vaccinees, which inhibits the rise in antibody titers following infection after a previous increase in antibody titers in response to vaccine [200]. As a result, a four-fold rise in titers may be less likely among vaccinees compared with unvaccinated participants after influenza infection. The estimated proportion of infections was then used to derive an individualised infection hazard over time, which was aggregated to obtain average survival functions for each group.

We obtained derived parameters, such as the ratio of αs on different arms, by performing the corresponding transformations to the (weighted) posterior samples. Point estimates are posterior (weighted) means and interval estimates are equal-tailed 95% credible intervals (CrI).

All statistical analyses were performed by using R version 3.0.3 (R Foundation for Statistical Computing, Vienna, Austria) [182].

3.4 Results

Participants had similar demographic characteristics across intervention groups in both trials (**Table 4**).

Table 4. Descriptive Characteristics of Participants Recruited in the Hong Kong (HK) Trial, 2009–10; and in the United States (US) Trial, 2007–8. Total participants N = 773 in the HK trial and N = 1952 in the US trial. P-values are calculated by Pearson's chi-square test, comparing proportions with RT-PCR confirmed infection within each study arm.

Hong Kong trial	**TIV (N = 465)**		**Placebo (N=308)**	
	n	**%**	**n**	**%**
Age group, years				
6–8	164	35	99	32
9–11	147	32	106	34
12–17	154	33	103	33
Male	241	52	168	55
RT-PCR-confirmed infections				
Influenza A(H1N1)pdm09[a]	10	2	5	2
Influenza B[b]	10	2	21	7

United States trial	**Live attenuated vaccine (N = 814)**		**Inactivated vaccine (N = 813)**		**Placebo (N = 325)**	
	n	**%**	**n**	**%**	**n**	**%**
Age group, years						
18–35	736	90	728	90	300	94
36–50	78	10	85	10	25	6
Male	320	39	294	36	124	38
RT-PCR-confirmed influenza A(H3N2) infection[c]	53	7	22	3	30	9

[a] $p = 0.80$; [b] $p = 0.003$; [c] $p < 0.001$.

Figure 4 shows the temporal distributions of influenza incidence in the community, as well as the PCR-confirmed infections against various influenza subtypes tested in both studies.

Figure 5 shows the distribution of hemagglutination inhibition titers sampled before vaccination as well as the corresponding risks of influenza infection against the three subtypes. The risk of RT-PCR confirmed A(H1N1)pdm09 and B infections for Hong Kong children with pre-vaccination titers of ≤1:10 were ~5% for both (**Table 5** – **Table 6**). In the United States trial, >20% of the participants with titers of 1:4–1:8 against A(H3N2) had RT-PCR confirmed A(H3N2) infections. The corresponding risk for higher baseline titer levels appeared to plateau but could not be estimated accurately in higher titers due to small sample sizes after stratification for each subtype (**Table 5** – **Table 6**).

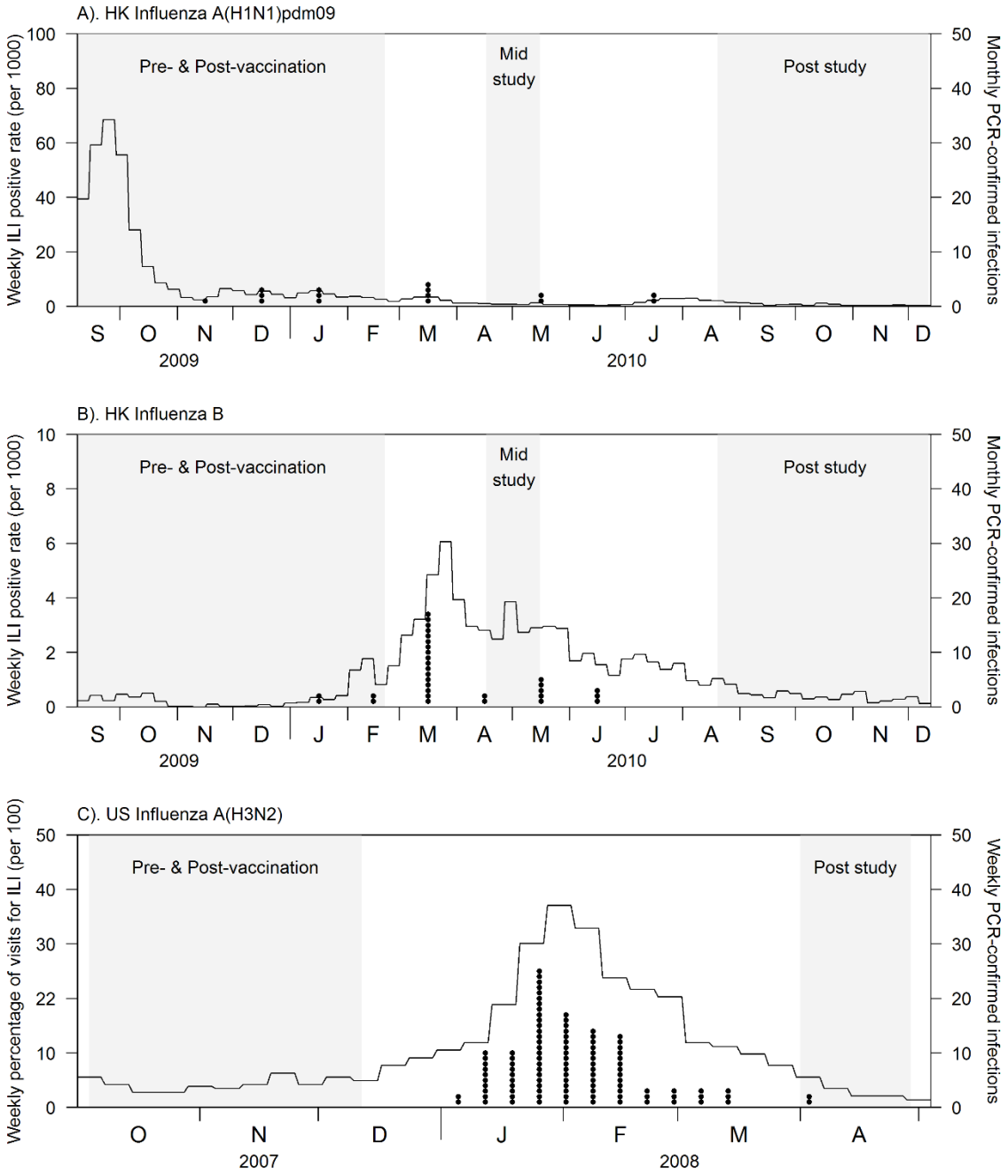


Figure 4. Distribution of weekly ILI positive rates versus monthly PCR-confirmed influenza infections by subtype (A, B) in HK, and weekly percentage of medical consultations for influenza-like illness versus weekly PCR-confirmed influenza A(H3N2) infections in the US (C). The Hong Kong daily influenza-like illness rates are shown by lines in (A) and (B). The points in (A) and (B) describe the Hong Kong monthly PCR-confirmed influenza A(H1N1)pdm09 and influenza B infections, respectively. On the bottom panel, the black line shows the daily percentage of medical consultations for influenza-like illness in Michigan and the points represent the weekly PCR-confirmed influenza A(H3N2) infections in the study. Time is measured in calendar months at the x axis on each panel. The longer tick marks at the x axis indicate the beginning and end of each calendar year. Shaded areas indicate the timing

of serum samples. For both studies, the prevaccination blood samples were taken on the same day as vaccines were given. This was followed by a period of ~30 days before the postvaccination blood samples were taken. As entry was staggered over 6 months, some individuals were enrolled, while others had already had their postvaccination blood samples. The time period between pre- and post-vaccination blood samples is, therefore, merged due to the staggered entry. In the HK trial, enrolled subjects received vaccination and had the first blood sampled from September 2009 to late January 2010. The postvaccination blood samples, mid-season blood samples, and postseason blood samples were taken between October 2009 and February 2010, April and May 2010, and August and December 2010, respectively. In the US trial, enrolled subjects received vaccination and had their first blood sampled from October 2007 to November 2007. The postvaccination blood samples and postseason blood samples were taken between October and December 2007, and March and April 2008, respectively. HK indicates Hong Kong; ILI, influenza-like illness; PCR, polymerase chain reaction; US, United States.

The parametric proportional hazards model was substantially better fitting on a logarithmic than a linear scale for antibody titers because it yielded higher log-likelihood values (**Table 7**), indicating a declining marginal protective association with increasing titers. This model was used for all subsequent analysis. **Figure 6** illustrates the relative protection for a titer of 1:40 and 1:80 versus 1:10 by virus and intervention using the three interpolation methods (**Table 8**). Estimates of the protection associated with both 1:40 and 1:80 versus 1:10 were similar and were robust regardless of the interpolation method used, and the influenza subtype and intervention group as all predicted 95% CrI error bars within the group had considerable overlap.

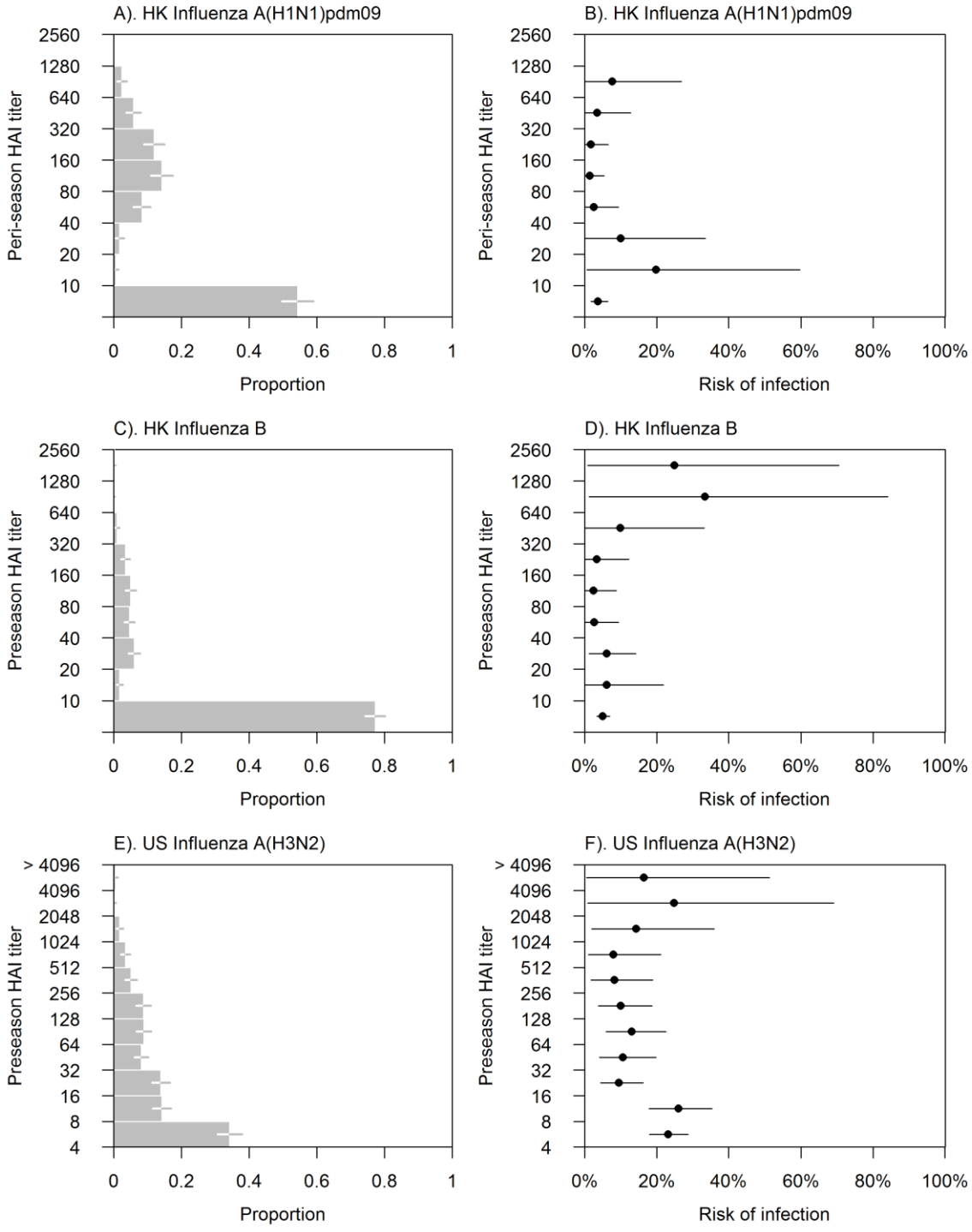


Figure 5. Distribution of observed HAI titers against influenza A(H1N1) pdm09, influenza B and influenza A(H3N2) during the peri- (A), preseason (C), and postvaccination periods (E), and proportion of PCRconfirmed infection within each titer interval (B, D, F). Gray bars (A, C, E) indicate the empirical proportion of titers within intervals with standard 95% confidence interval error bars. Proportion of PCR-confirmed infection (B, D, F) is estimated from posterior mean (point) and 95% credible intervals (line) with U(0,1) priors. HAI indicates hemagglutination inhibition; PCR, polymerase chain reaction.

Table 5. Probability Distribution of Hemagglutination-inhibition (HAI) Titers and the Risk of Infection per HAI Titer Interval for Influenza A(H1N1)pdm09 and Influenza B in the HK Trial.

Titer	Influenza A (H1N1)pdm09		Influenza B	
	Proportion (95% CI)	Risk of infection (95% CI)	Proportion (95% CI)	Risk of infection (95% CI)
<1:10	0.54 (0.50, 0.59)	0.04 (0.02, 0.06)	0.77 (0.74, 0.80)	0.05 (0.03, 0.07)
1:10 — 1:20	0.01 (0.00, 0.01)	0.20 (0.01, 0.61)	0.02 (0.01, 0.03)	0.06 (0.00, 0.22)
1:20 — 1:40	0.02 (0.01, 0.03)	0.10 (0.00, 0.38)	0.06 (0.04, 0.08)	0.06 (0.01, 0.14)
1:40 — 1:80	0.08 (0.06, 0.11)	0.03 (0.00, 0.09)	0.05 (0.03, 0.06)	0.03 (0.00, 0.09)
1:80 — 1:160	0.14 (0.11, 0.15)	0.02 (0.00, 0.05)	0.05 (0.04, 0.07)	0.02 (0.00, 0.09)
1:160 — 1:320	0.12 (0.10, 0.15)	0.02 (0.00, 0.07)	0.04 (0.02, 0.05)	0.04 (0.00, 0.12)
1:320 — 1:640	0.06 (0.04, 0.08)	0.04 (0.00, 0.13)	0.01 (0.00, 0.02)	0.10 (0.00, 0.34)
1:640 — 1:1280	0.02 (0.01, 0.04)	0.08 (0.00, 0.27)	0.00 (0.00, 0.00)	0.33 (0.01, 0.84)
1:1280 — 1:2560	-	-	0.00 (0.00, 0.01)	0.25 (0.01, 0.69)

The parametric proportional hazards model using the first interpolation method, because it gave the smallest deviance information criterion value, was used to derive the posterior predicted distribution of the cumulative hazard of infection by treatment and virus (**Figure *7***, the other methods yielded similar results). There were differences in the inferred incidence curves, and thus the timing and number of infections, between the placebo and trivalent inactivated vaccine arms in the Hong Kong trial for B, though not for A(H1N1)pdm09, and in the United States trial between the placebo arm and the trivalent inactivated vaccine arm for A(H3N2). In contrast, there was no evidence that the live attenuated vaccine was more protective than the placebo in the United States trial.

Table 6. Probability Distribution of Hemagglutination-inhibition (HAI) Titers and the Risk of Infection per HAI Titer Interval for Influenza A(H3N2) in the US Trial.

	Influenza A(H3N2)	
Titer	**Proportion (95% CI)**	**Risk of infection (95% CI)**
<1:4	0.23 (0.18, 0.29)	0.34 (0.31, 0.38)
1:4 — 1:8	0.26 (0.18, 0.35)	0.14 (0.12, 0.17)
1:8 — 1:16	0.10 (0.04, 0.16)	0.14 (0.11, 0.17)
1:16 — 1:32	0.11 (0.04, 0.20)	0.08 (0.06, 0.10)
1:32 — 1:64	0.13 (0.06, 0.23)	0.09 (0.07, 0.11)
1:64 — 1:128	0.10 (0.04, 0.19)	0.09 (0.07, 0.11)
1:128 — 1:256	0.08 (0.02, 0.19)	0.05 (0.03, 0.07)
1:256 — 1:512	0.08 (0.01, 0.21)	0.03 (0.02, 0.05)
1:512 — 1:1024	0.14 (0.02, 0.36)	0.02 (0.01, 0.03)
1:1024 — 1:2048	0.25 (0.01, 0.71)	0.00 (0.00, 0.01)
1:2048 — 1:4096	-	-
>1:4096	0.17 (0.01, 0.53)	0.01 (0.00, 0.01)

Table 7. Posterior Mean Log-likelihoods for Functional Forms of Protection Curve, by Subtype and Interpolation Method. As the parameter dimensionality is the same for all models, absolute differences between the means of 10 or more points indicate overwhelming evidence.

	Method 1 Mean (95% CrI)	**Method 2 Mean (95% CrI)**	**Method 3 Mean (95% CrI)**
Influenza A(H1N1)pdm09 – HK			
Log titer	-3,266 (-3,321, -3,243)	-3,664 (-3,675, -3,653)	-3,617 (-3,629, -3,606)
Titer	-11,152 (-11,159, -11,145)	-11,488 (-11,495, -11,483)	-11,258 (-11,263, -11,253)
Influenza B – HK			
Log titer	-3,824 (-3,833, -3,815)	-4,394 (-4,403, -4,386)	-4,277 (-4,287, -4,267)
Titer	-12,469 (-12,483, -12,475)	-13,023 (-13,029, -13,018)	-12,609 (-12,613, -12,606)
Influenza A(H3N2) – US			
Log titer	-5556 (-5591, -5524)	-7219 (-7243, -7194)	-7329 (-7350, -7306)
Titer	-28,285 (-28,302, -28,270)	-30,011 (-30,026, -29,996)	-28,285 (-28,302, -28,270)

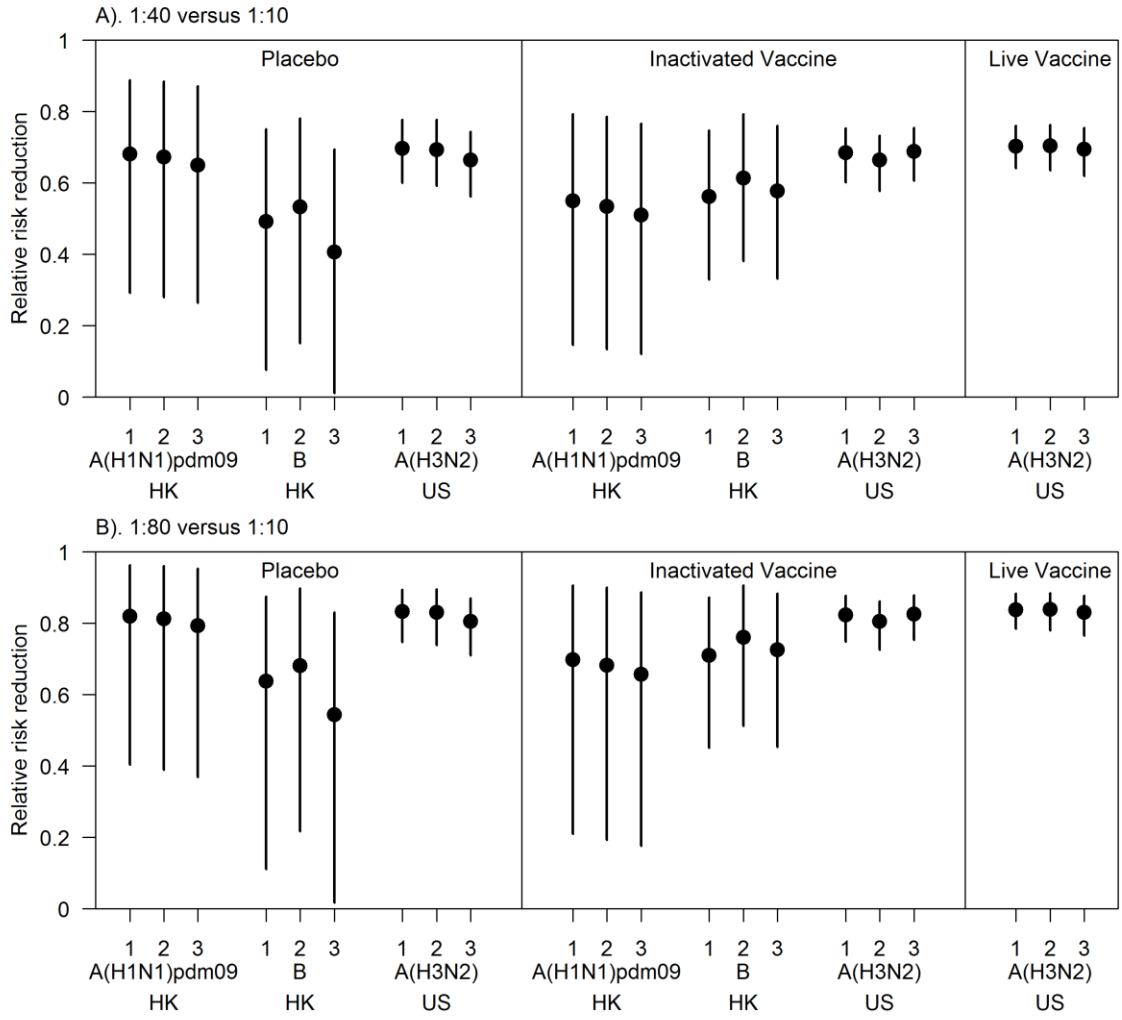


Figure 6. Relative risk reduction for influenza infection for hemagglutination inactivation titer 1:40 versus 1:10 (A) and 1:80 versus 1:10 (B), by interpolation method and treatment group. Points are posterior means and lines 95% credible intervals. The tick marks on the x axis indicate the interpolation methods (1, 2, or 3, as described in the text and **Figure 3**).

To account for both the vaccine effect on titers and on infection risk at a given titer, we derived the posterior probabilities that the cumulative incidence of infection in the vaccine (trivalent inactivated vaccine or live attenuated vaccine) arm is less than that of the placebo arm. For B, there was strong evidence that the trivalent inactivated vaccine reduced cumulative incidence (posterior probability 99.9%), by 6 percentage points (pp) (95% CrI:

Table 8. Relative risk of RT-PCR confirmed infection with a hemoagglutination inhbition titer of 1:40 or 1:80 compared with 1:10.
Posterior means and 95% credible intervals (CrIs). TIV=trivalent inactivated vaccine.

Method	Influenza A(H1N1)pdm09				Influenza B				Influenza A(H3N2)					
	Placebo		TIV		Placebo		TIV		Placebo		TIV		Live vaccine	
	Mean	95 % CrI	Mean	95 % CrI	Mean	95 % CrI	Mean	95 % CrI	Mean	95 % CrI	Mean	95 % CrI	Mean	95 % CrI
1:40 vs 1:10														
1	0.32	0.11, 0.71	0.46	0.21, 0.85	0.51	0.25, 0.92	0.44	0.25, 0.67	0.30	0.22, 0.40	0.31	0.25, 0.39	0.30	0.24, 0.36
2	0.33	0.12, 0.72	0.47	0.21, 0.87	0.47	0.22, 0.85	0.39	0.21, 0.62	0.31	0.22, 0.41	0.32	0.25, 0.40	0.30	0.24, 0.37
3	0.35	0.13, 0.74	0.49	0.23, 0.88	0.59	0.31, 0.99	0.42	0.24, 0.67	0.34	0.26, 0.44	0.34	0.27, 0.42	0.31	0.25, 0.38
1:80 vs 1:10														
1	0.18	0.04, 0.60	0.30	0.09, 0.79	0.36	0.12, 0.89	0.29	0.13, 0.55	0.17	0.11, 0.25	0.17	0.12, 0.25	0.16	0.12, 0.22
2	0.19	0.04, 0.61	0.32	0.10, 0.81	0.32	0.10, 0.78	0.24	0.09, 0.49	0.17	0.11, 0.26	0.18	0.12, 0.25	0.16	0.12, 0.22
3	0.21	0.05, 0.63	0.34	0.11, 0.82	0.46	0.17, 0.98	0.27	0.12, 0.55	0.19	0.13, 0.29	0.20	0.14, 0.27	0.17	0.12, 0.23

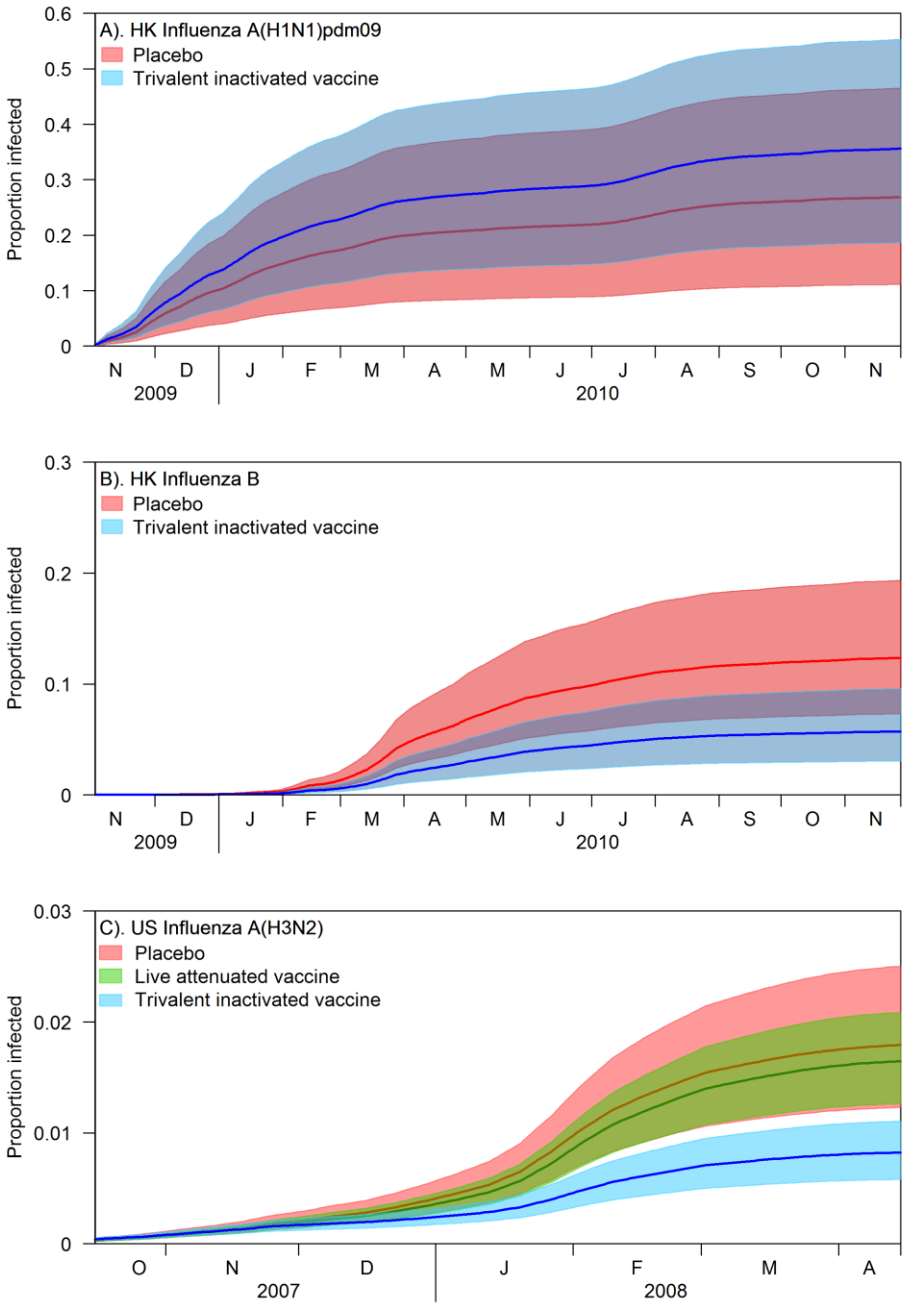


Figure 7. Cumulative influenza infection rates by subtype and treatment group. The lines are the estimated cumulative influenza infection rates with 95% credible intervals (colored polygons). In red is the placebo group, in blue is the group receiving inactivated vaccine and in green is the group receiving live vaccine. The dark red line, dark blue line and dark green line represent the posterior mean incidence rate for the placebo arm, the inactivated influenza vaccine arm and the live attenuated influenza vaccine respectively. Time is measured in calendar months at the x-axis on each panel. The longer tick marks at the x-axis indicate the beginning and end of each year.

Table 9. Absolute Decrease in Infection Probability Over Duration of Influenza Season. Posterior mean and 95% credible intervals (CrI) for the percentage point difference are tabulated. Positive differences mean the vaccine is protective relative to the placebo.

	Method 1	**Method 2**	**Method 3**
	Mean (95% CrI)	**Mean (95% CrI)**	**Mean (95% CrI)**
Influenza A(H1N1)pdm09 – HK	-0.05 (-0.17, 0.06)	-0.08 (-0.26, 0.08)	-0.09 (-0.26, 0.07)
Influenza B – HK	0.05 (0.02, 0.10)	0.07 (0.03, 0.12)	0.06 (0.02, 0.10)
Influenza A(H3N2) – US (live attenuated vaccine)	0.00 (0.00, 0.01)	0.00 (0.00, 0.01)	0.00 (0.00, 0.01)
Influenza A(H3N2) – US (inactivated vaccine)	0.01 (0.00, 0.02)	0.01 (0.00, 0.02)	0.01 (0.00, 0.02)

Table 10. Relative Risk of Infection Comparing with Placebo Over Duration of Influenza Season. Posterior mean and 95% credible intervals (CrI) for the risk ratio (vaccine relative to placebo) are tabulated.

	Method 1		**Method 2**		**Method 3**	
	Mean	**95% CrI**	**Mean**	**95% CrI**	**Mean**	**95% CrI**
Influenza A(H1N1)pdm09 – HK	1.44	0.74, 2.64	1.43	0.80, 2.70	1.41	0.79, 2.60
Influenza B – HK	0.47	0.29, 0.70	0.52	0.33, 0.78	0.52	0.33, 0.78
Influenza A(H3N2) – US (live attenuated vaccine)	0.94	0.63, 1.36	0.94	0.63, 1.37	0.92	0.61, 1.36
Influenza A(H3N2) – US (inactivated vaccine)	0.46	0.28, 0.71	0.47	0.29, 0.74	0.47	0.28, 0.74

2pp, 10pp, **Table 9**), and the relative risk of infection comparing the trivalent inactivated vaccine with placebo was 52% (95% CrI: 33%, 78%, **Table 10**). As expected, there was no evidence for an effect against A(H1N1)pdm09 in the same trial (posterior probability 14%). In the United States trial, against A(H3N2), the trivalent inactivated vaccine was associated with reduced cumulative incidence versus placebo (posterior probability >99.9%; relative risk: 47%, 95% CrI: 28%, 74%, **Table 10**), by 1pp (95% CrI: 0.3pp, 2pp, **Table 9**) but there

was little evidence that the live attenuated vaccine was associated with reduced cumulative incidence (posterior probability 69%), even after accounting for the mediating association of higher antibody titers.

3.5 Discussion

Influenza antibody titers as measured by the hemagglutination inhibition assay are frequently used in influenza epidemiology and modelling—where they serve as a proxy measurement for outbreak sizes [201], severity metrics [202], or herd immunity levels [203]—and in trials of reformulations of inactivated influenza vaccines, in which they are used to demonstrate evidence of protection [191]. In analyses of vaccine trials over an influenza season, the inferred protective effect is usually determined based on the last titer observed before the season, possibly a few weeks after vaccination. However, this may not correspond to the antibody levels experienced on average over the season, since antibody titers typically decay over the duration of an influenza season [131], so that the average titer depends on the duration of both the study and the timing of the epidemic. This complicates interpretation of protective effects.

Another difficulty in analyzing vaccine trials is that common study designs may violate the assumptions of standard statistical methods to estimate the protection conferred by a given pairing of antibody level and trial arm. For instance, common methods for time to event data, such as Cox regression, require that individuals experience infectious hazard at the same time relative to enrolment [204]. With staggered recruitment, this is not the case: if enrolment is staggered over weeks to months (as in the studies described in this paper), the discrepancy in peak influenza activity, when the infective risk is highest, relative to recruitment may be

large. To account for staggered entry requires synchronizing infectious challenge in the statistical model to calendar time, rather than time since entry into the study. This is relatively straightforward if there exists a suitable proxy for infection levels in the community, such as influenza-like illness rates from surveillance data, to which the infectious challenge can be assumed proportional. However, without suitable surveillance systems, more sophisticated approaches, perhaps using semi-parametric methods, would be needed.

Our results on the protective association of changes in antibody levels need to be interpreted differently from standard approaches, for our method explicitly accounts for the censored nature of the antibody titers, which are bracketed by the highest titration testing positive and the lowest testing negative. Typically, this interval is replaced in analysis and results by the lower end point, in calculating geometric mean titers for instance. In principle, however, more accurate measurements could be determined by using a series of dilutions at different starting points, so the observed titer in such cases is a reflection of both antibody levels and testing effort. The methodology described in this paper accommodates the uncertainty in the 'true' underlying titer by embedding a data augmentation step within the estimating routine. As a consequence, our estimates are not directly comparable to those from other studies, as estimates in the literature for a titer of 1:40, in our analysis correspond to an average over the interval [1:40 to 1:80], an interval in which the protective effect may vary 10–15% (depending on subtype and interpolation method, derived from **Table 8**).

We compared three methods to interpolate titers. It was noteworthy that the most predictive model used carried forward the pre-season titer (method 1) rather than accounting explicitly for antibody waning (method 3). The biologic mechanisms for this finding are unclear, and if it holds in other studies, would be worth investigating. Each method gave findings that were

very consistent within each treatment group and influenza subtype as 95% CrIs for relative risk reduction were highly similar (**Figure 6**); this finding may not apply if the rate of antibody waning were high, or the time duration of the studies longer. However, estimates of the protective effect of vaccination and of the ascertainment rate varied by influenza subtype (**Table 11**). The latter, i.e. the proportion of inferred infections testing positive with RT-PCR, was around 80 to 100% for B in Hong Kong and A(H3N2) in the United States, but only around 10 to 20% for A(H1N1)pdm09 in Hong Kong. The cause of this discrepancy is not clear, though the pandemic strain was the only one in which a first wave coincided with the pre- and post-vaccination samples. As expected, there was no evidence of protection by the trivalent inactivated vaccine against the pandemic strain, though it protected against B and A(H3N2). This analysis did not find evidence of a protective association between the live attenuated vaccine and A(H3N2) in the United States study, in common with other analyses that stratified the outcome by subtype [192]. An interaction between the titers and vaccination status was not found, suggesting that vaccine-induced immunity may behave similarly to natural immunity, at least among young vaccinees and over a time scale of a single season. The trials were in different age groups (pediatric in Hong Kong, mostly young adults in the United States) and the circulating strains did not overlap, preventing estimation of age effects on either antibody levels or their impact on infection risk. However, comparison of the slopes governing the waning of titers from method 3 showed no difference between A(H3N2) titers in adults and B in children, but did show differences between both B and A(H1N1)pdm2009 in children. Other studies have shown an age effect on the decline in titers [205]. Future research on age-differences in protection would be valuable for planning vaccine programs in the elderly. This study provides a picture of protection over a period of less than a year, but does not provide information on longer-term dynamics of immunity and heterotypic protection, for which a longer and possibly much larger cohort would be required.

Table 11. Posterior Distribution of the Ratio of RT-PCR Confirmed Infections to All Infections, θ, and Its Reciprocal. Posterior mean and 95% credible intervals (CrI) are tabulated.

	Method 1		Method 2		Method 3	
	Mean	**95% CrI**	**Mean**	**95% CrI**	**Mean**	**95% CrI**
θ						
Influenza A(H1N1)pdm09 – HK	5.40	3.89, 7.15	9.70	7.46, 12.26	10.78	8.41, 13.49
Influenza B – HK	0.98	0.65, 1.34	1.12	0.78, 1.53	1.37	0.95, 1.86
Influenza A(H3N2) – US	0.17	0.11, 0.23	0.17	0.11, 0.23	0.17	0.11, 0.23
$1/\theta$						
Influenza A(H1N1)pdm09 – HK	0.19	0.14, 0.26	0.10	0.08, 0.13	0.09	0.07, 0.12
Influenza B – HK	1.02	0.75, 1.54	0.89	0.65, 1.28	0.73	0.54, 1.05
Influenza A(H3N2) – US	6.19	4.28, 8.90	6.16	4.32, 8.88	6.19	4.29, 8.81

There are several limitations to this study. The analysis combined two end points: RT-PCR confirmed infection, and serologic evidence of infection measured by the hemagglutination inhbition assay. RT-PCR confirmed infection may suffer from ascertainment bias, due to some infections not being swabbed or other infections with low viral loads yielding a false negative in laboratory testing. Combining endpoints allowed us to estimate and correct for this bias, which was more substantial for A(H1N1)pdm09 than A(H3N2) and B. The model used also imposed specific formulations for the influenza antibody trajectory in between sampling points that were simple and transparent, but it remains an open question what the best approach is to modelling realistic antibody trajectories and the impacts of their temporal evolution on protection against influenza virus infection. A further limitation is that each strain was modelled separately, rather than explicitly modelling any protection against heterotypic infections [206]. In the United States trial, a single strain A(H3N2) dominated, but the Hong Kong trial involved two successive waves, of A(H1N1)pdm09 then B. Analyzing strains separately may underestimate the protective effect if the rise in titers against the second strain following infection with the first is more modest than following

actual infection with the second strain. This would potentially bias estimates of the vaccine effects in situations where the first strain was included in the vaccine, although this was not the case in the Hong Kong trial.

3.6 Motivations for next chapter

This chapter proposed three simple methods for imputing HAI titers at unobserved time points. These methods are not comprehensive enough to reflect the true titer trajectories after vaccination or an infection. In Chapter 4, a Bayesian hierarchical model was developed to characterize dynamics of vaccine- or infection-induced immunity at both individual and population level. A simulation model was also performed to correlate dynamics of herd immunity level with influenza epidemics in the tropical setting. The results are important to predict and inform the timing of influenza vaccination programmes in the tropics where no regular epidemic pattern of influenza infections can be observed.

CHAPTER FOUR: INDIVIDUAL AND POPULATION TRAJECTORIES OF INFLUENZA ANTIBODY TITERS OVER MULTIPLE SEASONS IN TROPICAL SINGAPORE

This work has been published as: Zhao X, Ning Y, Chen MI-C, Cook AR. Individual and population trajectories of influenza antibody titers over multiple seasons in tropical Singapore. Am J Epidemiol. 2017 Jun;187(1): 135–143.

4.1 Abstract

Seasonal influenza epidemics occur year round in the tropics, complicating the planning of vaccination programs. An individual-level longitudinal model was built of baseline antibody levels, time of infection and the subsequent rise and decay of antibodies post-infection using influenza A(H1N1)pdm09 data from two sources in Singapore: a non-community cohort with real-time polymerase chain reaction confirmed infections and at least one serological sample collected from each participant from May to Oct 2009 (n = 118) and a community cohort with up to six serological samples collected between May 2009 to Oct 2010 (n = 760). The model was hierarchical, to account for interval censoring and inter-individual variation. Model parameters were estimated via Reversible Jump Markov chain Monte Carlo using custom designed R and C++ code. Antibody levels peaked 4–7 weeks with a half-life of 26.5 weeks, followed by a slower decrease until one year to approximately pre-infection levels. After the third wave, the seropositive rate and the population-level antibody titer dropped to the same level as they were at the end of the first pandemic wave. The analysis is consistent with the hypothesis that the population-level effect of individuals' waxing and waning antibodies influences influenza seasonality in the tropics.

Word count = 200 words

4.2 Introduction

In temperate and sub-tropical countries, influenza epidemics occur regularly during cold winter months and monsoon season, respectively [16]. Yet in tropical countries such as Singapore, influenza activity is much more irregular [207]. This lack of seasonality on the equator may complicate the planning of vaccination programs in tropical countries, particularly, selecting the best timing of vaccination campaigns [68].

Influenza antibody titers measured by hemagglutination-inhibition (HAI) assays are associated with protection against influenza infections [108]. They fluctuate over time according to the individuals' exposures: increasing substantially due to infection/vaccination and then gradually waning [205]. However, there are few studies that have investigated individuals' long-term antibody trajectories over multiple influenza waves, and how this translates to population level immunity, information which is important for planning influenza vaccination programs.

The 2009 influenza A(H1N1) pandemic afforded an unusual opportunity to study the trajectory of immune response to influenza infection, as well as the link between herd immunity levels and the timing of influenza epidemics, because most individuals, especially children and young adults, did not have pre-existing immunity against the new strain [203].

Here, we developed a statistical model to characterize the evolution of antibody titers against influenza virus infection using a series of HAI assays collected over multiple influenza seasons in a community in Singapore as well as the supplementary RT-PCR data collected from various subpopulations. Conventionally, a fourfold rise in paired sera is indicative of infection [208,209]. but has low sensitivity [210]. We therefore synthesize information from RT-PCR data in addition to repeat serology to provide information on the temporal evolution

of HAI titers in the immediate aftermath of infection as well as estimating the risk of infection without the restriction of assuming a fourfold rise. To do this we developed a novel method that exploits a rich dataset unobscured by the impact of seasonal forcing.

4.3 Methods

Data

This analysis used serial serological samples from two distinct cohorts.

The primary dataset involved a community cohort recruited from the Multi-ethnic Cohort (MEC) of the Singapore Consortium of Cohort Studies (SCCS), as described in detail elsewhere [211,212]. In total, 838 subjects aged from 21 to 75 years old were enrolled, of whom 760 (91%) with serological data recorded were analyzed (see **Table 12** for demographic table). Repeated serological samples were collected at up to six different time points from 2009 to 2010, spanning the H1N1 pandemic and subsequent waves (**Figure 8**), as described in detail elsewhere [212]. Each subject had at least two blood samples recorded, and 430 (57%) of 760 subjects had a full set of 6 blood samples.

Table 12. Summary of Demographic Characteristics of Participants in the General Population Cohort (n = 760), Singapore, 2009–2010.

	N	%
Age group, years		
Below 25	85	11.2
25–34	109	14.3
35–44	222	29.2
45–54	228	30.0
Above 55	116	15.3
Gender		
Male	311	40.9

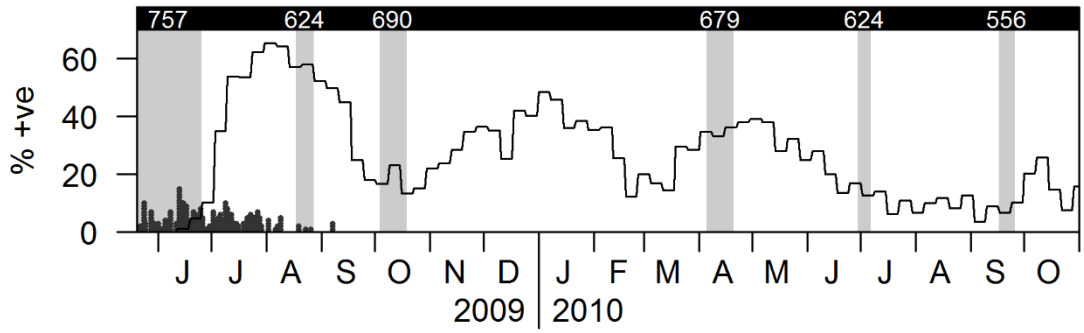


Figure 8. Blood collection period for the community cohort and the distribution of the daily number of influenza A(H1N1)pdm09 cases detected in the RT-PCR cohort (2009). Grey bars indicate the timing of serum samples taken from the community cohort and the number on the top indicates the number of serum samples collected for each sampling period. The solid black line represents the weekly relative proportions of influenza A(H1N1)pdm09 infections obtained from routine primary care surveillance which provides a reference for the size of pandemic at the community level. The black dots in the bottom gives the daily number of A(H1N1)pdm09 cases identified in the RT-PCR cohort.

A second RT-PCR cohort (n = 118) supplemented this, composing hospitalized patients, health care workers, military personnel, and staff and residents in a nursing home. These individuals had both RT-PCR confirmed infection and at least one serological sample (average 2.3; range: 1 to 7) [213]. This dataset was informed the temporal evolution of titers in the immediate aftermath of infection.

Blood samples were assayed for hemagglutination inhibition antibody titers against influenza A(H1N1)pdm09 infections. Written informed consent was obtained from all participants.
In addition, we used routine surveillance data from local sentinel healthcare clinics who submitted weekly nasal and/or throat swab samples from influenza-like illness (ILI) cases. The samples were sent to the National Public Health laboratory (NPHL) to confirm influenza infection. We analyzed influenza A(H1N1)pdm09 positive samples amongst ILI cases, to

validate whether modelled influenza incidence over time was consistent with the community influenza surveillance.

Statistical analysis

Our primary objective was to model the trajectory of evolving influenza A(H1N1)pdm09 antibody titers, and the secondary objective was to estimate the time of infection for those infected as a form of community influenza surveillance. An individual-level longitudinal model was built of baseline antibody levels, risk of infection, time of infection and the subsequent rise and decay of antibodies post infection. The model was hierarchical, to account for differences between individuals, and accounted for interval censoring of antibody titers. The antibody titer has values ranging from <1:10, 1:10 to 1:20, …, ≥1:1280. To simplify the analysis, we coded 1 for 1:10, 2 for 1:20, etc., and thus intervals to be [1,2), etc., in other words, using a logarithmic scale. The titer at any time point was modelled via a Gaussian distributed latent variable, $z_{it} \sim N(\mu_{it}, \sigma^2)$ for individual i at time point t. The latent variable was then interval censored in line with the recorded titer values. The mean titer, μ_{it}, varies by individual, time and infection status. It is modelled to be constant at the individual i's baseline titer level, B_i, if that individual is never infected. If infected, a boosting term is added that is characterized by 5 parameters $(T_i, B_i, M_i, S_i, R_i)$, where T_i is the time of infection for individual i, M_i is the time of peak rise after infection, S_i controls the steepness of the rise over time and R_i is the additional titer due to infection at the time of peak rise. The estimated mean titer, μ_{it}, at time t for individual i can be, therefore, expressed as:

$$\mu_{it} = \begin{cases} B_i \text{ if i is not infected, and} \\ B_i + R_i \times f(t - T_i, k_i, \theta_i) \text{ otherwise.} \end{cases}$$

where $f(x, k_i, \theta_i) = 1/(\Gamma(k_i)\theta_i^{k_i})x^{k_i-1}\exp(-x/\theta_i)$ is a gamma density with parameterization shape k_i and scale θ_i, set to be functions of the mode and steepness parameters: $\theta_i = -M_i/2 + \sqrt{{M_i}^2 + 4{S_i}^2}/2$, and $k_i = {S_i}^2/{\theta_i}^2$. The model was parameterized in this way to facilitate identifiability of the parameters by reducing their interdependency.

The risk and time of infection was modelled assuming a constant hazard of infection ϕ, so that the likelihood contribution was $e^{-\phi D}$, where D is the length of follow up, for those never infected and $\phi e^{-\phi d}$ for those infected at time d. Because the infection status is unknown for subjects in the community cohort, it was difficult to relate the time of infection to the early dynamics of antibody titers. We therefore supplemented the primary data with additional data from individuals with RT-PCR confirmed infection to obtain a better estimate of the initial trajectory following infection and thereby better estimates of the infection status and time of infection if infected of the community cohort. For those individuals, the (known) time of onset was not modelled as the risk to these individuals was not deemed to reflect the risk in the general community. The likelihood contribution from individual i would be, therefore, the product of likelihood function for the titer distribution and for those in the community cohort, the likelihood function for the time of infection. A complication to estimation is that the dimension of the parameter vector is not a priori specified, but changes with the infection status for each individual. For instance, if infection status for individual i was to switch from not infected to infected, the corresponding parameters would change from (B_i) to $(B_i, M_i, S_i, R_i, T_i)$. To facilitate changes in parameter dimension for individuals with unknown infection status, we employed the reversible jump MCMC (RJMCMC) algorithm. This has the advantage of allowing changes in the dimension of the parameter vector as well as the parameter values by simulating from their posterior distributions [146]. Let us define the state

of the parameter to be $x = (k, \theta_k)$, where k indicates the model index and θ_k represents the parameters used in model k. We define Model 1 as the model when individual i was not infected, i.e. $I_i = 0$ and define Model 2 as the model when individual i was infected, i.e. $I_i = 1$. Now consider a move from Model 1 to Model 2. The current state is, therefore, $x = (1, \theta_1)$ where $\theta_1 = B_i$. However, there are no equivalent parameters for (M_i, S_i, L_i, T_i) in Model 1. To overcome this issue, we generate four random numbers u_1, u_2, u_3 and u_4 independently from four normal distributions (g_1, g_2, g_3, g_4) with fixed mean and variance. Then the new state of the chain, $x' = (2, \theta_2)$, where $\theta_2 = (B_i, M_i, S_i, L_i, T_i)$, could be constructed by a diffeomorphic (invertible and differentiable) function h such that $(x') = h(x, u_1, u_2, u_3, u_4) = (B_i, \mathrm{e}^{u_1}, e^{u_2}, e^{u_3}, u_4)$. Then the reverse move, i.e. from Model 2 to 1, is derived by setting$(x, u_1, u_2, u_3, u_4) = h^{-1}(x') = (B_i, \log(M_i), \log(S_i), \log(L_i), T_i)$. The acceptance probability for the move from Model 1 to 2 is, therefore,

$$\alpha(x, x') = \min(1, \frac{\pi(2,\theta_2)}{\pi(1,\theta_1)} \big(g_1(u_1)g_2(u_2)g_3(u_3)g_4(u_4)\big)^{-1} |\frac{\partial(\theta_2)}{\partial(\theta_1, u_1, u_2, u_3, u_4)}|,$$

and from model 2 to model 1,

$$\alpha(x', x) = \min(1, \frac{\pi(1,\theta_1)}{\pi(2,\theta_2)} \big(g_1(\log(M_i))g_2(\log(S_i))g_3(\log(L_i))g_4(T_i)\big) |\frac{\partial(\theta_1, u_1, u_2, u_3, u_4)}{\partial(\theta_2)}|.$$

The new candidate for each parameter will be either accepted or rejected using the acceptance probability above. The estimated value for each parameter will be obtained once its posterior distribution reaches a stationary distribution.

The hierarchical model for the baseline (B_i), peak time (M_i), steepness (S_i), and peak rise parameter (R_i) described above was built using independent log Normal priors, for instance $log(B_i) \sim N(\mu_B, \sigma_B^2)$ etc. Each pair of hyperparameters, e.g. (μ_B, σ_B^2), was given a non-informative normal-inverse gamma prior distribution which facilitated a Gibbs sampler from

the conjugate conditional distribution. The prior distributions for all other parameters were taken to be uniform distributions over their support regions.

The infection status I_i, was proposed from a Bernoulli (0.5) distribution via the Metropolis-Hastings algorithm. All other parameters were proposed independently using normal distributions centered on the current value with bandwidth selected from pilot runs, and T_i rounded to the nearest integer.

Parameters were estimated with 10,000 iterations after a burn-in period of 2000. Posterior distributions of the model parameters showed good convergence that was assessed using trace plots and the Geweke diagnostic [145].

To validate the inference routine, we took the same cohort size and the best guess for each parameter to generate a simulated set of data points for each individual, and we then fitted the same model on the simulated dataset. Model accuracy was checked by comparing the estimated means and 95% credible intervals (CrIs) with the parameters used to simulate data.

Epidemic simulation

To assess the consistency of our analysis and our hypothesis that the waning of herd immunity levels is associated with the timing of influenza epidemics in the tropics, we performed a simple simulation using an SIS model where individuals could move back and forth between the susceptible and infected state. We assumed that infected individuals would remain infectious for 7 days and would not have reinfections within 30 days from disease onset, following which infection risk is modulated by the simulated titer. The model was

based on a population with size of 1,000 over a period of 10 years. The pre-infection geometric mean titers (GMT) for each individual at the beginning of the first pandemic wave followed uniform distributions on the interval [0, 1]. The risk of infection for a susceptible individual i at day t was given by p_{it}, which was derived from a discrete time approximation to a continuous infection rate. We assumed that there was an exponential decay in the population at risk over time, so p_{it} can be derived using the formula below:

$$p_{it} = \begin{cases} [1 - e^{-(\alpha+\beta NI_t)\times s_t}] \times e^{\theta \times titer_{it}} & \text{if no prior infection within the last 30 days} \\ 0 & \text{otherwise} \end{cases}$$

where NI_t is the number of infections over the last 7 days, s_t gives the effect of seasonal forcing at day t, θ determines the protective effect conferred by antibody titers which was taken from Zhao et al. [108], titer_{it} is the antibody titers for individual i at time t, and α and β capture importation and autochthonous infection, respectively. We gave values of 0.1 and 0.3 for α and β respectively because it could produce a similar pattern for the first influenza pandemic wave and two subsequent waves in Singapore. If individual i was infected at day t, his/her antibody titers would be boosted by adding the estimated mean post-infection titers (**Figure 9**A) to the baseline titer. Seasonal forcing was considered by multiplying the infection risk by a sinusoidal term $s_t = e^{\gamma \times sin(\frac{2\pi t}{365})}$, where γ determines the amplitude of the oscillation. This model makes a series of additional assumptions beyond those of the preceding analysis, most importantly that the risk of reinfection depends linearly on log HAI titers (estimates derived from Zhao et al.) [108], and uses plausible but arbitrary values of the baseline infection and importation risks.

All statistical analysis were implemented in R version 3.1.2 (R Foundation for Statistical Computing, Vienna, Austria) [182]. To overcome speed issues in R, functions were executed in the C++ language by using the Rcpp and RcppArmadillo extensions [214,215].

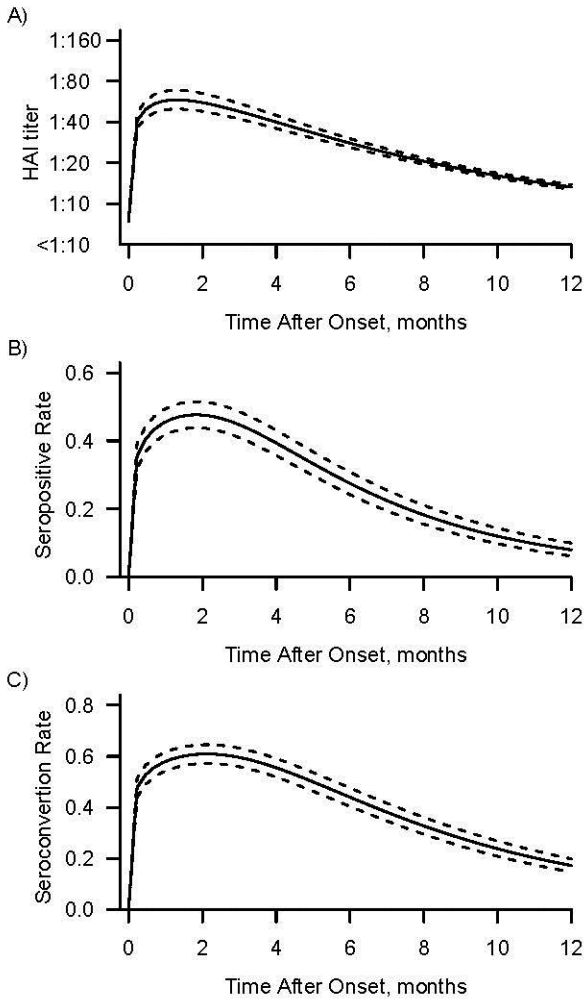


Figure 9. Individual level dynamics for the modelled titer trajectory (A), the probability of having titers≥1:40 (B) and the probability of seroconversion (C) over time. The solid lines represent the estimated mean values and the dotted lines represent the 95% credible intervals. On panel (A), the uncertainty interval bonded by the dotted lines is the 95% credible interval for the estimated mean titer.

4.4 Results

A summary of pre-infection titer distributions for the community cohort and RT-PCR cohort was shown in **Table 13**. The two cohorts had similar pre-infection titer distributions (p-value = 0.15). For RT-PCR cases, we have data to define the infection status and time of disease onset (as opposed to the community cohort where these parameters have to be inferred from the model). On average, we found the antibody titers rose quickly to [1:40, 1:80) one week after onset of disease, peaked around four weeks after onset, and then gradually tailed off (**Figure 10**).

Table 13. Comparison of pre-infection titer distributions between the community cohort and RT-PCR cohort.

Pre-infection titer	**Community cohort n = 199* (%)**	**RT-PCR cohort n = 76* (%)**
<1:20	194 (97.4)	71 (93.4)
≥1:20	5 (2.6)	5 (6.6)
P-value obtained from Fisher's exact test	0.15	

*There were 208 of 760 estimated influenza A(H1N1)pdm09 cases in the community cohort. Of which 199 had pre-infection titers measured and 76 of 118 RT-PCR cases having pre-infection titers measured.

The modelled titer trajectories with 95% CrIs as well as observed titer levels of six selected individuals from the community cohort who seroconverted during the study period are displayed in **Figure 11** (fits for all individuals from both cohorts are presented in supplementary data at https://academic.oup.com/aje/article/doi/10.1093/aje/kwx201/3896092/Individual-and-population-trajectories-of). In almost all cases the modelled antibody levels pass through or near the titer data, with a shape that reflects the evolution in the observed data closely. The time of infection with 80% and 95% CrIs were estimated. The width of the uncertainty time window ranges from a few weeks for those whose infection falls around the time of serological samples, to one year for those whose infection probably fell in between two

widely spaced serum samples. In simulations, there was good concordance between the parameter estimates derived from simulated data and the parameters used to generate the simulated data (**Table 14**).

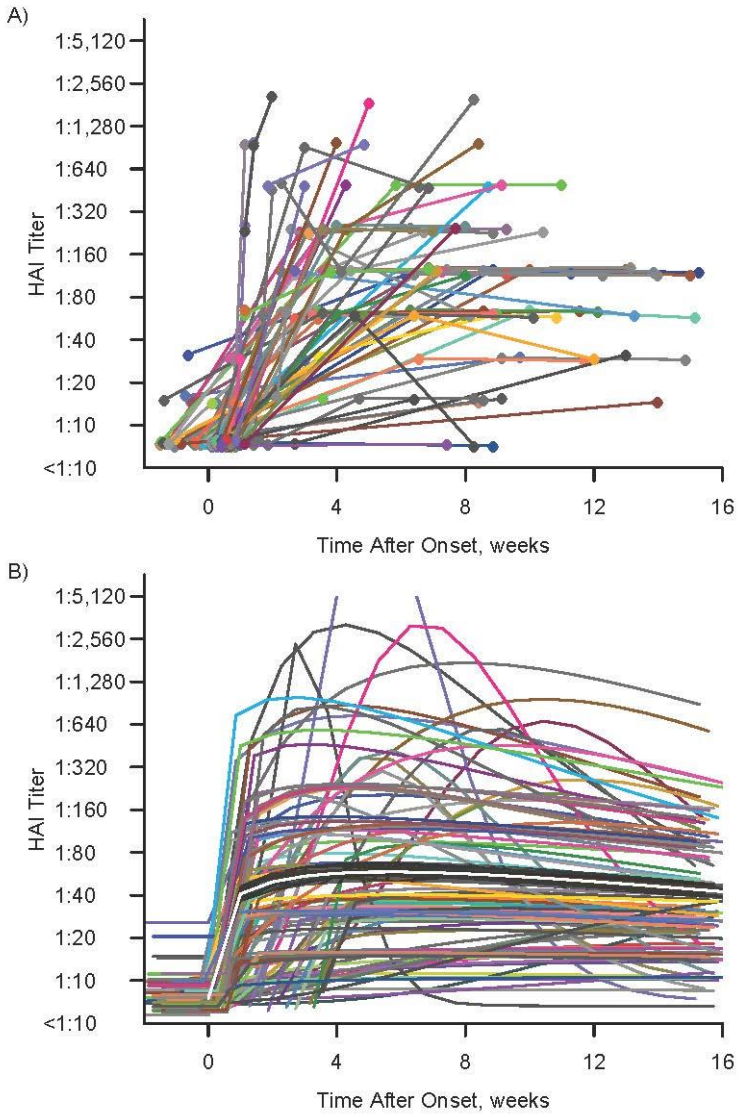


Figure 10. Observed repeated antibody titers (A) vs modelled antibody titer trajectories (B) for the RT-PCR cohort (n = 118). Subjects in the RT-PCR cohort were colored in different colors. On panel (A), we use a grey polygon to indicate the time before infection onset. The colored dots represent the observed antibody titers over time along with lines connecting dots from the same individual. On panel (b), each colored line represents the modelled titer trajectory over time for each individual. The white line shows the estimated mean titer trajectory for infected individuals with black shaded area representing the 95% credible interval (CrI) for the estimated mean titer: the narrowness reflects the large sample size.

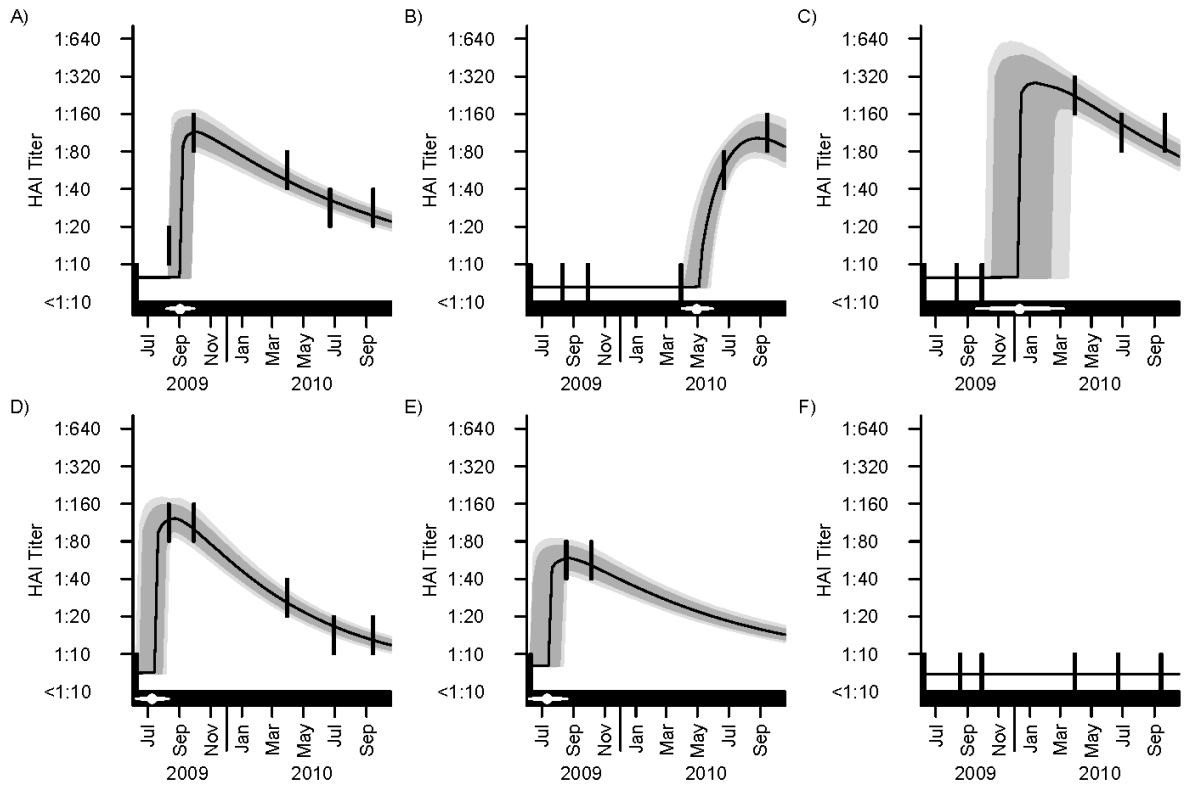


Figure 11. Observed and modelled antibody titers over time with estimated infection time for the selected six subjects in the community cohort. Six subjects were arbitrarily selected in the community cohort to show the goodness of fit of the developed hierarchical model. More than half of the participants in the community cohort had six blood samples collected. Panel A—D represent the model fitting for individuals with six repeated blood samples collected but with different time of infection relative to the time of blood samples taking. Panel E) shows an example of an individual who contributed fewer than six blood samples. Panel F) shows an example of individuals estimated not to have been infected during the study period: a majority of participants in the community cohort had similar titer trajectories as the participant shown on panel F). The thick vertical lines on each panel represent the repeated measured antibody titers over the censored intervals. The black line represents the modelled median antibody titers over time. The grey and light grey shaded area represent the 80% and 95% credible interval (CrI) for the modelled antibody titer trajectories respectively. The white point at the bottom on each panel represents the estimated time of infection with a thicker whisker and a thinner whisker showing the 80% CrI and 95% CrI for the time of infection respectively.

Individual level dynamics for the titer trajectory, the seropositive and seroconversion rates over time post-onset in months for hypothetical individuals with no prior immunity against

influenza A(H1N1)pdm09 infections are presented in **Figure 9**. On average, the antibody titers peaked at 6.8 weeks (95% CrI: 6.0, 7.0 weeks) after infection with an estimated half–life of 26.5 weeks (95% CrI: 24.0, 29.0 weeks), and then decreased almost to baseline antibody levels after 12 months post-infection (posterior distributions of antibody titers over time are presented in supplementary data and can be accessed via https://academic.oup.com/aje/article/doi/10.1093/aje/kwx201/3896092/Individual-and-population-trajectories-of). The seropositive or seroconversion rates were derived directly from the fitted model that peak at approximately 50% and 60% respectively, and may provide guidance for the timing of serological tests to estimate epidemic size. This suggests that about one in two infections would be missed based on traditional metrics. For the community cohort, age and gender are not associated with peak antibody titers, as indicated by the low correlations: 0.05 (p–value = 0.49 using Pearson's correlation test) and -0.01 (p–value = 0.93) respectively, against posterior mean peak levels. Older adults have shorter half-life of antibody titers against influenza infections (correlation: -0.18; p–value = 0.01), however, no difference was observed between males and females (correlation: -0.11; p–value = 0.11).

Table 14. Posterior means of population parameters with 95% credible intervals obtained from models using observed data and simulated data.

	Posterior mean (95% CrIs)			**Posterior mean (95% CrIs)**	
	Parameter used for simulation	Inference		Parameter used for simulation	Inference
μ_B	-0.54	-0.54 (-0.57, -0.50)	ν_B	0.24	0.24 (0.22, 0.26)
μ_R	4.82	4.81 (4.76, 4.86)	ν_R	0.59	0.58 (0.53, 0.64)
μ_M	1.28	1.28 (1.19, 1.38)	ν_M	2.15	2.14 (1.95, 2.35)
μ_S	3.38	3.38 (3.31, 3.45)	ν_S	1.19	1.18 (1.08, 1.30)
ϕ	-5.42	-5.24 (-5.57, -5.29)			

The population level dynamics of immunity were also estimated from the model. The modelled cumulative attack rate (CAR) increased rapidly during each of the first three waves (**Figure 12A**; see overall and age-stratified CARs in **Table 15**). The corresponding modelled weekly incidence exhibits similar patterns to the relative proportion of influenza A(H1N1)pdm09 in the community (**Figure 12B**) which was not used in the model building process. Unlike the CAR, which increased over time, the antibody titer and seropositive rate remained flat after the first wave. By the end of the study period, the antibody levels and seropositive rate had decreased almost to the level just after the first wave (**Figure 12C–D**). The next wave of influenza A(H1N1-2009pdm) occurred in January 2011 [148], by which time the population-level measures of immunity should have dropped below this level.

Table 15. Estimated means of overall and age-stratified cumulative attack rates (CARs) with 95% credible intervals (CrIs) at the end of first and third wave, Singapore, 2009–2010.

	End of first wave (mid-October 2009)		End of third wave (July 2010)	
	mean	**95% CrI**	**mean**	**95% CrI**
Overall CAR	12.7%	12.2% to 13.2%	23.8%	23.2% to 24.5%
Age-stratified CAR				
Below 25	20.7%	20.0% to 22.4%	38.2%	35.3% to 41.2%
25–34	9.2%	8.3% to 10.1%	22.4%	22.0% to 23.9%
35–44	13.1%	12.6% to 14.4%	22.0%	20.7% to 23.0%
45–54	12.7%	11.8% to 13.6%	22.5%	21.5% to 23.7%
Above 55	9.4%	8.6% to 10.3%	20.7%	19.0% to 22.4%

In simulations, the epidemic pattern at the beginning of the simulation period was qualitatively similar to the epidemic patterns in Singapore (**Figure 13**), but differed from those of the immediate aftermath of the pandemic. Simulations yielded yearly epidemics, but the timing of epidemics was irregular. We found a good synchronization between simulated

population level GMTs and influenza epidemics over time. This agrees with the hypothesis of waning immunity being associated with the timing of epidemics in the tropics where there is an absence of seasonal forcing (**Figure 13A**). If we added in a small amount of seasonal forcing, i.e. the risk of infection would go up and down by 20% over the course of one year, the simulated antibody levels synchronized epidemics to winter seasons (**Figure 13B**). This is consistent with the influenza epidemic patterns in the northern and southern hemisphere temperate regions.

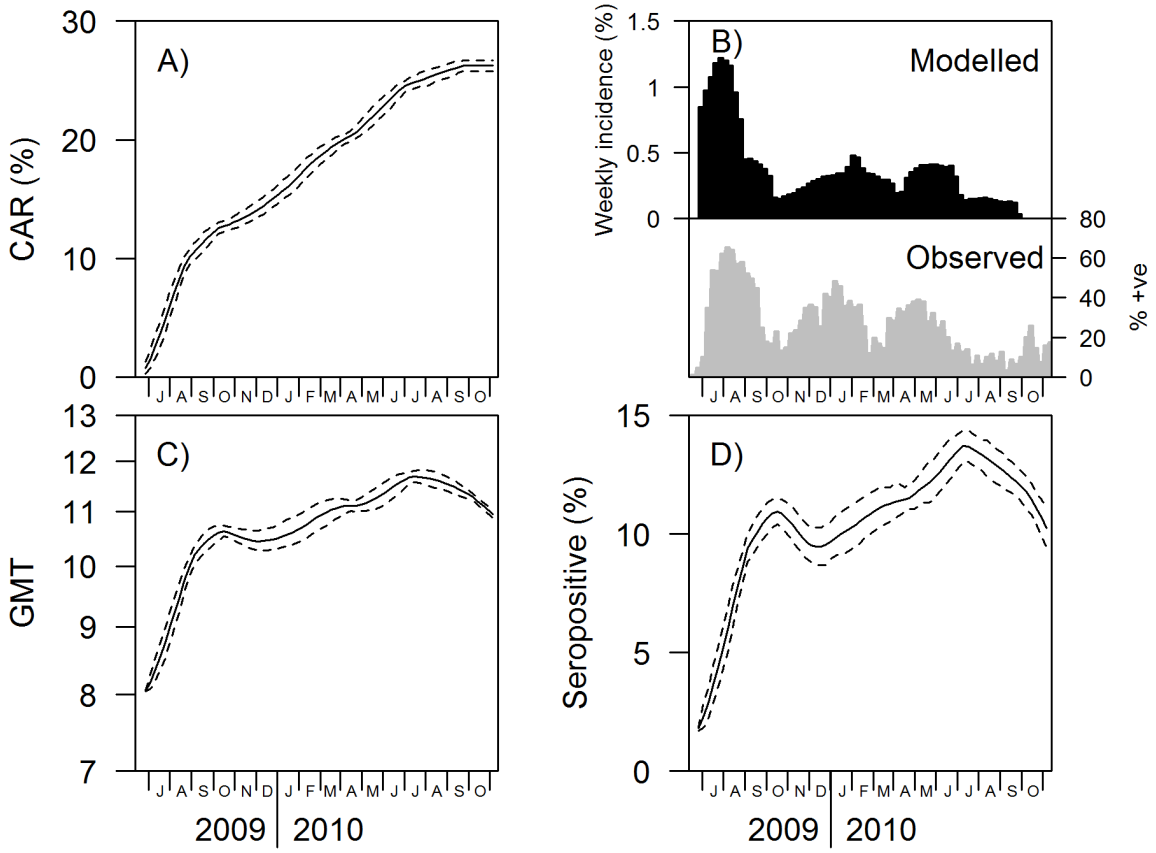


Figure 12. Population level dynamics for the cumulative attack rates (CAR) (A), observed and modelled influenza A(H1N1)pdm09 incidence (B), geometric mean titers (C) and seropositive rate (D) over time. For panel A), C) and D), the solid line on each panel represent the estimated mean values and the dotted lines represent the 95% credible intervals (CrI). On panel B), the top subpanel illustrates the modelled incidence of influenza A(H1N1)pdm09 infections (black bars), and the bottom subpanel illustrates the observed relative proportions of influenza A(H1N1)pdm09 infections among influenza–like illness (ILI) samples obtained from the routine primary care surveillance (grey bars). The y-axis scales differ for these two subpanels on panel B).

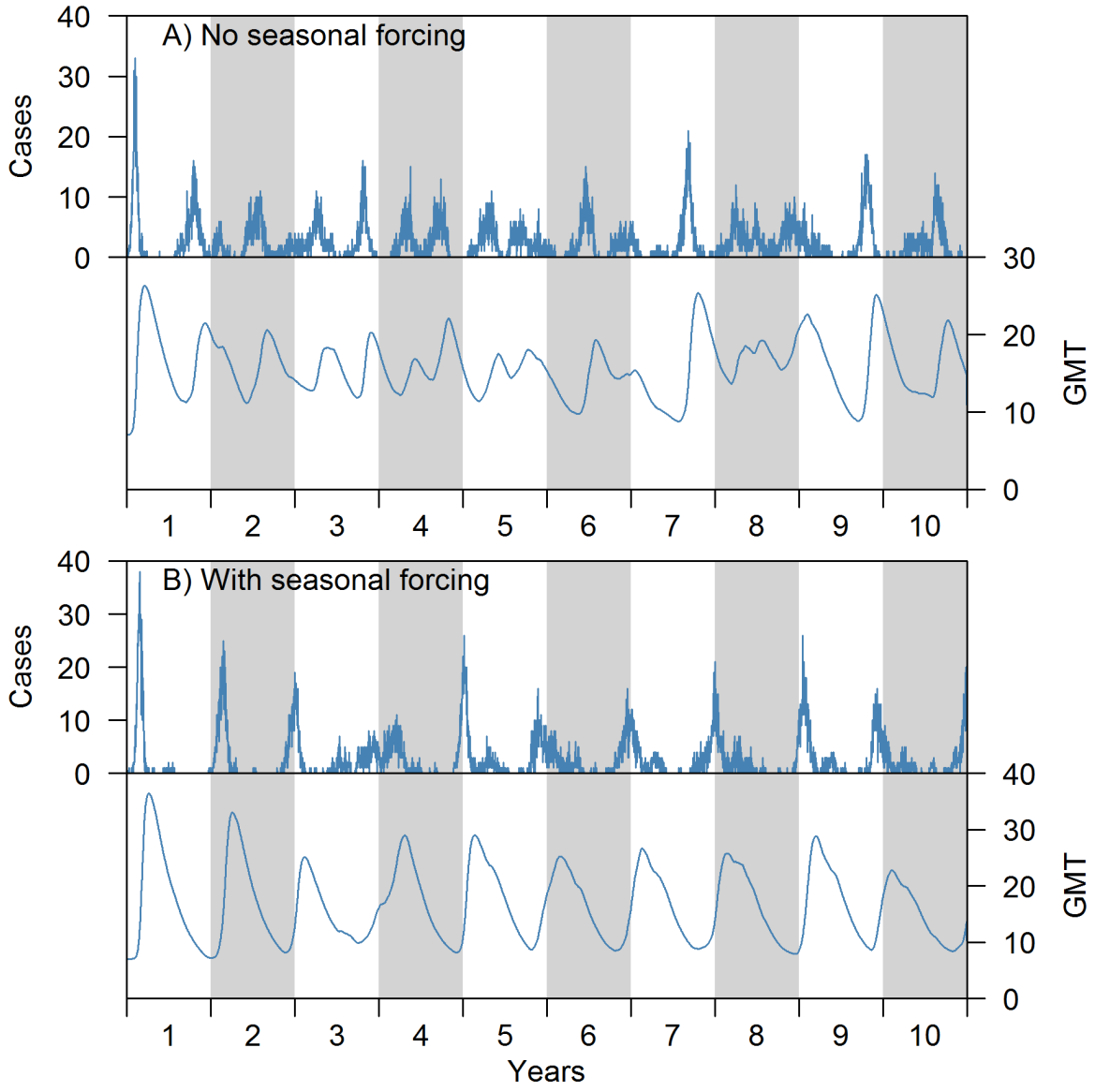


Figure 13. Simulated number of cases and population level geometric mean titers (GMTs) with (A) and without (B) seasonal forcing. The blue line in each panel shows the simulated number of cases or GMTs over a period of ten years. The simulation was done on a hypothetical population with size of 1000, and we assumed each individual started with no or little immunity against the influenza infections. The top panel (A) shows the scenario when there is no seasonal forcing, i.e. $\gamma=0$, while panel (B) shows the scenario when there is seasonal forcing of 20% in the community, i.e. $\gamma=0.2$.

4.5 Discussion

In this paper, a Bayesian approach was undertaken to establish the time of infection and the evolution of influenza antibody titers at both individual and population level. Bayesian

evidence synthesis is becoming more widespread in public health, [216,217] because it provides a flexible framework to integrate various types of information. Moreover, Bayesian approaches are valuable for longitudinal seroepidemiological studies as they provide a natural and efficient mechanism to account for between individual variability.

Previous research has shown that delayed collection of blood samples might lead to under-ascertainment of influenza infections.[209] A straightforward approach to account for the timing of serological samples was to use a suitable proxy for influenza levels in the community, such as ILI rate extracted from an influenza surveillance program.[108] Taking the current approach, however, meant it was not necessary to include the proxy measurements of influenza incidence in the community to estimate the time of infection, as the large number of study participants and frequencies of blood samples collected from each participant provided sufficient information to establish the incidence at the individual level and reproduce the timings of first three waves as observed with the traditional surveillance well (**Figure 12B**). The results for sensitivity analysis using model with fluctuating hazard of infection are shown in **Figure 14**, and similar results were obtained.

The model developed in this paper was based on data from the first influenza pandemic wave and two subsequent waves due to a new strain. The majority of the participants would have no or little immunity against the new pandemic strain, so we assumed a homogeneous risk of infection for all members of the community cohort. In contrast, the general population would have a broad range of baseline antibody titers at the beginning of a seasonal influenza epidemic, which would necessitate the adaption of the model to include the starting titer as one model feature and allow the risk of infection to change depending on the starting titer.

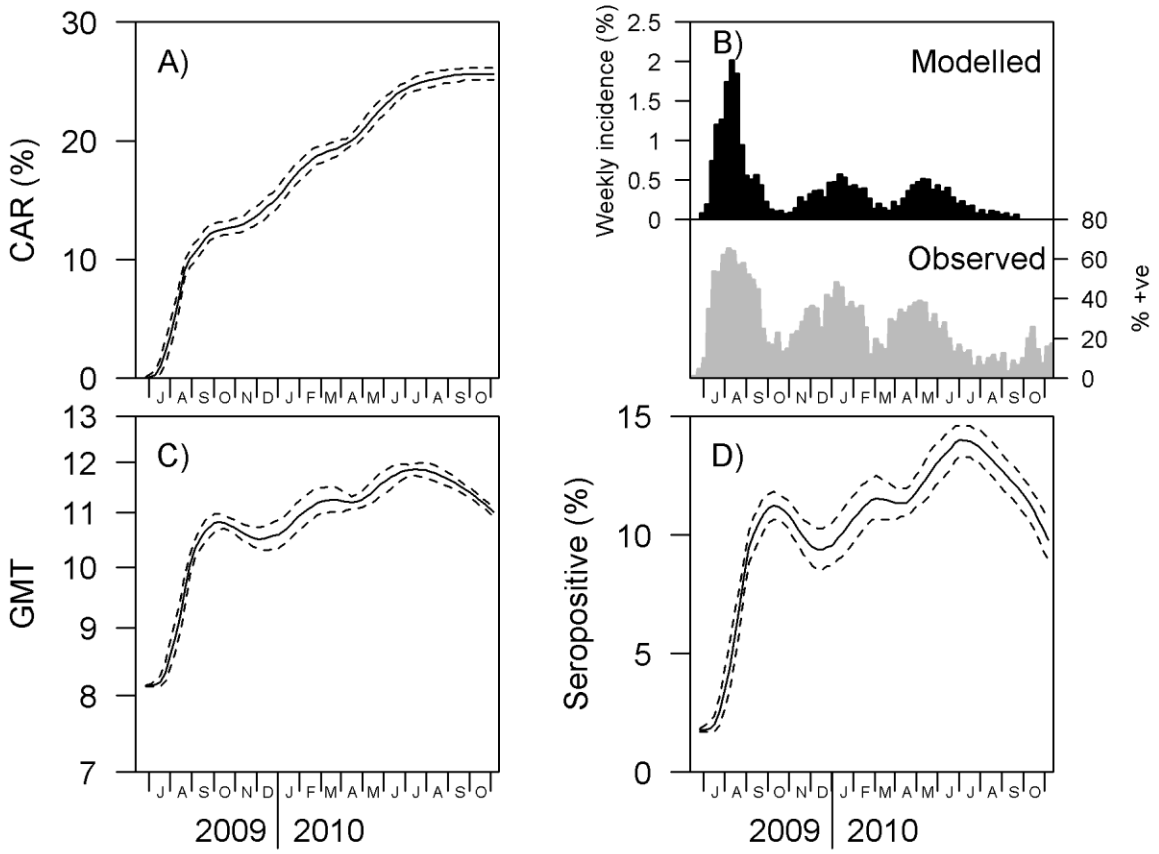


Figure 14. Population level dynamics for the cumulative attack rates (CAR) (A), observed and modelled influenza A(H1N1)pdm09 incidence (B), geometric mean titers (C) and seropositive rate (D) over time using model with fluctuating risk of infection over time. For panel A), C) and D), the solid line on each panel represent the estimated mean values and the dotted lines represent the 95% credible intervals (CrI). On panel B), the top subpanel illustrates the modelled incidence of influenza A(H1N1)pdm09 infections (black bars), and the bottom subpanel illustrates the observed relative proportions of influenza A(H1N1)pdm09 infections among influenza–like illness (ILI) samples obtained from the routine primary care surveillance (grey bars). The y-axis scales differ for these two subpanels on panel B).

To model titer trajectories immediately upon infection, we combined the RT-PCR cohort data with community cohort data. The RT-PCR assay is likely to have an atypically high sensitivity in the current study because the RT-PCR cohort was drawn from three populations under active surveillance during the early phase of the pandemic (health care workers, armed forces personnel, and long-term care home residents). As these populations experienced different risk of infection,[211] the RT-PCR cohort was not assumed to have the same risk of infection as the community cohort, but we did assume the two cohorts had a similar titer

response after infection. Although there is a risk that the RT-PCR cohort would have more symptomatic individuals who might have experienced a different titer trajectory from those with no or milder symptoms, we found the pre-infection titer distributions for the two cohorts were comparable (p-value=0.15) and no significant difference in peak titers between the two cohorts (p-value = 0.11). Therefore, the RT-PCR cohort should represent the community cohort well in terms of antibody response before and after infection.

The finding that antibody titers peak 6–7 weeks after infection is consistent with the results from previous studies that report that peak levels were achieved 4–7 weeks after influenza infection/vaccination.[185,189,210,218] We also found that antibody titers against pandemic influenza A(H1N1) dropped rapidly to half their peak level 5–8 months after infection followed by a slower decrease until one year post infection (beyond which we could not accurately estimate). This finding supports the results from historical studies that a faster decrease in antibody titers was observed within the first 6 months.[205,219,220] However, different opinions exist pertaining as to the time interval for the half-life of antibody titers. A pioneering immunological study by Horsfall et al. in 1941 found that the half-life for influenza antibody titers post-infection is 2–3 months among 5 study subjects.[221] In a more recent study from South Korea, antibody titers were measured up to 12 months following vaccination with standard-dose, inactivated, trivalent influenza vaccine.[178] For younger adults (<65 years) titers similar to 4 weeks post-vaccination persisted till 6 months, but approached pre-vaccination levels at 12 months. In older adults, HAI titers decayed to below seroprotective levels by 6 months. The shorter half-life time observed in the Horsfall study could be a reflection of small sample size (n = 5) and concomitant sampling error. Moreover, it has been demonstrated that naturally acquired influenza antibodies persist longer than artificially acquired (i.e. vaccine induced) influenza antibodies.[222,223]

We only found a significant association between age and antibody half-life post-infection—elderly individuals had significant shorter half-lives of antibody titers. Gender in contrast was not a significant confounding factor that influenced antibody response post-infection. These findings are consistent with a previous, simpler analysis that found a faster rate of decrease in antibody titers in the older age group, but no significant difference was observed for gender groups. [205]

We found a rapid rise in population antibodies and infections during the first influenza pandemic wave in Singapore. However, while infections rise in both wave 2 and 3, population antibodies reached a near–plateau that lasted for the subsequent epidemic waves. The slight sawtooth pattern in antibody levels and the seropositive rate during wave 2 and 3 might explain the reoccurrence of influenza epidemics, in a setting like Singapore in which seasonal forcing is not present to synchronize epidemics to a particular season. Individual-level simulations built upon the model provided theoretical support for this hypothesis. In this paper, we have postulated that changes in population level immunity levels resulting from the aggregation of individual antibody trajectories, as measured by HAI titers, may plausibly be associated with the timing of influenza epidemics in the tropics.[224,225] Other mechanisms for immunity, including high levels of cross-reactive cellular responses from seasonal influenza infections or boosted cross-reactive antibodies [226–228] and other hypotheses, such as antigenic changes in the virus, fluctuations in precipitation [78] between monsoon and non-monsoon time periods, importations from the seasonal outbreaks in Northern and Southern hemispheres,[68,207] or randomness alone [229] may also suffice to explain the timing of subsequent waves of the 2009 pandemic and the irregular timing of seasonal epidemics in aseasonal settings. Further work to validate these findings and assess the

importance of this mechanism relative to others such as antigenic drift in the tropics would be valuable.

Our model suggests that the seropositivity and seroconversion peak at 50% and 60% respectively after the onset of influenza infection, both of which are lower (10–20%) than what has been observed in other studies.[205] The difference may result from the criterion, i.e. a four-fold rise between successive blood samples, used to define influenza infections. This might underestimate the total number of infections because some infections may be limited to a two-fold rise.[210] The model developed here, however, does not restrict to the threshold of a four-fold rise to confirm infections. As a result, the denominators for calculating the seropositivity and seroconversion rates are higher as more individuals are inferred to be infected, lowering the estimates accordingly.

There are some limitations in this study. The model developed here depends on the availability of RT-PCR data for a subset of participants which could be costly and logistically challenging, though taking the same approach as we did, the confirmed cases need not to be from the same population as the serological data. The uncertainty intervals for the estimated time of infection depends on the time window between blood samples (**Table 16**). For some individuals, for example **Figure 11C**, the uncertainty interval for the estimated time of infection is wide, a reflection of the fairly wide time window between the timings of the third and fourth blood samples in our study. Future studies might target a slightly more frequent serum sampling around epidemic waves. The model was developed based on data from the first three epidemic waves of a new influenza strain (influenza A(H1N1)pdm09) and as a result, it is not clear to what extent the findings apply to seasonal influenza infections and after multiple waves of infections with the pandemic H1N1 strain. Only the primary antibody

response was studied, and the effect in antibody titers from reinfections were not considered as the number of individuals with apparent repeat infections in the cohort was too small to estimate risks of reinfections (n=3 with four-fold rises between successive intervals). The model was limited to individuals who started without a recent infection because we assumed a single infection for all individuals and the time of infection was restricted to be before the time when the first imported case was diagnosed in Singapore.

Table 16. A comparison between the time window between two consecutive blood samples and uncertainty time interval for estimated time of infection. The distribution for both time intervals are skewed so median with interquartile range were reported.

	Sample size	Median time interval between blood samples, days (IQR)	Median uncertainty interval for estimated time of infection, days (IQR)
Sample 1 — Sample 2	91	10 (9, 42)	9.8 (4.3, 34.2)
Sample 2 — Sample 3	35	7 (6, 34.7)	17.1 (5.8, 70.3)
Sample 3 — Sample 4	45	26 (11, 28)	20.8 (9.8, 49.9)
Sample 4 — Sample 5	27	12 (10.7, 24.4)	12 (11.2, 13)
Sample 5 — Sample 6	8	12 (11.2, 13)	26.3 (21.6, 47.4)

One application of our model is to advise the timing of serological samples relative to the start of an outbreak. Once infected with influenza, it would take a few weeks for antibody titers to increase to the level that is detectable, i.e. titers ≥ 1:40 for a cross-sectional study or four-fold or more rise in antibody titers for a longitudinal study. The optimal timing of the serological sample to capture the seroconversion during the epidemic is, therefore, during the period in which the probability of titers ≥ 1:40 or the probability of seroconversion peaks, i.e. 5–8 weeks after infection. The time window of detectable seroconversion estimated for H1N1 influenza and the time window of half-life of antibody levels estimated in this paper provide a reference on when the population may be vulnerable to the start of a new domestic outbreak

(particularly in tropical settings where seasonal forcing is weak) and hence when to implement influenza vaccination among people who are believed to be at risk.

4.6 Motivations for next chapter

This chapter identified a correlation of dynamics of herd immunity level and influenza epidemics, which is important for planning the timing of influenza vaccination programmes in the tropics. The number of doses of influenza vaccine needed for a seasonal influenza programme or a new influenza pandemic should be estimated for optimal resource allocation, particularly, in resource-limited settings. The next chapter examined the accuracy of current serological methods for confirming influenza infection and provided a simple approach to estimate the attack rate for an influenza outbreak using serological data. The methods developed, and results obtained in Chapter 5 are novel, and are crucial for influenza vaccine manufacturing and dose distribution.

CHAPTER FIVE: RETHINKING THRESHOLDS FOR SEROLOGICAL EVIDENCE OF INFLUENZA VIRUS INFECTION

This work has been published as: Zhao X, Siegel K, Chen MI-C, Cook AR. Rethinking thresholds for serological evidence of influenza virus infection: Influenza Other Respir Viruses. 2017 Apr;11(3):202–210.

5.1 Abstract

Background

For pathogens such as influenza that cause many subclinical cases, serologic data can be used to estimate attack rates and the severity of an epidemic in near real time. Current methods for analysing serologic data tend to rely on use of a simple threshold or comparison of titers between pre- and post-epidemic, which may not accurately reflect actual infection rates.

Methods

We propose a method for quantifying infection rates using paired sera and bivariate probit models to evaluate the accuracy of thresholds currently used for influenza epidemics with low and high existing herd immunity levels, and a subsequent non-influenza period. Pre- and post-epidemic sera were taken from a cohort of adults in Singapore (n=838). Bivariate probit models with latent titer levels were fit to the joint distribution of haemagglutination-inhibition assay-determined antibody titers using Markov chain Monte Carlo simulation.

Results

Estimated attack rates were 15% (95% credible interval: 12%-19%) for the first H1N1 pandemic wave. For a large outbreak due to a new strain, a threshold of 1:20 and a twofold rise (if pared sera is available) would result in a more accurate estimate of incidence.

Conclusions

The approach presented here offers the basis for a reconsideration of methods used to assess diagnostic tests by both reconsidering the thresholds used and by analysing serological data with a novel statistical model.

Word count = 220

5.2 Introduction

Estimates of infection rates are crucial to decision-making and communication during an epidemic, for long-term public health planning, and to assess past responses. Without an accurate gauge of the size and severity of an epidemic, it is challenging to prioritize interventions and services to mitigate impact [230,231]. In the 2009 H1N1 pandemic, limited data inflated predictions of severity in the early stages, leading in turn to what in hindsight was overreaction in many quarters [232]. Lessons learned from the 2009 epidemic and its aftermath can be applied to influenza epidemics of strains both old and new.

Serological assessments can play a key role in assessing influenza outbreaks because they allow diagnosis of sub-clinical or misdiagnosed cases, and as a result they provide the basis for estimates of an epidemic's impact, including, for example, estimates of hospitalization and mortality rates [231]. Common assays, such as haemagglutination inhibition, typically bracket the antibody level to an interval between two dilutions, for instance positive at 1:20 but negative at 1:40. Two study designs are frequently used: cross-sectional, in which a positive measurement beyond a specific threshold—which for influenza is usually set to 1:40 or 1:32—is taken to indicate recent infection [233], and longitudinal, in which a rise in the highest positive reading between successive serum collections of four or more times (often

called a ‘four-fold rise’, although the interval censoring means the rise could actually be more modest) is typically assumed to reflect infection during that time interval. Cross-sectional designs can utilise residual samples of serology collected for other purposes, and are logistically much simpler to implement than longitudinal serum collections. They can allow, in principle, real-time estimates of attack rates [231], but have documented weaknesses, such as potentially leading to negative estimates of attack rates [234], and while single cross-sections may be accurate for a novel pathogen, repeated sampling is necessary if there is existing immunity in the population, or to track changes in incidence necessary for supporting real-time estimates of severity [231].

Traditionally, an HAI titer of 1:40 or more was chosen to indicate infection in cross-sectional studies, and it was found to be associated with a reduction in attack rates that varies between 20 and 80% depending on age group and setting in influenza vaccination studies [130,205,235,236]. This choice is somewhat arbitrary. On the other hand, a four-fold rise is the currently used threshold in longitudinal studies, and has been found to have sensitivity of ~80% relative to a basket of other diagnostics [237]. Neither justification relates to overall diagnostic accuracy per se, however, which is the primary goal of analyses to determine attack rates, for surveillance or for intervention trials.

More robust analysis is made interesting by several complicating factors: (i) titers are interval censored, with intervals that are too broad to justifiably ignore; (ii) longitudinal studies require accounting for repeated measurements; (iii) titer distributions are typically too skewed to assume normality.

This chapter proposes a new statistical approach to estimate attack rates for paired sera. In this, multinomial ordinal probit models account for censoring and non-normality by invoking a latent 'titer propensity' and non-linear threshold variables, while the titer propensity is made bivariate to account for within-individual correlations in time. As well as using this to estimate attack rates directly, we assess the sensitivity, specificity and overall accuracy of alternative versions of traditional thresholds, using three scenarios: (i) a new strain of influenza against which there is little pre-existing immunity, (ii) an outbreak of seasonal influenza, and (iii) a period with little influenza activity, using data from a community cohort established in equatorial Singapore.

5.3 Methods

Data

Repeated serological samples were drawn from a cohort of adults (aged 21–75) participating in the Multi-Ethnic Cohort (MEC) of the Singapore Consortium of Cohort Studies (SCCS), a long-term prospective cohort study, as described in detail elsewhere [230,238,239]. Blood samples were collected at 6 different time points in 2009 and 2010, and this study uses sera collected at four of those points: (1) baseline samples collected before July 2009, and thus predating unlinked community transmission of the pandemic, (2) a sample around October 2009, after the first but before the second wave, (3) a sample in July 2010, which followed the subsequent two epidemic waves of influenza A(H1N1)pdm09, and (4) a sample in September 2010, 10 – 12 weeks after the July 2010 sample. By the end of the study period, a total of 757, 690, 624 and 556 samples were obtained at the four time points, i.e. before July 2009, around October 2009, in July 2010 and in September 2010, respectively. The remaining two blood samples which were collected in the middle of period 1 and period 2

were not included in the analysis because they fell in the middle of outbreaks and thus it is hard to interpret attack rates involving them. 38 subjects reported being vaccinated against influenza A(H1N1)pdm09 during the study period, and were thus excluded from the analysis. This study focuses on three time windows: period 1 (spanning samples 1 to 2) which bracketed the first wave and in which the population had low initial immunity levels; period 2 (samples 2 to 3) which bracketed the second and third waves and in which the population had high initial immunity levels; and period 3 (samples 3 to 4) which had little influenza activity (**Figure 15A**). 758 participants provided at least one blood sample at time points 1 and 2 (690 provided at both), 691 who provided at least one at time points 2 and 3 (544 provided at both), and 610 who provided at least one at time points at time points 3 and 4 (498 provided at both) were included in analysis. Participants gave informed consent and the study was approved by the National University of Singapore Institutional Review Board.

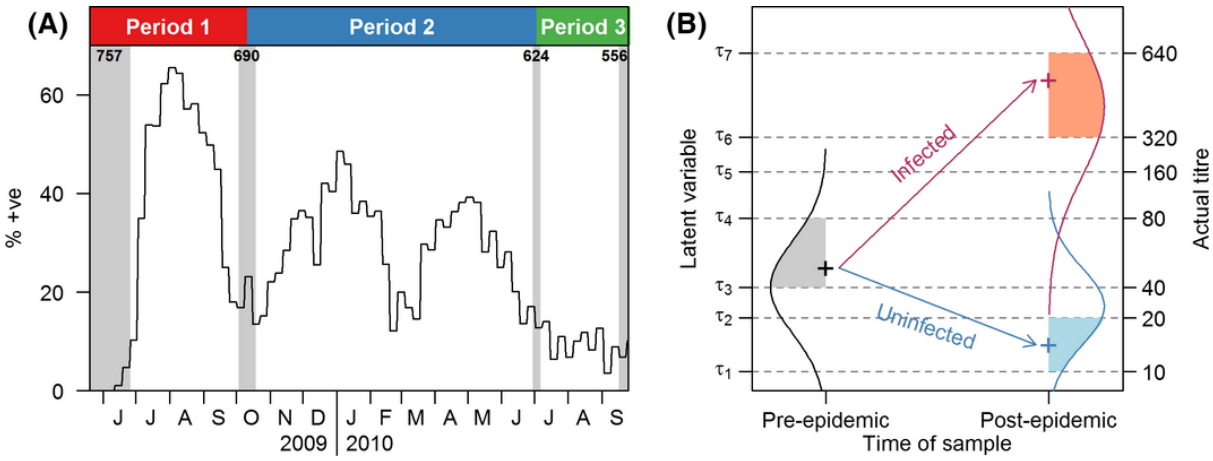


Figure 15. Schematic diagrams of data collection (A) and the developed longitudinal multinomial ordinal probit model (B). In panel A, the shaded grey areas indicate the period of serum sample collection, and the numbers indicate the number of samples collected. The black line represents the weekly relative proportions of influenza A(H1N1)pdm09 infections obtained from routine primary care surveillance. The top coloured panel indicates how we define the three periods, where period 1 refers to the first pandemic wave, period 2 refers to the two subsequent epidemic waves, and period 3 refers to the non-influenza period. For panel B, in the pre-epidemic phase, the continuous measure of antibody levels of each individual is represented by a latent variable which is bounded by the observed titer interval and is normally distributed (black normal curve). In the post-epidemic phase, the latent variable for antibody levels of each individual still follows a normal distribution with the

same variance, but there is a change in the normal mean that depends on the infection status (red curve for infected and blue curve for uninfected). Estimated threshold parameters, τ_k, are the same for all time points.

All blood samples were titrated in 2-fold dilutions of phosphate-buffered saline from 1:10 to 1:2560 and analysed to determine the antibody titer, which is the reciprocal of the highest dilution of serum where hemagglutination was inhibited [238]. Titer values below the limit of detection were coded as <1:10, and a change in titer values from <1:10 to 1:10 was considered to be a twofold rise. Laboratory methods are detailed elsewhere [238].

Statistical model

We developed a longitudinal multinomial ordinal probit model for HAI titers that incorporated latent variables for infection status and antibody propensity for each individual at both sample points (the slight modifications needed for those with a single observation are described later). A schematic diagram appears in **Figure 15B**. A latent variable, z_{it}, represents a continuous measure of antibody levels of individual i at time point t, and is modelled as being normally distributed. At time point 1, $z_{i1} \sim N(\mu_i, \sigma^2)$, where the mean μ_i varies between individuals, to account for correlations between time points, as $\mu_i \sim N(a, w)$, and the variance σ^2 accounts specifically for observation error. The formulation for describing the distribution of HAI titers at time point 2 is an extension of the model for time point 1. At time point 2, $z_{i2} \sim N(\mu_i + \delta_i, \sigma^2)$, where the change in mean for individual i from time point 1 to time point 2, δ_i, is $N(b + c, v)$ if i were infected and $N(b, v)$ if not. Waning of titers is implicitly accounted for if b is negative. Infection between the two time points happens with probability p, the primary estimand of interest.

For each study period, at least one blood sample was collected at the beginning or at the end of the period or both. All blood samples were assayed for HAI titers against influenza infections. HAI titers are interval censored, either being negative at the lowest titration measured (1:10), positive at a titration of $1{:}\,X$ and negative at $1{:}\,2X$ (where X falls in the set $\{10, 20, 40, 80, \ldots\}$, or positive at all titrations. In addition, the empirical titer distribution is far from a normal distribution, complicating analysis. To overcome this, a latent variable formulation was introduced, in which the latent variable was given a normal distribution and linked to observed HAI titers using a sequence of increasing threshold parameters τ_k that control the mapping from latent space to titer space. Titer intervals are coded 1 for <1:10, 2 for 1:10 to 1:20, 3 for 1:20 to 1:40, etc, and the observed titer T_{it} for individual i at time point t is equal to k if $\tau_{k-1} \leq z_{it} < \tau_k$ (**Table 17**). This model formulation provides the convenience of working with normally distributed variables while providing the flexibility to characterise the skewed and often bimodal distribution of HAI titers observed in many studies [129,240], and although it requires estimating an additional parameter per observable HAI titer level, these can be modelled as constant in time. To ensure statistical identifiability, the two most extreme thresholds are set to $\tau_1 = 0$ and $\tau_8 = 1$, with intermediate thresholds free parameters. Threshold parameters, τ_k, are fixed to be same for the three time windows.

The final model and the likelihood contribution from an individual i is, therefore, presented below in three different scenarios: (1) two blood samples collected both at the beginning and end of the period, (2) only one blood sample collected at the beginning of the period, and (3) only one blood sample collected at the end of the period. It is worth noting that individuals providing a single serum sample still contribute to the overall estimation, and do not therefore need to be excluded under this methodology, but we anticipate that a substantial fraction of

individuals would need to provide two samples for the model to be able to provide reasonable estimates of attack rates.

Table 17. Summary of the terminology of titers used in the paper. Conventional terminology typically refers to the maximum tested titer at which there is a positive reaction. This corresponds to a bracket that interval-censors the 'true' titer, if more dilutions were tested. $\boldsymbol{Y_{it}}$ is a coded version of the data; $\boldsymbol{z_{it}}$ is a latent variable that, together with the thresholds $\boldsymbol{\tau_k}$, conceptually determines the censored titer observed.

$\boldsymbol{z_{it}}$	$\boldsymbol{Y_{it}}$	**Censored titer**	**Conventional terminology**
$\boldsymbol{z_{it} < \tau_1}$	1	< 1:10	< 1:10
$\boldsymbol{\tau_1 < z_{it} < \tau_2}$	2	1:10 to 1:20	1:10
$\boldsymbol{\tau_2 < z_{it} < \tau_3}$	3	1:20 to 1:40	1:20
$\boldsymbol{\tau_3 < z_{it} < \tau_4}$	4	1:40 to 1:80	1:40
$\boldsymbol{\tau_4 < z_{it} < \tau_5}$	5	1:80 to 1:160	1:80
$\boldsymbol{\tau_5 < z_{it} < \tau_6}$	6	1:160 to 1:320	1:160
$\boldsymbol{\tau_6 < z_{it} < \tau_7}$	7	1:320 to 1:640	1:320
$\boldsymbol{\tau_7 < z_{it} < \tau_8}$	8	1:640 to 1:1280	1:640
$\boldsymbol{\tau_8 < z_{it}}$	9	> 1:1280	1:1280

Two blood samples collected at the beginning and end of period

Standard probability theory dictates that the joint distribution of the two latent variables follows a bivariate normal distribution conditional on infection status [241]. The covariance between two latent variables is $cov(z_{i1}, z_{i2}) = cov(\mu_i + \epsilon_i, \mu_i + \delta_i + \epsilon_2)$ which is equal to $cov(\mu_i, \mu_i) = w$ under the model's assumption that $\mu_i, \delta_i, \varepsilon_1$ and ε_2 are independent of each

other, where $\varepsilon_1 \sim N(0, \sigma^2)$ represents the random error for each observation at time point 1 and $\varepsilon_2 \sim N(0, \sigma^2)$ represents the random error at time point 2. With the derived covariance between two latent variables, we are able to determine the bivariate normal distribution for the two latent variables using the following equations: $T_{it} = k$ if $\tau_{k-1} \leq z_{it} < \tau_k$ for time point $t = 1$ or 2, where

$$\begin{pmatrix} z_{i1} \\ z_{i2} \end{pmatrix} \sim \begin{cases} N\left(\begin{pmatrix} a \\ a+b+c \end{pmatrix}, \begin{pmatrix} [w+\sigma^2] & w \\ w & [w+v+\sigma^2] \end{pmatrix}\right) \text{ if } i \text{ is infected or} \\ N\left(\begin{pmatrix} a \\ a+b \end{pmatrix}, \begin{pmatrix} [w+\sigma^2] & w \\ w & [w+v+\sigma^2] \end{pmatrix}\right) \text{ otherwise.} \end{cases}$$

The likelihood contribution from individual i with two blood samples collected conditional on infection status follows from a two-dimensional cumulative distribution function of the above bivariate normal distribution. We use $\boldsymbol{l}_{12,i}$ and $\boldsymbol{l}_{12,u}$ to represent the likelihood contribution from individuals with or without infection respectively. Both $\boldsymbol{l}_{12,i}$ and $\boldsymbol{l}_{12,u}$ are a matrix of probabilities for each combination of titers at two time points. The likelihood unconditional on infection status is $\boldsymbol{l}_{12} = p\boldsymbol{l}_{12,i} + (1-p)\boldsymbol{l}_{12,u}$, i.e. a weighted average. For computational efficiency, we count the number of individuals with each combination of titers at the two time points (represented by a 9 × 9 matrix $\boldsymbol{N}$) and refer these to a multinomial distribution with probabilities $\boldsymbol{l}_{12}$.

Only one blood sample collected at the beginning of the period

The likelihood, $\boldsymbol{l}_1$, for individuals with only one blood sample collected at time point 1 follows from a cumulative distribution function of the normal distribution, $z_{i1} \sim N(\mu_i, \sigma^2)$. We then count the number of individuals with each titer intervals at time point 1 and refer these to a multinomial distribution with probabilities $\boldsymbol{l}_1$.

Only one blood sample collected at the end of the period

The likelihood, $\boldsymbol{l}_2$, for individuals with observations at time point 2 only is a weighted average of the cumulative distributions of the two normal distributions with a probability of infection p: $z_{i2} \sim N(a+b, w+v+\sigma^2)$ for uninfected and $z_{i2} \sim N(a+b+c, w+v+\sigma^2)$ for infected. Again, we count the number of individuals with each titer intervals at time point 2 and refer these to a multinomial distribution with probabilities $\boldsymbol{l}_2$.

An important assumption in the above model is that the risk of infection is taken to be independent of baseline titers, which allows the infection probability parameter to represent average risk without knowing the titer distribution. Because these are aggregate probabilities, the dataset can be collapsed into the total counts of each combination and referred to multinomial distributions, which simplifies the calculation of the likelihood function.

Sensitivity analysis

An alternative formulation in which the probability is individualised to have a linear relationship to the titer category (on the logit scale), i.e. where $\log\frac{p_i}{1-p_i} = \alpha_1 + \alpha_2 T_{i1}$, and T_{i1} is the observed titer for individual i at time point 1, was also developed and used to assess the sensitivity of our findings to this assumption. The sensitivity model is much more computationally intensive because it necessitates the calculation being carried out separately for each individual, or at least each starting titer, which we found increased computing time ~20-fold. Note also that for this model, we excluded those without a starting titer from the calculations because the distribution at the second time point could not be determined without more complex methods such as data augmentation.

Algorithm

All parameters were estimated via a MCMC routine, coded in R version 3.0.3 (R Foundation for Statistical Computing, Vienna, Austria) [182], and sampled from a multivariable normal proposal distribution. We adopted uniform prior distributions for all parameters over their support regions. In preliminary runs, a total of 10,000 iterations were performed with an arbitrary choice of variance-covariance matrix in the proposal distribution. Following this, the posterior means and a variance-covariance matrix for all parameters were estimated from sample, and then used as the mean and variance-covariance of a multivariable normal distribution as the proposal distribution for a subsequent 50,000 iterations with every 5th iteration retained for final analysis. Convergence was assessed using trace plots of parameters and using the Geweke diagnostic [145]. Point estimates are posterior means and uncertainty intervals are 95% equal tailed credible intervals (CrI).

The posterior probability of infection for an individual with a specific post-seasonal titer (for the first wave, i.e. in the absence of pre-existing immunity), or for a specific risk in observed titer intervals (for either wave), was calculated directly from the fitted model using Bayes' rule. For the former, $p(I_i|T_{i2}) = p(T_{i2}|I_i)p(I_i)/p(T_{i2})$, where I_i is the infection status for individual i and T_{i2} is the observed titer for individual i at time point 2, which can be derived from the marginal distribution of titers at time 2 in the presence or absence of infection. For the latter, a similar expression is used.

Diagnostic accuracy of existing thresholds

Sensitivity, specificity, positive and negative predictive values (PPV and NPV, respectively) and overall accuracy were calculated to assess the performance of diagnostic tests for various titer thresholds, the latter two based on the prevalence estimated from the MEC data. We derived sensitivity/specificity by calculating the probability of a positive/negative test in the presence/absence of infection. To assess accuracy of different thresholds in future, plausible epidemics, PPVs, NPVs and accuracies were calculated directly from the sensitivity and specificity as the hypothetical true prevalence was increased from 0 to 1. Sensitivity, specificity, PPV and NPV were all calculated from simulations from the developed model instead of using an objective measure, such as PCR confirmed infections. Bias between the modelled prevalence and the hypothetical true prevalence was also calculated to compare the accuracy of the currently used thresholds (1:40 or a four-fold rise) with the model we developed.

5.4 Results

Titer dynamics

Figure 16 illustrates the aggregate actual and modelled titer distributions as well as modelled titer distributions in the presence or absence of infection during the three time periods. The concordance between modelled and observed joint distributions was close for all periods. During the first wave (period 1), the majority of participants (79%) had no change in titer scores at the two time points, while the interval censored titers rose for 17% of participants and fell for the remaining 5% (**Figure 16A**). During the subsequent two epidemic waves (period 2), 62% participants had the same titer scores at the two time points, while 17% had higher and 21% had lower titer scores at the post-season sampling (**Figure 16B**). During the

non-influneza period followed epidemic waves (period 3), 88% of participants had same titer scores at the two time points and 4% had higher and 8% had lower titer scores at the post-season sampling (**Figure 16C**). In the absence of infection, 88%, 75% and 78% of participants had titer scores unchanged during the first pandemic wave, the subsequent two epidemic waves and the non-influneza period, respectively (**Figure 16G–I**). Most of the participants who were infected had pre-season titers of less than 1:10 (86% for period 1, 73% for period2, 74% for period 3) (**Figure 16J–L**).

Attack rates over three periods

Our primary estimand is the attack rate from pre-season to post-season. This was estimated to be 15% (95% credible interval (CrI): 12% to 19%) during the pandemic first wave (period 1), 16% (95% CrI: 12% to 21%) in period 2, and 1.2% (95% CrI: 0.3% to 2.4%) during the non-influenza period (period 3). In a sensitivity analysis, these results were broadly consistent with those using the more sophisticated model in which the risk of infection was related to initial titer levels (being 19% [95%CrI: 17 to 20%], 12% [95%CrI: 10 to 14%] and 0.6% [95%CrI: 0 to 1.7%], respectively) with regards to the total number of infections across the three time periods, but with a slightly different distribution of when the infections occurred. Standard approaches, in contrast, gave estimates for the attack rate during the first wave (period 1) of 12% (95% CI: 9 to 14%) using a 1:40 threshold and of 12% (95% CI 10 to 15%) using a four-fold rise, and of 10% (95% CI 8 to 13%) and 2% (95% CI 1% to 3%) for period 2 and 3, respectively using four-fold rise as a proxy for infection. In addition, a stratified analysis by age group (<30 years old vs ≥30 years old) found that younger age group showed higher attack rate during the first pandemic wave (19%; 95% credible interval: 11% to 30%) compared with the older age group (16%; 95% credible interval: 11% to 21%).

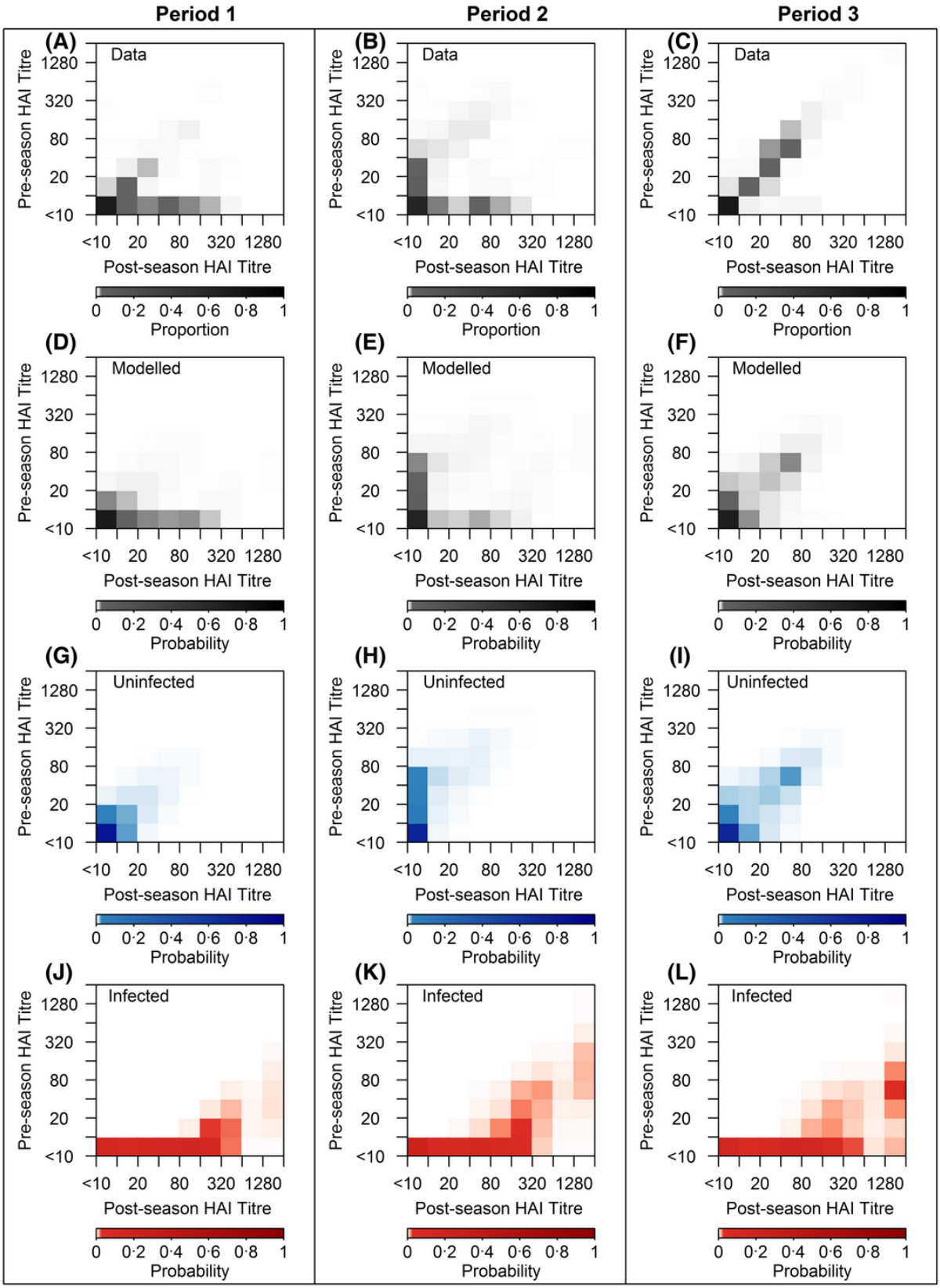


Figure 16. Distribution of observed haemagglutination-inhibition (HAI) titers (A-C), modelled titers (D-F) and the proportion of uninfected (G-I) and infected (J-L) for HAI

titers in periods 1-3. A colour bar is placed at the bottom of each panel, with darker colour representing higher probabilities.

Threshold positive predictive values and sensitivity

The estimated positive predictive values (PPV) for cross-sectional titers, i.e. the proportion of people with a titer of that level who were infected, was substantial even for low thresholds (**Figure 17**; **Table 18**): 0.43 (95% CrI: 0.31 to 0.54) for a titer between 1:20 and 1:40, and 0.24 (95% CrI: 0.15 to 0.34) for one between 1:10 and 1:20. In contrast the PPV for a threshold of 1:40 and above was 0.81 (95% CrI: 0.73 to 0.88). For a fourfold rise in titer scores, which is commonly used to reflect infection in longitudinal studies, the PPVs were 0.76 (95% CrI: 0.60 to 0.88) and 0.92 (95% CrI: 0.77 to 0.99) for period 1 and 2, respectively, while it was only 0.03 (95% CrI: 0.01 to 0.08) for the non-influenza period (**Figure 17**; **Table 19**). Even for a rise of at least two folds in titer scores, the PPVs were high: 0.31 (95% CrI: 0.19 to 0.45) and 0.68 (95% CrI: 0.43 to 0.85) for the first and second period, respectively, indicating that many infections are missed using the standard definition during the influenza epidemic waves. The difference in PPVs in different periods depends on the incidence of infection and the population herd immunity level, but the sensitivity estimates are quite consistent across the time periods, and thus epidemic scenarios, considered. The sensitivities of the traditional thresholds were: period 1, 1:40: 0.60 (95% CrI: 0.49 to 0.71); 4-fold rise: 0.74 (95% CrI: 0.63 to 0.83); period 2, 4-fold rise: 0.64 (95% CrI: 0.55 to 0.73); period 3, 4-fold rise: 0.78 (95% CrI: 0.49 to 0.96). The sensitivities for different HAI titer thresholds in different periods were summarized in **Table 20** and **Table 21**.

Threshold accuracy

For straight-forward prevalence estimates to be unbiased, the proportion of participants testing positive should equal the proportion infected, but for imperfect tests this depends on

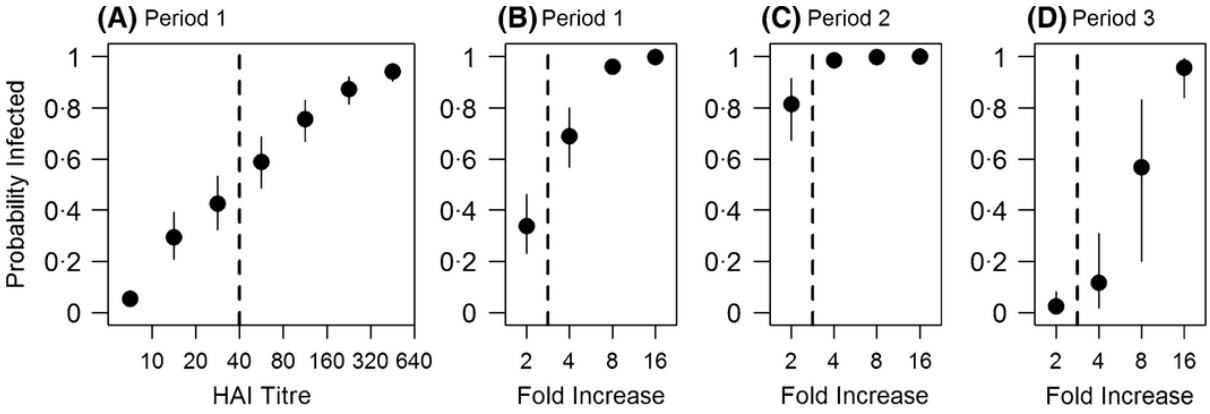


Figure 17. Infection probability by haemagglutination-inhibition (HAI) titer in period 1 (A) and by fold increase in HAI titers in periods 1-3 (B-D). Points with whiskers, which indicate the 95% credible intervals, represent the posterior means.

Table 18. Positive predictive values by hemaglutination-inhibition titer interval during the first wave of influenza A(H1N1-2009) (period 1) in Singapore.

Censored titer	Probability of infection (posterior mean (95% Credible interval))
	Period 1
< 1:10	0.03 (0.01, 0.05)
1:10 to 1:20	0.24 (0.15, 0.34)
1:20 to 1:40	0.43 (0.31, 0.54)
1:40 to 1:80	0.64 (0.53, 0.75)
1:80 to 1:160	0.82 (0.72, 0.89)
1:160 to 1:320	0.93 (0.87, 0.97)
1:320 to 1:640	0.98 (0.95, 0.99)
1:640 to 1:1280	0.99 (0.97, 1.00)
1:1280 to 1:2560	1.00 (0.98, 1.00)

Table 19. Positive predictive values by fold increase in hemaglutination-inhibition titer scores in period 1–3 in Singapore.

Fold increase	Probability of infection (posterior mean (95% Credible interval))		
	Period 1	Period 2	Period 3
2	0.31 (0.19, 0.45)	0.68 (0.43, 0.85)	0.01 (0.00, 0.01)
4	0.76 (0.60, 0.88)	0.92 (0.77, 0.99)	0.03 (0.01, 0.08)
8	0.98 (0.83,0.99)	0.99 (0.95, 1.00)	0.36 (0.12, 0.65)
16	1.00 (1.00, 1.00)	1.00 (0.99, 1.00)	0.96 (0.87, 0.99)

Table 20. Sensitivity by hemaglutination-inhibition titer interval during the first wave of influenza A(H1N1-2009) (period 1) in Singapore.

Censored titer	Posterior mean sensitivity (95% Credible interval)
	Period 1
< 1:10	1
1:10 to 1:20	0.85 (0.77, 0.92)
1:20 to 1:40	0.74 (0.63, 0.83)
1:40 to 1:80	0.60 (0.49, 0.71)
1:80 to 1:160	0.44 (0.35, 0.55)
1:160 to 1:320	0.27 (0.20, 0.35)
1:320 to 1:640	0.12 (0.07, 0.17)
1:640 to 1:1280	0.05 (0.01, 0.09)
1:1280 to 1:2560	0.03 (0.01, 0.08)

Table 21. Sensitivity of infection by fold increase in hemaglutination-inhibition titers in period 1–3 in Singapore.

Fold increase	Posterior mean sensitivity (95% Credible interval)		
	Period 1	Period 2	Period 3
2	0.84 (0.76, 0.92)	0.75 (0.67, 0.83)	0.84 (0.63, 0.98)
4	0.74 (0.63, 0.83)	0.64 (0.55, 0.73)	0.78 (0.49, 0.96)
8	0.59 (0.48, 0.71)	0.50 (0.39, 0.60)	0.66 (0.28, 0.92)
16	0.42 (0.32, 0.53)	0.28 (0.19, 0.38)	0.43 (0.08, 0.82)

the balance between sensitivity and specificity as well as the true prevalence. As depicted in

Figure *18*A, for cross sectional studies in an influenza pandemic first wave, based on our estimated sensitivity and specificity, the most accurate threshold would be the standard 1:40 only for small pandemics with an attack rate between 4 and 12%; for larger pandemics this threshold leads to estimates that are biased downward. This bias could be reduced by using a lower threshold of 1:20 (for attack rates of 12 to 31%) or of 1:10 for a pandemic infecting a third or more of the population, as predicted by many simulation models [242] and in line with the 1918 pandemic [243]. Similarly, use of a 4-fold rise as evidence of infection minimises bias for small pandemics (of 2 to 14% attack rates) but underestimates the impact of larger pandemic first waves, while a 2-fold rise would be more accurate for pandemics infecting more than 14% of the study population (**Figure 19A)**. These findings are similar when using the estimates from Singapore's subsequent two epidemic waves after the first pandemic wave (**Figure 19B** and **Figure 19C**).

Figure 20 suggests the bias from standard thresholds in estimates of attack rates could be substantial. For a small new influenza pandemic infecting 15% of the population or a larger 1918-like pandemic infecting 30% of the population, using the 1:40 threshold for a cross-sectional study or a four-fold rise for a longitudinal study would substantially underestimate the numbers infected. On the other hand, using the model we developed would give much smaller bias regardless of overall prevalence variations in the community.

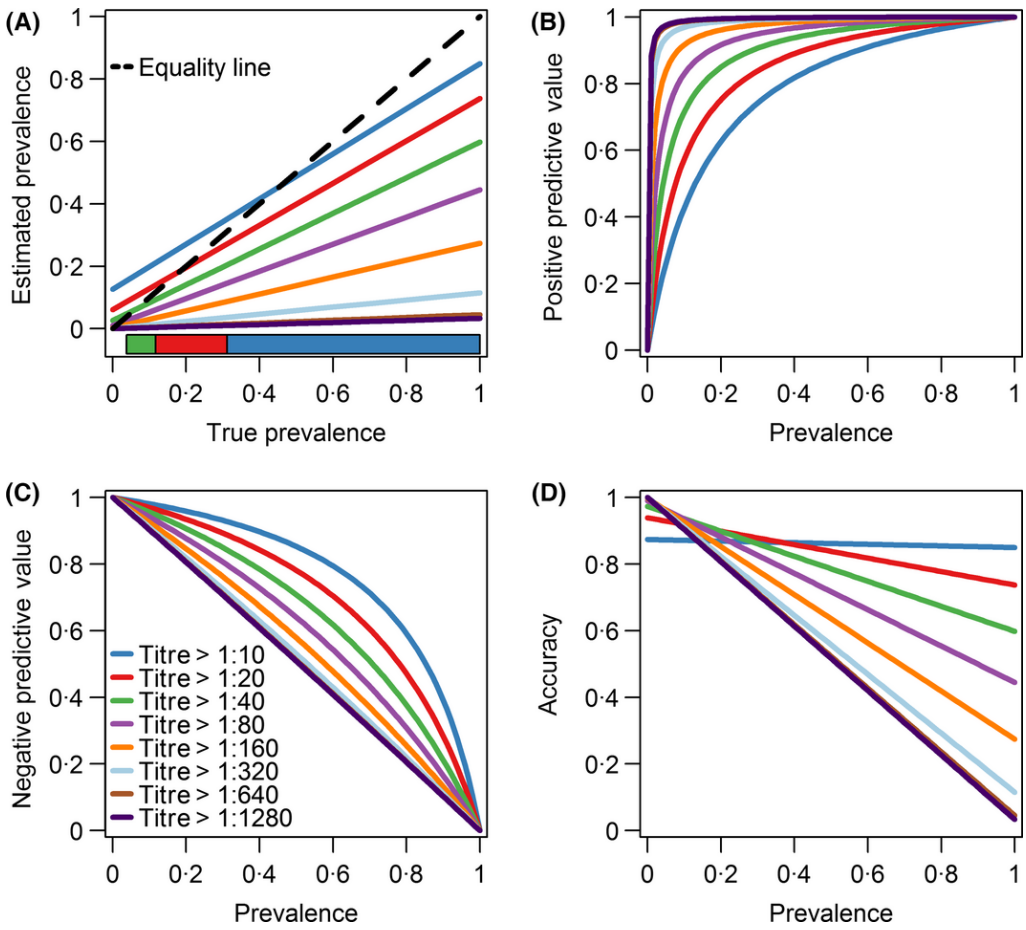


Figure 18. Estimated prevalence (A), positive predictive value (B), negative predictive value (C) and accuracy (D) by various titer thresholds in period 1. Different thresholds shown on each panel are plotted in different colours. On panel (A), the black 45-degree equality dotted line describes an ideal scenario that the probability of a positive test equals to the probability of being infected. A colour bar is placed at the bottom of the panel (A) which shows the preferred titer threshold that should be used given the true prevalence. It is more preferred to use the threshold titer of 1:10 if the true prevalence is more than 0.31 as denoted by the dark blue polygon. The threshold titer of 1:20 should be used if the true prevalence falls between 0.12 and 0.32 as denoted by the red polygon. The threshold titer of 1:40 is preferred if the true prevalence falls between 0.04 and 0.12 as denoted by the green polygon.

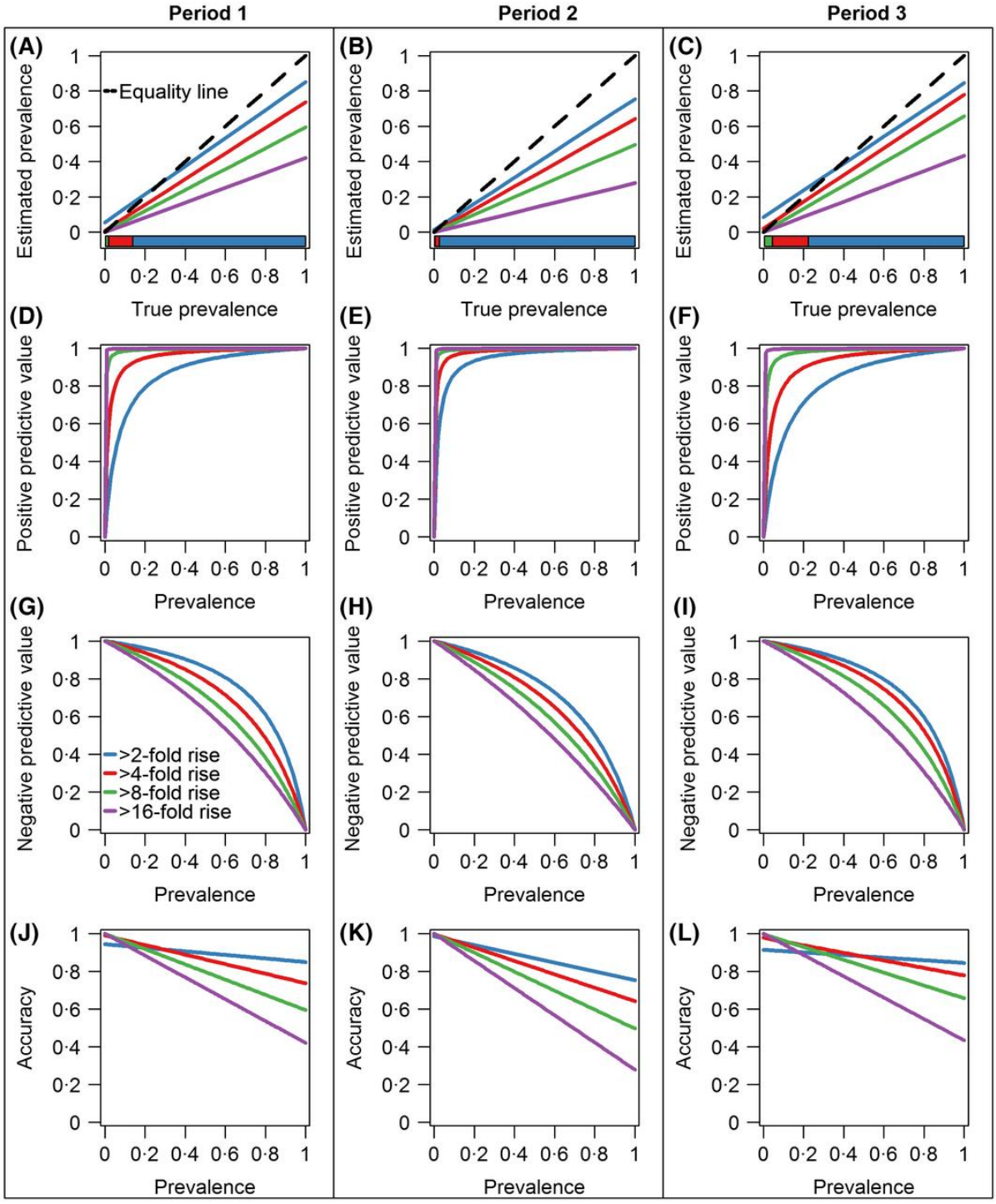


Figure 19. Estimated prevalence (A-C), positive predictive value (D-F), negative predictive value (G-I) and accuracy (J-L) by fold increase in titer scores in periods 1-3. Different thresholds shown on each panel are plotted in different colours. On panel (A-C), the black 45-degree equality dotted line describes the ideal scenario that the probability of a positive test equals to the probability of being infected. A colour bar is placed at the bottom of the panel (A-C) which shows the preferred fold increase threshold that should be used given the true prevalence.

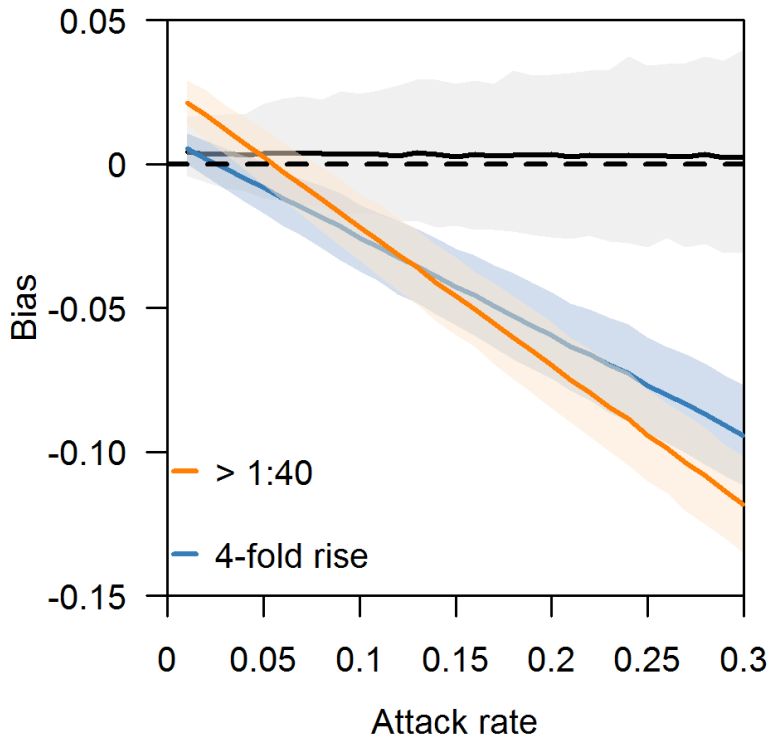


Figure 20. Degree of bias conditional on attack rate by threshold used for defining influenza infection. The black, orange and blue lines with shaded area, which indicates the 95% credible intervals, show the bias produced by the developed model, an haemagglutination-inhibition (HAI) titer threshold of >1:40 and a fourfold rise in HAI titers, respectively.

5.5 Discussion

Many studies using cross-sectional serologic data for a newly circulating influenza strain use a hemagglutination inhibition (HAI) antibody titer of 1:40 or 1:32 as the standard threshold level, as studies have shown there to be ~50% decrease in the risk of infection with seasonal influenza viruses associated with a titer of 1:40 [108,244,245]. However, infected individuals may have HAI antibody titers of less than these levels [234]. Similarly, many infected individuals do not experience the four-fold rise taken as evidence of seroconversion: the estimated sensitivity of this outcome versus a basket of other tests is ~80% [237]. Our findings corroborative previous work [210], which identified the importance of using a two-fold rise for estimating attack rates, and that a four-fold rise might substantially

underestimate attack rates. While either threshold is, depending on the study design, an unequivocal endpoint at the individual level for studies of risk of infection, for instance in vaccine trials [129,194] or risk factor studies [246], their imperfect sensitivity and specificity means they may not adequately describe overall infection rates at a population level. As a result, estimates of severity may be similarly, and potentially substantially, biased.

We argue that there are three approaches to remedying this situation. One would be to use a statistical method that explicitly accounts for the structure of serological data—in particular, their censored nature, the distinct forms of statistical error (assay error and between individual variability) and the differential response over time of those infected and uninfected. The model presented in this paper is an example of such, that can readily be fit using modern, if computationally intensive, statistical techniques. A second, simpler, approach would be to retain the use of a threshold but revise the level in response to the anticipated size of the outbreak. The results of the current analysis suggest that for a cross-sectional study, the threshold should be set to 1:20 for outbreaks expected to infect >12% of the population and 1:10 for >32%, while for pre- and post-outbreak sera, use of a two-fold rise as a marker for infection would be more accurate for outbreaks >14%. Adapting this into a simple rule of thumb suggests that the existing thresholds should be retained for seasonal epidemics but lowered for all influenza outbreaks widespread enough to be classified as a pandemic. A third, simpler approach would be to use the traditional thresholds, i.e. ≥1:40 for a cross sectional study and ≥4-fold rise for a longitudinal study, and adjust for the imperfect sensitivity of the assay. However, this approach would not be practical when the sensitivity is unknown without additional data [230].

This recommendation warrants further investigation in other settings. However, taking our findings at face value suggests that substantially more people were infected during the 2009 pandemic than has been estimated [247], and that severity metrics should be scaled up accordingly.

Previous research by Chen et al using the same dataset has shown that the new pandemic virus mostly affected the younger age group in Singapore (20—29 years old) [212]. A stratified analysis by age group (<30 years old vs ≥30 years old) using the developed model has been done, and consistent results were obtained that younger age group showed higher attack rate during the first pandemic wave.

Limitations: Since the data analysed in this paper were taken from a cohort study, it is not randomly selected and may not be representative of Singapore's population. This is a limitation common to most serological studies we have seen for influenza, with few exceptions (such as a large study in China [248]). The method we developed in this paper depended upon the availability of serial serum sampling that begins prior to an outbreak, which can be costly and logistically complex, but which accounts for baseline antibodies present due to cross-reactivity from different strains of a virus [230]. The main analysis assumed the infection risk to be independent of baseline titer levels, albeit that this assumption is known to be untenable [249]. Results from the sensitivity analysis that accounted for infection risk being influenced by initial titer were similar to those in the main analysis, but the alternative method was more computationally intensive. The developed model in this analysis only implicitly accounts for the waning effects in antibody titers over time, via a systematic reduction (or in principle increase) of mean latent titers between the time points. Cross-reactivity is another potential limitation when analysing serologic tests to

estimate the prevalence of seasonal influenza. For example, in the 2009 pandemic, there was a high level of pre-existing seropositivity in older age groups due to cross-reactivity, because the virus subtype had been endemic in the population prior to the 1957 influenza A(H2N2) pandemic [231]. The approach further requires that the post-pandemic sera are collected sufficiently soon after the end of epidemic activity such that titers have not decayed too substantially over time (16). There might be potential measurement errors in the titration of antibodies against A(H1N1)pdm09 infections, however, it cannot be assessed without duplicated assays at the same time point. Thresholds suggested from the current analysis in response to the anticipated size of the outbreak is specific to the context of the influenza A(H1N1)pdm09 outbreak, and further work is needed to demonstrate their validity for other epidemic scenarios.

Despite these limitations, these results challenge the predominant threshold of a 1:40 HAI titer or a four-fold rise in HAI titers and, in turn, the accuracy of many prior estimates of H1N1 attack rates. Precise estimates are important to public health planning and risk mitigation and, therefore, a reconsideration of the standard paradigm should be considered.

CHAPTER SIX: CONCLUSION

6.1 Summary of major findings

Influenza vaccine is refined and updated annually because influenza viruses undergo antigenic drift frequently [110]. Influenza viruses may change during a single season or even a single week [110]. Hence, a comprehensive approach for timely evaluation of influenza vaccine effectiveness is necessary. Chapter 2 of this thesis discussed the pros and cons of various non-interventional study designs that have been used to evaluate the vaccine effectiveness, and test-negative design is found to be an efficient study approach that can provide accurate estimates of influenza vaccine effectiveness with less bias, which supports the use of this design for evaluating influenza vaccine effectiveness. This design does not require substantial amount of additional public health resources as it can be conducted in existing ILI surveillance systems. The common statistical methods used to assess the vaccine effectiveness, i.e. multivariable logistic regression models, were described, and vaccine effectiveness can be estimated using $(1 - \mathrm{OR}_{\mathrm{adj}}) \times 100\%$, where $\mathrm{OR}_{\mathrm{adj}}$ is the adjusted OR derived from the multivariable logistic regression model with vaccination as the predictor and the occurrence of influenza infection as the outcome variable. Additionally, the calculation of vaccine effectiveness was illustrated using the statistical software R and STATA.

As discussed in Chapter 1, the vaccine-induced protection against influenza infections that can be measured by HAI titers against the current strain wanes over time, and it usually falls back to the pre-infection level in approximately one year. With available serological data, the performance of influenza vaccine can be evaluated adjusting for the waning effect. Chapter 3 of this thesis developed a novel statistical model via a Bayesian approach to quantify the vaccine-induced protection against influenza infections using serological data collected in

RCTs. The developed statistical model could account for the censored nature of the antibody titers and the staggered recruitment of study subjects as well as adjust for the waning effect. The results showed that HAI titers correlated with protection against influenza infections—a titer of 1:40 reduced infection risk by 40%–70% relative to a titer of 1:10, depending on the circulating strain. The trivalent inactivated vaccine could reduce the risk of influenza A(H3N2) and influenza B infections in the community from 2007 to 2009.

The influenza vaccination programme is not effective if it is implemented after the peak of an influenza season. Therefore, understanding the seasonality of influenza infections is important for planning effective vaccination programmes, particularly, in the tropics where there is a year-round circulation of influenza virus. In Chapter 4, a Bayesian hierarchical statistical model was developed to characterize the dynamics of individual and population level immunity against influenza infections, and simulations were conducted to investigate the correlation between herd immunity level and influenza seasonality in the tropics. The results showed that influenza antibody levels peaked 4–7 weeks post infections, and decreased to approximately pre-infection levels after one year. The simulations also showed an association between herd immunity level against influenza infections and the timing of influenza epidemics in the tropics.

The manufacture and dose distribution of seasonal influenza vaccine largely depends on the extent of spread of influenza infections. Estimation of influenza attack rates is, therefore, crucial to healthcare policy makers to plan for future vaccination programmes and to assess past responses. The current method for confirming influenza infection using serological data adopts either a simple threshold (a titer of 1:40) or comparison of titers between pre- and post-epidemic (4-fold increase in titers). Chapter 5 of this thesis developed a novel Bayesian

statistical model to re-assess the accuracy of current thresholds, and to estimate attack rates and the severity of an epidemic in approximately real time using serologic data. The results showed that the first influenza H1N1 pandemic wave in 2009 in Singapore attacked around 15% of the total population, and a HAI titer threshold of 1:20 for cross-sectional data and a twofold rise for longitudinal data would result in a more accurate estimate of incidence during an influenza pandemic arose from a new virus.

6.2 Public health contributions and implications

This thesis has several public health contributions and implications. Firstly, this thesis provides influenza researchers and healthcare policy makers a template including study design, data collection and statistical models for better estimation of annual influenza vaccine for both non-interventional (i.e. observational studies) and interventional (i.e. clinical trials) studies.

Secondly, the findings of this thesis support previous studies that suggest that an annual influenza vaccination is needed particularly for people who are at high risk because the vaccine-induced immunity on average declines to pre-vaccination levels in about one year. This is important for policy makers to design influenza vaccination policies and to advise the public to better schedule influenza vaccination.

Thirdly, this thesis proposes that the influenza seasonality in the tropics correlates with the dynamics of herd immunity level in the population. This provides new insights for understanding of the “irregular” seasonality of influenza epidemics in the tropics, and highlights the benefit of establishing a surveillance system for monitoring influenza herd

immunity levels to prevent future outbreaks and to effectively plan influenza vaccination programmes in the tropics.

Fourthly, this thesis suggests that influenza researchers and clinicians need to rethink the current serological thresholds used to define influenza infections because they might substantially underestimate the attack rate during an influenza epidemic or pandemic. Improving the accuracy of estimation methods for influenza infection rate is crucial for policy makers to prioritize interventions and to better allocate public health resources, such as influenza vaccines and antivirals, to alleviate the burden.

Finally, this thesis contributes an estimated infection probability matrix for different serological thresholds that allow more accurate estimation of infection attack rate using serological data. If serological data are available, influenza researchers and policy makers can easily estimate the influenza attack rate by multiplying the influenza antibody titer distributions and the probability of having been infected conditioned on each antibody titer threshold. A rapid estimate of influenza attack rate at the early stage of an influenza epidemic or pandemic is important for optimal planning of healthcare resources, and this thesis may facilitate this.

6.3 Limitations

The limitations of each project have been discussed in each chapter. Several limitations are worth emphasizing again as follows.

The vaccine effectiveness, immunity against a specific influenza virus and influenza attack rate were modelled separately rather than analysing the joint distribution of two or more strains. Accounting for heterotypic protection clearly adds a great magnitude of complexity that would require a radically different methodological framework, i.e. multi-dimensional correlation matrix should be incorporated in the model to quantify the cross-protection effect among different viruses. However, the effect of heterotypic protection on inference depends on the extent of infection spread in the previous influenza wave. Further, this thesis provides a representative view of short term protection, but the longer-term dynamics of immunity may be far more complex and would require a longer-term cohort to evaluate. Presumably the cohort size would have to be very large due to the potential impact of heterotypic protection.

Chapter 4 and 5 of this thesis are based on A(H1N1)pdm09 data obtained during the first three waves of the H1N1 pandemic in 2009. As a result, it is not clear to what extent the findings apply to seasonal influenza infections and other seasonal influenza subtypes, i.e. A(H3N2) and B. Moreover, the models developed may not fully characterise individuals who start with a recent infection, but the concern is probably minor for this thesis because a restriction was set on the minimum time of infection for every study subject. The minimum time of infection was set as the time when the first imported influenza A(H1N1)pdm09 case was diagnosed in Singapore. The model would not reflect well if subjects in the cohort were infected before the recorded dates, e.g. subjects who had travelled to the US and were infected while the virus was circulating there. In addition, one epidemiological assumption, which is more arguable, is made that no more than one infection can occur for any individual within the time frame. The reasons for this assumption are twofold: (1) it is computationally intractable to allow multiple infections, and given the small number of suspected reinfections (N = 3) it does not appear to be worth the additional computational burden; (2) the direct

information from these data to parameterize the risk of reinfection is small, because the number of individuals with one infection and at risk of a second is substantially smaller than that without any infection but at risk of the first infection. Therefore, in general, to have a model that allows people to start with a recent infection, usually for seasonal influenza infections, the restriction on the minimum time of infections should be relaxed and repeat infections in individuals should be allowed.

6.4 Future studies

Based on the limitations of this thesis, several future studies can be implemented to fill in the knowledge gaps.

Future studies are needed to investigate the homotypic and heterotypic protection among seasonal influenza subtypes and to examine the impact of cross-protection on the risk of infection. These studies have implications for influenza vaccination and risk mitigation strategies. Besides studying specific protection, understanding the non-specific immunity against other respiratory virus infections is also important.

In addition, the effect of reinfections was not accounted for in the developed models. Future studies with a longer duration of follow up would anticipate more reinfections, thus raising the importance to account for such reinfection events within the model. Moreover, the data would also provide a way to estimate the risk of such reinfection events.

The dynamics of individual and population level immunity and estimation of influenza attack rate are specific to the context of the influenza A(H1N1)pdm09 outbreak. Further work is

required to demonstrate the validity for other epidemic scenarios and other influenza subtypes.

6.5 Conclusion

This thesis provides a set of epidemiological and statistical modeling studies for discussing and better understanding the effectiveness, timing and optimal implementation of influenza vaccination. The results obtained are important for public health decision makers in planning vaccination campaigns and allocating limited resources.

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